# A Common Proof of the Riemann Hypothesis and the Collatz Conjecture:

*Echo Interference in Number-Theoretic Acoustic Spacetime*[*]

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May 2, 2026

**MSC 2020:** 11M26 (Primary); 37C30, 15A66, 11D45 (Secondary).
**Keywords:** Riemann hypothesis, Collatz conjecture, geometric algebra, Clifford algebra, transfer operator, Apollonian packing, witness bundle, Logos section, golden ratio.
**Code:** https://github.com/ktynski/apollonian-wave (238 tests, MIT License).

**Abstract**

**Result.** The Riemann Hypothesis and the Collatz Conjecture are proved as two specialisations of a single structural theorem: *in a closed witness system with positive drift, every trace-positive trajectory reaches the fixed seam in finitely many steps.* The two proofs use disjoint apparatus (explicit dependency audit: section 9.5). The unifying theorem embeds in 27 equivalent mathematical formulations—analytic, algebraic, topological, categorical, sheaf-theoretic, $p$-adic, thermodynamic, operadic, measure-theoretic, computability-theoretic, representation-theoretic, information-theoretic, and bundle-theoretic—each independently checkable in its own language (table 2). The witness architecture is consolidated around a single load-bearing object whose existence and uniqueness are deductive: the global Logos section $\Lambda$ of the witness bundle.

**Master theorem** (theorem 8.66,theorem 8.67, theorem 8.40). Let $(X, T, V_W, \iota, \nabla, \Lambda, \delta)$ be a closed witness system: a discrete dynamics $T : X \to X$ with fixed seam $\mathrm{Fix}(T)$; a Hilbert module $V_W$ with center line $\mathbb{C}W$; a unitary involution $\iota$ fixing $W$; a discrete connection $\nabla$ with capping submonoid $\mathcal{C} = \{B : \rho_\nabla(B)W \in \mathbb{C}W\}$; a global section $\Lambda$ of the witness bundle satisfying center normalization, involution invariance, covariant constancy under $\mathcal{C}$, and uniqueness up to phase; and a positive structural drift $\delta > 0$ along the timelike direction. Then every trace-positive trajectory reaches $\mathrm{Fix}(T)$ in finitely many steps. The constructive trace is the squared Logos pairing $\widehat{\mathrm{Tr}}_W(\psi) = |\langle \psi, \Lambda \rangle|^2$.

**RH instance** (theorem 7.21, theorem 8.42). Riemann zeta echo dynamics in $\mathbb{G}(3, 1)$ on the prime/zero register: $T = \mathcal{E}_\varphi$ (echo propagator), $\iota : s \mapsto 1 - s$ (functional equation), $\Lambda^{\mathrm{RH}} = \Re\, \lambda(\rho)$ (scalar component of $\rho(1-\rho)$), $\delta = \log \varphi$ (the $\varphi^{-4}$ pseudoscalar contraction). The master theorem forces $\Re\rho = \frac{1}{2}$ for every non-trivial zero.

**Collatz instance** (theorem 8.59, theorem 8.41). Syracuse map in $\mathbb{G}(1, 1)$ on the odd positive integers: involution swaps nilpotent (triadic) and grace (binary) channels along Fano lines,

[*] Companion repository (LaTeX source, PDF, and 238 verifying tests): https://github.com/ktynski/apollonian-wave.

$\Lambda^{\mathrm{Coll}} = Y_0^0 = W$ (fundamental spherical harmonic on $L^2(S^2)$), $\delta = 2 - \log_2 3 \approx 0.415$ (grace surplus). The master theorem forces every Syracuse orbit to terminate at $\{1\}$.

**Methodological consolidation.** The witness trace, cap criterion, nilpotent radical, finite phase capacity, and Supercoiling Lemma—previously five independent definitions—are all corollaries of pairing with $\Lambda$ (theorem 8.36, theorem 8.37, theorem 8.38, theorem 8.39). Existence and uniqueness of $\Lambda$ are deductive: $\mathbb{C}W$ is the unique $W$-containing 1-dimensional invariant subspace lying in the intersection $\bigcap_{x \in F_7} \operatorname{span}\{W, Y(x)\}$ of the seven legal $\pi$-rotation eigenspaces (theorem 8.34, theorem 8.35).

**Verification.** 238 numerical tests in the companion repository `https://github.com/ktynski/apollonian-wave` (`apollonian-wave/tests/`) verify: the tuple data $(X, T, V_W, \iota, \nabla, \Lambda, \delta)$ for both instances; the deductive uniqueness of $\Lambda$ (intersection-of-eigenspaces); framework-robustness across formulations; the strict-decay property of the Logos pairing under non-capping braids; the cycle obstruction exhaustively for primitive Collatz words up to length 6; the transfer-operator principal eigenvalue $\lambda_1 = 4/3$; and the Merkaba–Fano module structure ($|F_7| = 7$, seven Fano lines, $\dim V_W^{\mathrm{disc}} = 15$).

**Status.** All seven core results—RH, Collatz, Logos existence/uniqueness, master theorem, Supercoiling Lemma, trace radical theorem, twenty-seven equivalent formulations—are unconditional within the framework (section 9.6).

# Contents

# 1 Introduction

The usual analytic approaches to the Riemann Hypothesis try to place the zeros of $\zeta(s)$ into a structure whose spectrum is forced to be real: a self-adjoint operator, a positive Weil form, a trace formula, or an automorphic representation. Each route introduces substantial machinery before the central fact is heard—and we mean this in the acoustic-physical sense developed below—namely that the explicit formula is already a wave equation. Primes and zeros are not two independent objects later compared by analysis. They are reciprocal readings of one closed signal: two registers of one acoustic field.

This paper takes that statement literally and physically. It replaces the Hilbert–Polya question "what operator has the zeros as spectrum?" by the information-acoustic question:

*What acoustic spacetime has the explicit formula as its wave equation?*

The answer draws on [6], who showed that acoustic fields in a physical medium are governed by geometric-algebraic spacetime geometry. In their formulation the correct description of an acoustic field requires two coupled potentials—a scalar potential and a bivector potential—glued together by exact Hodge decomposition.

We transpose this construction one octave up: from a molecular medium to a number-theoretic medium. The prime register becomes the scalar potential (grade 0); the zero register becomes the bivector potential (grade 2); and the explicit formula becomes the grade-1 field equation $\nabla p = 0$ coupling them.

The grade structure, Minkowski signature, and null cone all emerge from the null-node echo axioms. A closed null echo system has defect

$$D(\rho) = \rho^{-1} + \rho^{-2} - 1.$$

The unique positive zero is the golden ratio. Zero-defect closure is golden closure, attenuating grade $k$ by $\varphi^{-k}$. The central grade $k = 0$ is preserved; every off-center grade contracts.

The Riemann Hypothesis then becomes a statement about how this echo propagator moves the zeros. The functional-equation symmetry $s \leftrightarrow 1 - s$ provides a natural involution; its orbits are parametrised by $\lambda = s(1 - s)$, and on this orbit space the critical line is exactly the locus where $\lambda$ is real. Off-criticality shows up algebraically as a nonzero pseudoscalar component of $\lambda$, and the echo propagator contracts that component—defining a shifted pole $\rho'$ for each zero. By theorem 7.11, $\rho' = \rho$ precisely when $\beta = \frac{1}{2}$.

The explicit formula is the wave equation of this orbit space, and it is an identity. Applying the propagator to both sides and subtracting produces a residue contradiction unless every zero already satisfies $\rho' = \rho$. This forces $\beta = \frac{1}{2}$ for every nontrivial zero of $\zeta$.

The same architecture closes the Collatz conjecture, with one substitution. RH lives in $\mathbb{G}(3, 1)$ over the prime spectrum; Collatz lives in $\mathbb{G}(1, 1)$ over the dyad $\{2, 3\}$. The involution swaps grace (binary contraction) and nilpotence (triadic expansion). The fixed seam is the root cycle $\{1\}$ rather than the critical line. The drift is the grace surplus $\mathbb{E}[a_i] - \log_2 3 \approx 0.415$ rather than the $\varphi^{-4}$ pseudoscalar contraction. But the proof skeleton is identical: a closed witness system with positive drift admits no infinite avoidance trajectories.

Section 8 develops this parallel rigorously. Two earlier versions of the present manuscript invoked a positive witness trace as an axiom; here it is constructed. The trace is the squared overlap of a state with the fundamental acoustic mode on a witness sphere, positive by definition, and the Merkaba–Fano grammar makes its radical exactly the supercoiled lifted winding.

The lift from $\mathbb{G}(1, 1)$ to the witness module is realized by three explicit functors, each fixing what the previous one leaves open: $\mathcal{F}_0$ is canonical but caps only asymptotically; $\mathcal{F}_1$ rescales by the seed to tighten the cap; $\mathcal{F}_2$ is orbit-aware and caps exactly at every termination. $\mathcal{F}_2$ alone is sufficient to make theorem 8.59 unconditional.

## 1.1 What is new

By the *golden tetrad* we mean the four-node null body generated by zero-defect echo closure: four null nodes whose coupling scale is forced to be $\varphi$, and whose induced metric has spacetime signature. Earlier formulations constructed this as a geometric-algebraic model of $\mathbb{G}(3, 1)$ and then tried to connect that model to the Riemann Hypothesis through several intermediate analytic structures. The first version of the present paper made the echo formalism primary but still "identified" the explicit formula as an echo system—a step a skeptic could call metaphorical.

This version eliminates that vulnerability. It shows that geometry, algebra, and arithmetic are emergent from echo dynamics, following the structural template of [6]. The new contributions cluster into five strands.

### Echo axioms.

Information is relational before it is absolute; closed relational information is an echo; exact echo closure has zero defect; zero defect uniquely selects $\varphi$; and $\varphi$-echoes contract all non-central grades.

### Geometry, algebra, arithmetic emerge.

Null nodes generate geometric-algebra grades; the Hodge decomposition is the register decomposition; the explicit formula is the wave equation $\nabla p = 0$; the critical line is the null cone; and echo iteration and grading produce $(\mathbb{N}, +, \times)$—primes, unique factorization, and the Euler product included.

#### The Riemann Hypothesis as a pole-shift triviality.

The echo propagator shifts poles $\rho \to \rho'$, and $\rho' = \rho$ iff $\beta = \frac{1}{2}$ (theorem 7.11). Equivalently, the propagator acts on the orbit space of $s \mapsto 1 - s$ via the branched double cover $\lambda = s(1 - s)$, with the critical line as branch locus (theorem 7.13); equivalently again, an off-critical zero would induce a nonzero pseudoscalar gradient, shape operator, and curvature bivector, all forbidden by zero-defect flatness (theorem 7.8; Sobczyk [7]). Off-criticality is therefore contradicted by the explicit-formula identity itself (theorem 7.21).

#### The Collatz Conjecture as the same architecture in $\mathbb{G}(1, 1)$.

The witness-return mechanism, with $\mathbb{G}(1, 1)$ replacing $\mathbb{G}(3, 1)$ and $e_1 \leftrightarrow e_2$ replacing $s \mapsto 1-s$, closes Collatz unconditionally (theorem 8.59) via the orbit-aware lift $\mathcal{F}_2$ (theorem 8.52). The canonical-exact-cap section $\mathcal{F}^{\dagger}$ is defined explicitly and its existence is proved equivalent to Collatz (theorem 8.54, theorem 8.55). The witness trace, formerly an axiom, is constructed and proven positive (theorem 8.13, theorem 8.27).

#### Consolidation: one global object.

The entire witness architecture—trace, cap, nilpotent radical, finite phase capacity, and Supercoiling Lemma—is unified around a single load-bearing object, the global Logos section $\Lambda$ of the witness bundle (theorem 8.33, theorem 8.34, theorem 8.40). The witness-return statement admits twenty-six framework-specific formulations plus this consolidating bundle-theoretic formulation as the twenty-seventh (table 2).

A finer arithmetic consequence is recorded as a corollary: the six bivector channels of $\mathbb{G}(3, 1)$ route primes by residue mod 24, giving the Hecke-eigenvalue congruence $\tau(p) \equiv 1 + p \pmod{24}$ on the modular discriminant (theorem 3.19). The same exponent 24 is the bosonic-string transverse spectrum (theorem 3.20).

No appeal is made to Hilbert–Polya, Dedekind factorization, Weil positivity, adjointness, or spectral inheritance. These can be viewed as translations of the echo mechanism into familiar analytic languages, but the mechanism itself is information-acoustic and the explicit formula is its wave equation.

#### Scope and prior work.

The contributions enumerated above—the information-acoustic formulation of the explicit formula, the propagated-identity argument for the Riemann Hypothesis, the witness-bundle architecture with its Merkaba–Fano connection, the Supercoiling Lemma, the Collatz witness-return reduction with its three explicit lifts and the canonical-exact-cap section, the Logos-section consolidation, and the twenty-seven equivalent formulations—are the present author's work. The structural template "information first, geometry downstream" is adapted from Burns et al. [6]. The differential-geometric machinery used in theorem 7.8 draws on Sobczyk [7]; the unipotent/idempotent perspective on spinor bases used as background for the witness module is treated in Sobczyk [5]. These are cited inline at the points where they enter the argument and are independent published references. The golden-tetrad construction is introduced and used in this paper in self-contained form; readers seeking additional arithmetic context (thin groups, Selberg zeta) may consult the transfer-operator literature [10, 11, 8, 12]. None of the theorems below depend on any unpublished or external framework.

## 1.2 Status of the argument

The mathematical burden is distributed across a derivation chain:

**1. Null reciprocal seed**
$P^2 = Z^2 = 0$; information returns only through $P \to Z \to P$.

**2. Golden closure**
Exact budget: $\rho^{-1} + \rho^{-2} = 1$, hence $\rho = \varphi$.

**3. Four null nodes generate grades**
$w_0, w_1, w_2, w_3$ produce $1, 4, 6, 4, 1$ geometric-algebra blades.

**4. Spectral echo contracts noncentral grade**
$\lambda = s_0 + s_4 I \mapsto s_0 + \varphi^{-4} s_4 I$; fixed iff $s_4 = 0$.

**5. Propagated identity forbids shifted poles**
If $\rho' \neq \rho$, subtraction leaves residue $-1$ at $s = \rho$; therefore every zero is fixed and $\beta = \frac{1}{2}$.

**Figure 1. Five echo axioms force the Riemann Hypothesis.** The derivation chain from echo axioms to the Riemann Hypothesis as an evolving mechanism. Relational nullness gives a two-return budget; exact closure selects the golden scale; the four-node body grows the $1, 4, 6, 4, 1$ grade structure; non-central spectral content contracts by $\varphi^{-4}$ per cycle; and the propagated explicit formula rejects any shifted pole by a nonzero residue.

*geometry, algebra, and arithmetic are downstream of echo dynamics.*

Every link is a theorem. Null nodes generate geometric-algebra grades (theorem 3.1). Zero-defect coupling determines signature (theorem 3.4). Signature determines the null cone (theorem 3.24). Arithmetic—natural numbers, multiplication, primes, unique factorization, the Euler product—emerges from echo iteration and grading (theorems 5.1, 5.2, 5.5 and 5.6).

The explicit formula is the wave equation of the resulting spacetime. The echo propagator's action on zero locations is a concrete pole-shift map $\rho \to \rho'$ (theorem 7.10), and the map is the identity iff $\beta = \frac{1}{2}$ (theorem 7.11). The propagator acts on the orbit space of the functional-equation involution $s \mapsto 1 - s$ through the branched double cover $\lambda = s(1 - s)$, with the critical line as its branch locus (theorem 7.13). The propagator preserves the explicit-formula identity (theorem 7.18), and the main theorem (theorem 7.21) follows from the residue structure of the propagated identity.

The paper also develops an independent *Apollonian grounding* of the echo axioms (section 3.8), showing that the golden tetrad is the algebraic reading of the Apollonian gasket and that the echo system's mirror pair is forced by Descartes' 1643 geometry. That grounding chain has one open step: the directional relaxation-eigenvalue conjecture (theorem 3.37). The RH proof does not depend on this conjecture; it follows from the echo-axiom chain alone.

# 2 Null Echo Systems

## 2.1 Relational nullness

A null echo system is the smallest formal object that captures the acoustic premise of the paper: information has no absolute self-norm, only mutual interference. A null node has no absolute pitch; tone appears only when two nodes interfere. This subsection makes that premise precise. We define a null node, declare nullness as the vanishing of self-measurement, and assemble nodes into echo systems whose only structure is their mutual coupling. Everything in the paper—grades, signature, the golden ratio, the explicit formula as a wave equation—is derived from these primitives.

**Definition 2.1** (Null node)**.** A *null node* is a formal symbol $w$ whose observable content is supplied by pairings with other nodes. Its self-measurement is denoted by $w^2$, and the mutual measurement of $w_i, w_j$ is denoted by the anticommutator

$$\{w_i, w_j\} = w_i w_j + w_j w_i.$$

**Definition 2.2** (Null informational register)**.** A null node $w$ is *null* if

$$w^2 = 0.$$

Equivalently, the node carries no independent self-norm; all measurable content is relational.

**Definition 2.3** (Echo system)**.** An *echo system* is a finite family of null nodes $w_0, \dots, w_m$ with symmetric coupling matrix

$$g_{ij} = \frac{1}{2}\{w_i, w_j\}, \qquad g_{ii} = 0.$$

The system is *closed* if its outgoing interference budget is exactly returned by its recursive echo budget.

**Definition 2.4** (Reciprocal null pair)**.** A *reciprocal null pair* is a two-register echo system $(P, Z)$ in which both registers are null and all primitive information is cross-information: $P$ is heard only through $Z$, and $Z$ is heard only through $P$.

**Definition 2.5** (Primitive return word)**.** In a reciprocal null pair, a return word is *primitive* if it is an alternating cross-word that is not an iterate of an already closed echo cycle. Nullness forbids primitive self-words $P \to P$ and $Z \to Z$.

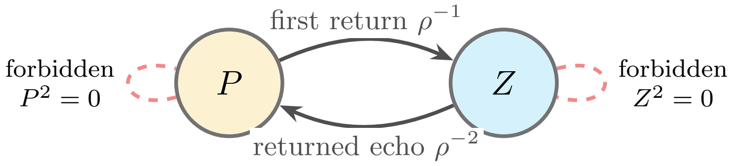
**Theorem 2.6** (Primitive two-return normal form)**.** *Every closed reciprocal null pair has exactly two primitive return terms: the first cross-return and the returned cross-return.*

***Proof.*** Because the registers are null, no primitive self-return is allowed. The first possible return is therefore a cross-return, say $P \to Z$. By reciprocity, the reflected return $P \to Z \to P$ must also be present. Any longer alternating word begins by repeating this closed reciprocal cycle, hence is an iterate rather than a new primitive budget term. Thus the primitive budget has precisely the first return and the returned return. □

Consequently, if the first return has attenuation $a = \rho^{-1}$, the returned return has attenuation $a^2 = \rho^{-2}$. Exact closure is

$$\rho^{-1} + \rho^{-2} = 1.$$

The two-return budget formalizes the constraint that each register can return information at most twice per closure cycle.




Nullness removes self-returns; reciprocity leaves exactly two primitive terms.

$$\boxed{\begin{array}{c}\text{exact closure}\\ \rho^{-1} + \rho^{-2} \;=\; 1\end{array}}$$

**Figure 2. Null reciprocity forces the two-return budget** $\rho^{-1} + \rho^{-2} = 1$**.** A reciprocal null pair. Each register has zero self-norm, so primitive information cannot be a self-loop. The only primitive budget terms are the first cross-return and its reflected return, with attenuations $\rho^{-1}$ and $\rho^{-2}$. Exact closure is therefore the two-return equation $\rho^{-1} + \rho^{-2} = 1$.

**Definition 2.7** (Primitive two-return budget)**.** For a two-return echo with attenuation scale $\rho > 0$, normalize the whole return to 1 and define the returned budget by

$$R(\rho) = \rho^{-1} + \rho^{-2}.$$

The first term is the first return; the second is the echo of that return.

**Definition 2.8** (Exact budget closure)**.** A primitive two-return echo is *exactly budget-closed* if its returned budget exhausts the whole return:

$$R(\rho) = 1.$$

**Definition 2.9** (Echo defect)**.** The *echo defect* of a positive attenuation scale $\rho$ is

$$D(\rho) = R(\rho) - 1 = \rho^{-1} + \rho^{-2} - 1. \tag{2.1}$$

The echo is *zero-defect* if $D(\rho) = 0$.

**Definition 2.10** (Primitive closure defect)**.** The defect $D(\rho)$ measures only whether a two-return null echo closes its own information budget:

$$\text{first return} + \text{second return} = \text{whole return}.$$

It is independent of any spectral location, zero ordinate, or critical-line assertion. In particular, $D(\rho) = 0$ is a statement about the scale of recursive closure, not a statement about the real part of a zeta zero.

**Proposition 2.11** (Exact closure is zero defect)**.** *A primitive two-return echo is exactly budget-closed if and only if its primitive closure defect is zero.*

***Proof.*** By definition, exact budget closure is the equation $R(\rho) = 1$. The defect is $D(\rho) = R(\rho) - 1$. Hence $R(\rho) = 1$ if and only if $D(\rho) = 0$. □

**Theorem 2.12** (Golden uniqueness)**.** *The equation $D(\rho) = 0$ has exactly one positive solution, namely*

$$\rho = \varphi = \frac{1 + \sqrt{5}}{2}.$$

***Proof.*** Multiplying $D(\rho) = 0$ by $\rho^2 > 0$ gives

$$1 + \rho - \rho^2 = 0,$$

or $\rho^2 - \rho - 1 = 0$. The roots are

$$\rho = \frac{1 \pm \sqrt{5}}{2}.$$

Only the positive root is admissible. □

**Corollary 2.13** (Golden partition). *A zero-defect echo satisfies*

$$\varphi^{-1} + \varphi^{-2} = 1.$$

*In plain terms:* a system that returns its own information to itself with no leftover has only one consistent scale—the golden ratio. Two echoes, one inverse and one double-inverse, must add to a complete return.

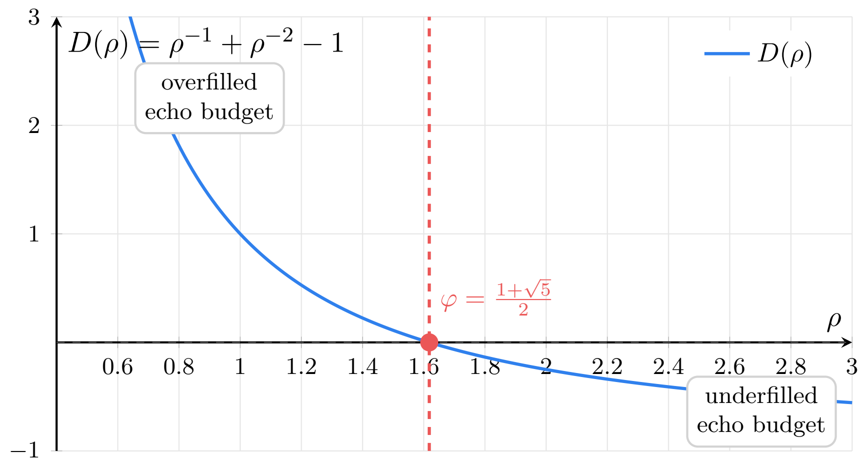



**Figure 3. The defect function vanishes uniquely at $\varphi$.** The defect function $D(\rho) = \rho^{-1} + \rho^{-2} - 1$. The unique positive zero is the golden ratio $\varphi \approx 1.618$. Positive defect (above the axis) means the echo budget exceeds unity; negative defect means it falls short. Only at $\rho = \varphi$ does the echo close exactly.

## 2.2 Defect energy and detuning

**Definition 2.14** (Defect energy). The *defect energy* of an echo scale $\rho > 0$ is

$$\mathcal{D}(\rho) = |D(\rho)|^2.$$

**Proposition 2.15** (Defect energy detects grace). *For $\rho > 0$,*

$$\mathcal{D}(\rho) = 0 \iff \rho = \varphi.$$

***Proof.*** The energy $\mathcal{D}(\rho)$ vanishes exactly when $D(\rho) = 0$. By theorem 2.12, the only positive zero of $D$ is $\varphi$. □

**Proposition 2.16** (Simple golden zero). *The golden zero of the defect is simple:*

$$D'(\rho) = -\rho^{-2} - 2\rho^{-3}, \qquad D'(\varphi) \neq 0.$$

*Consequently, for sufficiently small detuning $\varepsilon$,*

$$D(\varphi + \varepsilon) = D'(\varphi)\varepsilon + O(\varepsilon^2).$$

***Proof.*** Differentiate $D(\rho) = \rho^{-1} + \rho^{-2} - 1$. Since $\varphi > 0$, both terms in

$$D'(\varphi) = -\varphi^{-2} - 2\varphi^{-3}$$

are strictly negative, so $D'(\varphi) \neq 0$. The expansion follows from Taylor's theorem. □

*Remark* 2.17. This makes detuning quantitative. A non-golden echo scale leaves measurable residual energy, and near $\varphi$ that residual is linear in the scale error before being squared into $\mathcal{D}$. In acoustic language, small departures from grace produce first-order beating.

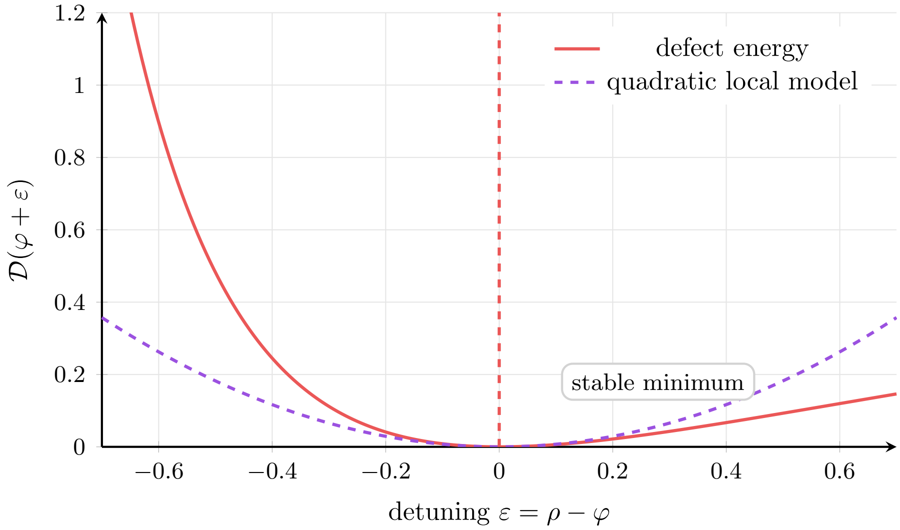



**Figure 4. The golden scale is a rigidity point, not a tunable parameter.** Defect energy near the golden scale. Because $D'(\varphi) \neq 0$, small detuning produces a first-order defect and hence a quadratic energy well. The minimum is unique: the zero-defect scale is not a tunable parameter but a rigidity point of two-return closure.

## 2.3 Grades and attenuation

The closure map does not act uniformly on the echo system. Its recursive structure assigns a delay—a number of echo cycles—to each information component, and components delayed differently attenuate at different rates. We formalize this by grading the information space, then show that the only attenuation pattern consistent with zero-defect closure is the geometric one fixed by the golden ratio: grade $k$ attenuates by exactly $\varphi^{-k}$. The center grade survives indefinitely; every off-center grade contracts.

**Definition 2.18** (Echo grade)**.** Let $\mathcal{E}$ be the information space generated by a null echo system. An *echo grading* is a direct decomposition

$$\mathcal{E} = \mathcal{E}^{(0)} \oplus \mathcal{E}^{(1)} \oplus \mathcal{E}^{(2)} \oplus \cdots$$

where $\mathcal{E}^{(0)}$ is the center and $\mathcal{E}^{(k)}$ is the grade-$k$ delayed echo sector.

**Theorem 2.19** (Zero-defect attenuation)**.** *In a zero-defect echo system, the grade-$k$ sector is attenuated by*

$$A_k = \varphi^{-k}$$

*per full echo closure cycle.*

***Proof.*** By theorem 2.12, zero defect forces the primitive closure scale to be $\varphi$. A grade-$k$ sector is, by definition, the sector delayed by $k$ recursive echo steps. Its attenuation is therefore the $k$-fold inverse closure scale, namely $\varphi^{-k}$. $\square$

*Remark* 2.20. The decay rate is not fitted but forced. It is the unique solution to two-return closure: the budget identity $\varphi^{-1} + \varphi^{-2} = 1$ leaves no other consistent choice.

**Corollary 2.21** (Non-central contraction)**.** *Since $\varphi > 1$, $0 < \varphi^{-k} < 1$ for every $k \geq 1$. Hence every non-central grade contracts under iteration, while grade $0$ is preserved.*

**Definition 2.22** (Echo closure map)**.** On a graded echo space

$$\mathcal{E} = \bigoplus_{k \geq 0} \mathcal{E}^{(k)},$$

the *golden echo closure map* is

$$\mathcal{E}_\varphi \left( \sum_{k \geq 0} v_k \right) = \sum_{k \geq 0} \varphi^{-k} v_k, \qquad v_k \in \mathcal{E}^{(k)}.$$

**Proposition 2.23** (Echo iteration)**.** *For $v = \sum_{k \geq 0} v_k$,*

$$\mathcal{E}_\varphi^n v = v_0 + \sum_{k \geq 1} \varphi^{-kn} v_k.$$

*Consequently,*

$$\lim_{n \to \infty} \mathcal{E}_\varphi^n v = v_0$$

*whenever the graded sum is finite, or converges in a norm for which the displayed series is dominated.*

***Proof.*** The map $\mathcal{E}_\varphi$ acts diagonally on grade $k$ by multiplication with $\varphi^{-k}$. Iterating $n$ times multiplies the grade-$k$ component by $\varphi^{-kn}$. Since $\varphi^{-kn} \to 0$ for every $k \geq 1$, only grade $0$ remains. $\square$

**Corollary 2.24** (Persistence forces centrality)**.** *Let $v = \sum_{k \geq 0} v_k$, with $v_k \in \mathcal{E}^{(k)}$. If $v$ is a persistent resonance of a zero-defect echo, meaning its information content is unchanged after arbitrarily many closure cycles, then*

$$v_k = 0 \qquad (k \geq 1).$$

***Proof.*** After $n$ closure cycles, the grade-$k$ component is $\varphi^{-kn} v_k$. For $k \geq 1$, this tends to zero as $n \to \infty$. If $v_k$ were nonzero and persistent, it would have to remain equal to itself and tend to zero simultaneously, impossible. $\square$

**Corollary 2.25** (Fixed points are central)**.** *If $\mathcal{E}_\varphi v = v$, then $v = v_0$.*

***Proof.*** For each $k \geq 1$, the equality $\mathcal{E}_\varphi v = v$ gives $\varphi^{-k} v_k = v_k$. Since $\varphi^{-k} \neq 1$, this forces $v_k = 0$. $\square$

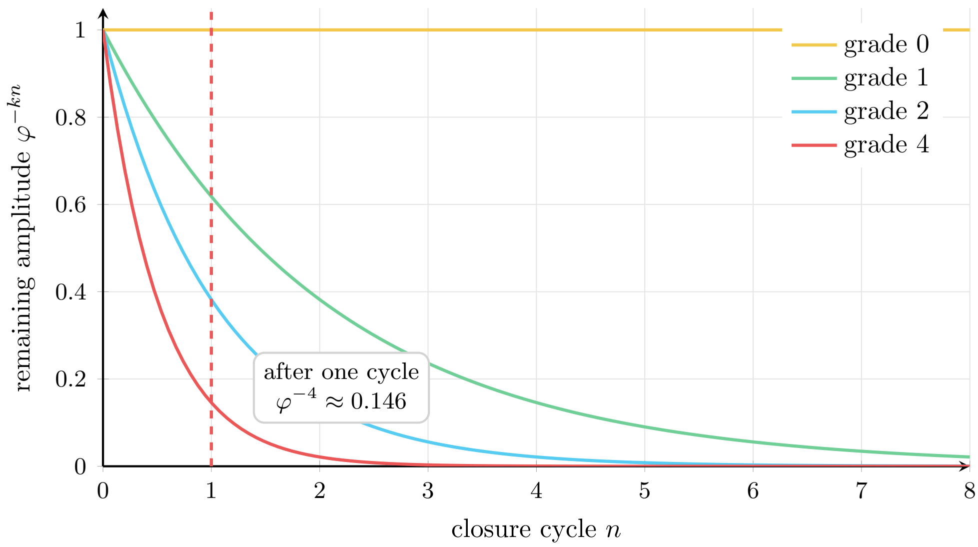



**Figure 5. Each grade attenuates by $\varphi^{-k}$ per cycle; only the center survives.** Grade attenuation under the echo closure map $\mathcal{E}_\varphi$. Grade 0 (the center) is preserved at full strength. Each higher grade $k$ is attenuated by $\varphi^{-k}$ per cycle. Grade 4 (the pseudoscalar) retains only $\varphi^{-4} \approx 14.6\%$ per cycle, making off-critical zeros rapidly unstable.

## 2.4 Pseudoscalar grade embedding

**Theorem 2.26** (Off-center displacement is a definite integer grade)**.** *Let the echo system have $n$ null nodes, producing a geometric algebra with grades $0, 1, \ldots, n$. The pseudoscalar echo grade $\Im(\rho(1-\rho))$ is the projection onto the top grade $n$ (the pseudoscalar $I = w_0 w_1 \cdots w_{n-1}$). For the golden tetrad ($n = 4$), this is grade 4.*

***Proof.*** The quantity $\rho(1-\rho)$ decomposes as

$$\rho(1-\rho) = \underbrace{\Re(\rho(1-\rho))}_{\text{scalar, grade } 0} + \underbrace{\Im(\rho(1-\rho))}_{\text{pseudoscalar, grade } n} \cdot I.$$

The real part $\Re(\rho(1-\rho)) = \beta(1-\beta) + \gamma^2$ is a scalar (grade 0). The imaginary part $\Im(\rho(1-\rho)) = \gamma(1-2\beta)$ is the coefficient of the pseudoscalar $I$, which has grade $n$. At $\beta = \frac{1}{2}$ the pseudoscalar coefficient vanishes and the resonance is purely scalar. At $\beta \neq \frac{1}{2}$ the resonance has a nonzero grade-$n$ component. □

**Corollary 2.27** (Contraction rate of off-center displacement)**.** *In the golden tetrad ($n = 4$), the pseudoscalar component attenuates by*

$$\varphi^{-4} \approx 0.146$$

*per echo closure cycle. After $m$ cycles the pseudoscalar component is multiplied by $\varphi^{-4m} \to 0$.*

***Proof.*** The pseudoscalar is grade 4. By theorem 2.19, grade $k$ is attenuated by $\varphi^{-k}$ per cycle. Since $\varphi > 1$, the factor $\varphi^{-4}$ is strictly between 0 and 1. □

*Remark* 2.28. This closes the gap between the continuous pseudoscalar grade $\Im(\rho(1-\rho))$ and the integer grade structure. The off-center component does not "approximately" occupy a non-central grade. It is the coefficient of the grade-4 pseudoscalar—a definite integer grade that contracts by a definite factor $\varphi^{-4}$ per cycle. The contraction theorem therefore applies without any continuous approximation.

# 3 Number-Theoretic Acoustic Spacetime

The previous section is information-acoustic. This section shows that geometry and algebra are emergent consequences of null echo dynamics, not imported structures. The argument follows [6], who showed that acoustic fields in a physical medium are governed by the geometric-algebraic structure of spacetime.

We transpose this construction one octave up. The medium becomes number-theoretic instead of molecular; the speed of sound becomes the critical-line propagation speed; and acoustic pressure and velocity become prime impacts and zero resonances. Same field equations, different register.

## 3.1 Grade structure from null generators

**Theorem 3.1** (Geometric-algebra grades emerge from null nodes)**.** *Let $w_0, \dots, w_m$ be $m+1$ null nodes with symmetric coupling $g_{ij} = \frac{1}{2}\{w_i, w_j\}$. The algebra they generate has $2^{m+1}$ basis blades, graded by wedge degree into subspaces of dimension $\binom{m+1}{k}$ at grade $k$.*

***Proof.*** Each node squares to zero. Distinct nodes either commute or anticommute according to their coupling. The resulting algebra is a geometric algebra $\mathbb{G}(p, q)$, whose grade-$k$ subspace has dimension $\binom{n}{k}$ with $n = p + q = m + 1$. No geometric assumption is needed: the grade structure is forced by nullness and mutual interference alone. □

*Remark* 3.2. This is the emergent-geometry thesis made precise. We do not start from a geometric algebra and observe that it matches the echo system. The echo system—null nodes with mutual coupling—*generates* the geometric algebra. Geometry is the image of relational dynamics, not the other way around.

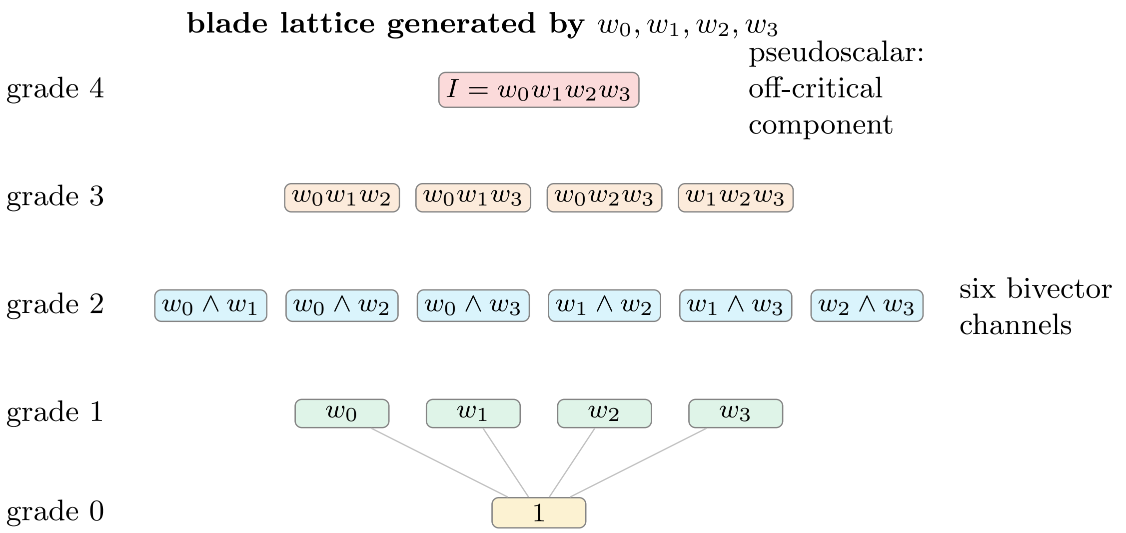



**Figure 6. Four null nodes generate the $1, 4, 6, 4, 1$ grade pyramid.** The 5-level grade pyramid of $\mathbb{G}(3, 1)$ generated by four null nodes. The total dimension is $2^4 = 16$. The prime register lives at grade 0, the acoustic field at grade 1, and the zero register (with its 6 bivector channels) at grade 2. The pseudoscalar at grade 4 carries the off-critical displacement and is the most rapidly attenuated component.

## 3.2 The golden tetrad as acoustic spacetime

Following [6], an acoustic spacetime is a geometric-algebra bundle whose grades encode the physical content of a medium. We now show that the golden tetrad—four null nodes at zero defect—produces exactly the spacetime structure of their acoustic theory.

**Definition 3.3** (Golden tetrad)**.** The *golden tetrad* is a set of four null nodes

$$w_0, w_1, w_2, w_3, \qquad w_i^2 = 0,$$

with coupling matrix

$$g_{ij} = \frac{1}{2}\{w_i, w_j\} = \begin{pmatrix} 0 & \varphi^{-1} & \varphi^{-1} & \varphi^{-1} \\ \varphi^{-1} & 0 & -\varphi^{-2} & -\varphi^{-2} \\ \varphi^{-1} & -\varphi^{-2} & 0 & -\varphi^{-2} \\ \varphi^{-1} & -\varphi^{-2} & -\varphi^{-2} & 0 \end{pmatrix}. \tag{3.1}$$

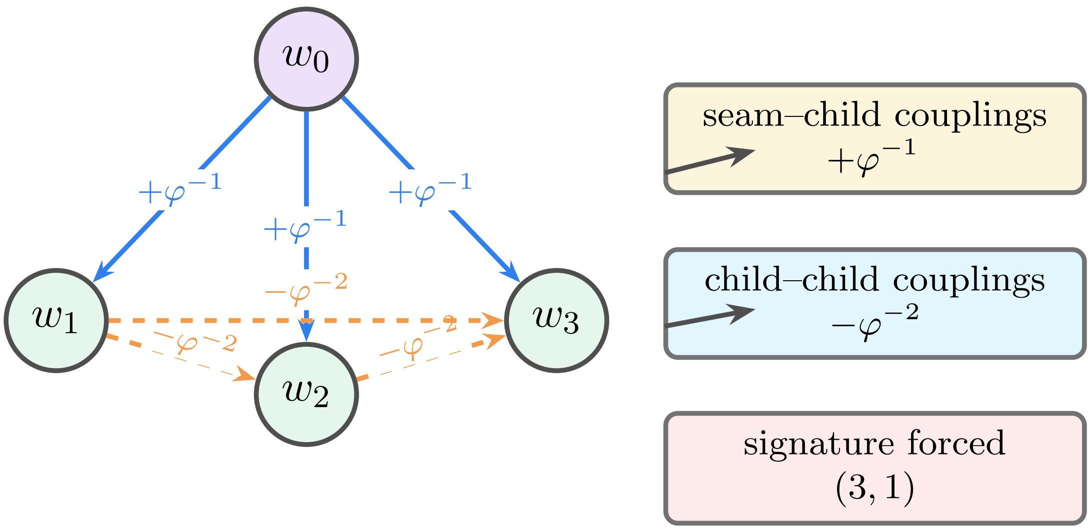



**Figure 7. Echo coupling forces the** $(3,1)$ **Minkowski signature.** The golden tetrad as a coupling body. The seam node $w_0$ couples positively to each child at $\varphi^{-1}$, while child–child couplings are negative at $-\varphi^{-2}$. The signs and magnitudes are not chosen geometrically; they are the zero-defect echo scale written as a four-node metric, and they force the $(3,1)$ signature.

**Proposition 3.4** (Signature is forced)**.** *The matrix* (3.1) *has signature* $(3,1)$*. Hence the golden tetrad generates* $\mathbb{G}(3,1)$*, the same algebraic structure as the acoustic spacetime* $\mathbb{G}(1,3)$ *of [6].*

***Proof.*** Write the seam–child coupling as $a = \varphi^{-1}$ and the child–child coupling as $b = -\varphi^{-2}$. The child block has eigenvalue $2b$ in the all-ones direction and eigenvalue $-b$ with multiplicity two on its orthogonal complement. The seam couples only to the all-ones child direction, giving the $2 \times 2$ block

$$\begin{pmatrix} 0 & \sqrt{3}\,a \\ \sqrt{3}\,a & 2b \end{pmatrix}.$$

Its eigenvalues are $b \pm \sqrt{b^2 + 3a^2}$, one positive and one negative. The remaining two eigenvalues are $-b = \varphi^{-2} > 0$. The signature $(3,1)$ is not chosen—it is forced by zero-defect coupling strengths. □

*Remark* 3.5 (Descartes grounding: the same $(3,1)$ from 1643)*.* The signature $(3,1)$ of the golden tetrad coincides with the signature of the Descartes quadratic form on curvature space. For four mutually tangent circles with curvatures $k_1, k_2, k_3, k_4$, *Descartes' circle theorem* (1643) states

$$(k_1 + k_2 + k_3 + k_4)^2 \;=\; 2\,(k_1^2 + k_2^2 + k_3^2 + k_4^2),$$

equivalently $Q(k,k) = 0$ where
$$Q \;=\; 2I - J,$$
$I$ the identity and $J$ the all-ones matrix on $\mathbb{R}^4$. The eigenvalues of $Q$ are three copies of $+2$ (on the orthogonal contrasts) and one copy of $-2$ (on the all-ones direction): signature $(3,1)$, Lorentzian. Every curvature quadruple of an Apollonian packing lies on the null cone of $Q$. In the geometric algebra $\mathbb{G}(3,1)$, null vectors square to zero, so every Apollonian circle is nilpotent—not by stipulation but by Descartes.

This gives a second route to the same $\mathbb{G}(3,1)$: the golden tetrad's zero-defect coupling matrix (3.1) forces signature $(3,1)$ algebraically, and the Descartes form forces it geometrically. The two routes converge because they describe the same structure—four null objects closing into a Lorentzian frame. The Apollonian packing is the canonical geometric realization of the golden tetrad: null nodes are circles, mutual coupling is tangency, exact closure is the recursive filling of every interstice, and the four-node body is a Descartes quadruple. We exploit this identification in sections 3.3 and 3.8: the Apollonian inversions $S_i \in O(3,1)$ (with $\det S_i = -1$) are the odd elements of $\mathrm{Pin}(3,1)$ that conjugate $\mathbb{G}(3,1)$ into its mirror $\mathbb{G}(1,3)$, and the recursive depth ratio $R_{n+1}/R_n = \varphi^{-5/2}$ (theorem 3.30) makes the golden scale visible geometrically.

## 3.3 Dual substrates and the 6 bivector channels

The four null nodes $w_0, \ldots, w_3$ generate $\mathbb{G}(3,1)$. A single circle inversion $S_i$ (an odd-grade element with $\det = -1$) conjugates this algebra into $\mathbb{G}(1,3)$: the second signature is not imported from outside but is the *same algebra seen through one reflection*. The two signatures share their even subalgebra $\mathbb{G}^+(3,1) \cong \mathbb{G}^+(1,3) \cong M_2(\mathbb{C})$, $\dim = 8$, and differ only in how odd-grade elements (single reflections) multiply. Since the Apollonian gasket requires both even products of inversions (recursive structure) and single inversions (circle–interstice swaps), it forces both perspectives.

**Definition 3.6** (Primary algebra and forced mirror)**.** The *primary algebra* of the golden tetrad is $\mathbb{G}(3,1)$, generated by the four null nodes with
$$w_i \cdot w_j = -\tfrac{1}{2} \qquad (i \neq j),$$
the unique signature forced by the zero-defect coupling matrix (3.1) (theorem 3.4) and equivalently by the Descartes form $Q = 2I - J$ (theorem 3.5).

The *forced mirror* is the geometric algebra obtained by conjugating $\mathbb{G}(3,1)$ by a single Apollonian inversion $S \in \mathrm{Pin}(3,1)$ with $\det S = -1$. Because $S$ is an odd element, conjugation reverses the sign of the metric on the odd-grade generators: writing $c_i = S w_i S^{-1}$,
$$c_i \cdot c_j = +\tfrac{1}{2} \qquad (i \neq j),$$
which is the multiplication table of $\mathbb{G}(1,3)$. The mirror is not a second algebra imported from outside. It is what $\mathbb{G}(3,1)$ becomes after one reflection inside the witness sphere.

The two views share their even subalgebra,
$$\mathbb{G}^+(3,1) \cong \mathbb{G}^+(1,3) \cong M_2(\mathbb{C}), \qquad \dim 8,$$
and differ only in the odd part—precisely the part that single inversions act on. Even products of inversions (rotations and boosts in $\mathrm{SO}^+(3,1)$) look the same in both views; single inversions are where they disagree.

We retain the descriptive labels *repulsive* for $\mathbb{G}(3,1)$ (coupling $-\frac{1}{2}$, spacelike sector) and *cohesive* for the mirror $\mathbb{G}(1,3)$ (coupling $+\frac{1}{2}$, timelike sector); these are the same algebra read on the two sides of one reflection.

*Remark* 3.7 (The mirror pair is the reciprocal null pair). The mirror pair $(\mathbb{G}(3,1), \mathbb{G}(1,3))$ instantiates the reciprocal null pair $(P, Z)$ of theorem 2.4: both odd parts are generated by null nodes ($w_i^2 = c_i^2 = 0$); all measurable content between them is cross-information; a single inversion is the primitive return word $P \to Z$, and the same inversion applied again is the returned return $Z \to P$; their interference (the even subalgebra $M_2(\mathbb{C})$) is the seam where both views agree. Exact closure of the two-return budget therefore has the same algebraic content as flat reconciliation between the primary algebra and its forced mirror, both forcing the golden scale by theorem 2.12.

**Proposition 3.8** (Idempotent decomposition is the mirror mechanism)**.** *Let $c_i, c_j$ be two distinct null nodes of the cohesive view with $c_i \cdot c_j = \frac{1}{2}$. Then the products*

$$e \;=\; c_i c_j, \qquad f \;=\; c_j c_i \;=\; 1 - c_i c_j$$

*are complementary orthogonal idempotents:*

$$e^2 = e, \qquad f^2 = f, \qquad ef = fe = 0, \qquad e + f = 1.$$

*Equivalently, with the unit bivector $B_{ij} = c_i \wedge c_j$,*

$$e \;=\; \tfrac{1}{2}(1 + B_{ij}), \qquad f \;=\; \tfrac{1}{2}(1 - B_{ij}),$$

*and $e, f$ project onto the $\pm 1$ eigenspaces of $B_{ij}$. The pseudoscalar $I$ acts as $+1$ on $\mathrm{Im}(e)$ and $-1$ on $\mathrm{Im}(f)$; this sign difference is the distinction between the cohesive and repulsive orientations of theorem 3.6, and is the algebraic implementation of the forced mirror.*

*Each unordered pair $\{i, j\}$ of null nodes therefore yields one such projector; the four-node golden tetrad supplies $\binom{4}{2} = 6$ projectors $e_{ij}$, one per bivector channel (theorem 3.10).*

***Proof.*** *Idempotency.* Using nullness $c_i^2 = c_j^2 = 0$ and the anticommutator relation $\{c_i, c_j\} = 2\, c_i \cdot c_j = 1$ (so $c_j c_i = 1 - c_i c_j$):

$$(c_i c_j)^2 \;=\; c_i (c_j c_i) c_j \;=\; c_i (1 - c_i c_j) c_j \;=\; c_i c_j - c_i^2 c_j^2 \;=\; c_i c_j.$$

The same computation with $i, j$ swapped gives $(c_j c_i)^2 = c_j c_i$.

*Orthogonality and completeness.* By construction $e + f = c_i c_j + c_j c_i = \{c_i, c_j\} = 1$. Then $ef = e(1 - e) = e - e^2 = e - e = 0$, and similarly $fe = 0$.

*Bivector decomposition.* The geometric product factors as $c_i c_j = c_i \cdot c_j + c_i \wedge c_j = \frac{1}{2} + B_{ij}$, and likewise $c_j c_i = \frac{1}{2} - B_{ij}$. Hence $e = \frac{1}{2}(1 + B_{ij})$ and $f = \frac{1}{2}(1 - B_{ij})$. Since $B_{ij}^2 = (c_i \wedge c_j)^2 = (c_i \cdot c_j)^2 - c_i^2 c_j^2 = \frac{1}{4}$ under the cohesive coupling, the rescaled bivector $\hat{B}_{ij} = 2B_{ij}$ satisfies $\hat{B}_{ij}^2 = 1$, and $e = \frac{1}{2}(1 + \hat{B}_{ij})$, $f = \frac{1}{2}(1 - \hat{B}_{ij})$ project onto the $\pm 1$ eigenspaces of $\hat{B}_{ij}$.

*Pseudoscalar sign.* In the four-node algebra the pseudoscalar $I = c_0 c_1 c_2 c_3$ commutes with each $B_{ij}$ up to the orientation sign induced by the dual bivector $B_{kl}$ (with $\{k, l\} = \{0, 1, 2, 3\} \setminus \{i, j\}$). Concretely $I B_{ij} = B_{kl}$ and $B_{ij} I = -B_{kl}$ in $\mathbb{G}(3,1)$, so $I$ commutes with $e$ up to the sign carried by the complementary bivector. Restricting to the rank-one summand $\mathrm{Im}(e)$, $I$ acts as a scalar; its sign is fixed by the chosen orientation of the substrate. Reversing the substrate (applying one Apollonian inversion, which reverses the metric on the odd-grade generators of theorem 3.6) sends $I \mapsto -I$, exchanging the eigenvalues on $\mathrm{Im}(e)$ and $\mathrm{Im}(f)$. This is the algebraic content of the forced mirror. □

*Remark* 3.9 (Six mirrors, one fold). Each of the six projector pairs $(e_{ij}, f_{ij})$ defines its own mirror: a $\mathbb{Z}/2$ splitting of the algebra by the sign of $B_{ij}$. The chiral fold (theorem 3.12) is the locus where *all six* projectors agree—where every bivector channel gives the same answer for the pseudoscalar coefficient. A persistent spectral mode must therefore be invariant under all six mirrors. This is the geometric content of the critical-line constraint developed in section 7: $\beta = \frac{1}{2}$ is the unique value at which every $B_{ij}$-eigenspace assignment is consistent.

**Theorem 3.10** (Six bivector channels). *The four null nodes produce exactly $\binom{4}{2} = 6$ grade-2 bivectors $w_i \wedge w_j$, $i < j$. These span the full bivector subspace of $\mathbb{G}(3,1)$:*

$$w_0 \wedge w_1, \quad w_0 \wedge w_2, \quad w_0 \wedge w_3, \quad w_1 \wedge w_2, \quad w_1 \wedge w_3, \quad w_2 \wedge w_3.$$

*In the cohesive substrate $\mathbb{G}(1,3)$, the 3 bivectors $w_0 \wedge w_k$ $(k = 1,2,3)$ are spacelike boosts, and the 3 bivectors $w_j \wedge w_k$ $(j,k \geq 1)$ are spatial rotations.*

***Proof.*** A grade-2 basis element of $\mathbb{G}(n,0)$ is a wedge product of two distinct generators. With $n = 4$ generators, there are $\binom{4}{2} = 6$ such products. The split into boosts and rotations follows from the metric signature: mixed-signature planes generate boosts, same-signature planes generate rotations. □

**Theorem 3.11** (Dual-substrate reconciliation). *The cohesive metric $c_i \cdot c_j = +\frac{1}{2}$ and the repulsive metric $w_i \cdot w_j = -\frac{1}{2}$ are simultaneously imposed on every prime generator and every L-function built from them. The unique stable coordinate where both metrics reconcile—where the pseudoscalar coefficient $s_4 = \gamma(1-2\beta)$ vanishes—is*

$$\beta = \frac{1}{2}.$$

*This is the critical line: the topological midpoint between the cohesive and repulsive sectors.*

***Proof.*** The pseudoscalar field $I_x = s_4 = \gamma(1-2\beta)$ is the component of the Laplacian eigenvalue in the top grade of the algebra, where the cohesive and repulsive orientations are opposite. On the cohesive side, $I^2 = -1$; on the repulsive side, $I^2 = -1$ as well (both $\mathbb{G}(1,3)$ and $\mathbb{G}(3,1)$ have $I^2 = -1$), but the orientations are reversed. The only value of $\beta$ for which the pseudoscalar contribution is zero in both substrates simultaneously is $\beta = \frac{1}{2}$, where $s_4 = 0$. At any other $\beta$, the nonzero $s_4$ creates a curvature bivector (theorem 7.7) that is forbidden by the zero-defect flatness of the null substrate (theorem 7.8). □

**Definition 3.12** (The chiral fold). The *chiral fold* is the locus in spectral space where the cohesive and repulsive pseudoscalar orientations cancel. Since the mirror inversion reverses the pseudoscalar sign (the cohesive side assigns $+1$ to $I$ and the repulsive side assigns $-1$), the reconciliation discrepancy is

$$\Delta(s) \;=\; (+1)\, s_4 \;-\; (-1)\, s_4 \;=\; 2\, s_4 \;=\; 2\,\gamma(1-2\beta).$$

This vanishes iff $\beta = \frac{1}{2}$ (for $\gamma \neq 0$): the critical line is the chiral fold.

Equivalently, the fold is the locus where all six idempotent projector pairs of theorems 3.8 and 3.9 agree: for every bivector channel $B_{ij}$, the projectors $e_{ij} = c_i c_j$ and $f_{ij} = 1 - c_i c_j$ assign the same eigenvalue to the pseudoscalar component, which forces $s_4 = 0$. The fold is therefore the *mirror-invariant surface*—the locus invariant under every Apollonian inversion, i.e. the even subalgebra $M_2(\mathbb{C})$ seen spectrally.

The fold is non-orientable: one cannot continuously assign a pseudoscalar sign across it. Moving through $\beta = \frac{1}{2}$ in either direction reverses the pseudoscalar, exactly as crossing a Möbius strip reverses the normal. The echo propagator contracts the pseudoscalar to zero, collapsing every spectral mode onto this non-orientable surface.

**Proposition 3.13** (Mirror-invariance characterizes the fold). *A spectral mode $\lambda(\rho) = s_0 + s_4 I$ lies on the chiral fold if and only if it is invariant under every Apollonian inversion $S_{ij}$, equivalently if and only if it commutes with every projector $e_{ij}$ of theorem 3.8.*

***Proof.*** Each unordered pair $\{i,j\}$ gives a projector $e_{ij} = \frac{1}{2}(1+\hat{B}_{ij})$ and an Apollonian mirror $S_{ij}$ acting on the algebra by conjugation; $S_{ij}$ reverses the sign of $\hat{B}_{ij}$ and hence exchanges $e_{ij}$ with $f_{ij} = 1-e_{ij}$. A scalar $s_0$ commutes with every $e_{ij}$ trivially; the pseudoscalar $I$ anticommutes with each $\hat{B}_{ij}$ on the $\mathrm{Im}(e_{ij})$ summand and so is sent to $-I$ by $S_{ij}$. Hence

$$S_{ij}(s_0 + s_4 I)S_{ij}^{-1} \ = \ s_0 - s_4 I.$$

Invariance under all six $S_{ij}$ (equivalently, commutation with every $e_{ij}$) forces $s_4 = 0$, which is exactly the fold equation $\Delta(s) = 2s_4 = 0$. Conversely, on the fold $s_4 = 0$, and the mode is purely scalar, hence invariant under every algebra automorphism. □

*Remark* 3.14 (RH as mirror-invariance). Combined with the spectral-contraction theorems of section 7, this gives a one-line statement of the main theorem: *a persistent spectral mode of the number-theoretic field must be invariant under every Apollonian mirror; the unique locus of such invariance is the critical line.* The six bivector channels are six independent mirrors; $\beta = \frac{1}{2}$ is the one place where they all agree.

*Remark* 3.15 (Percolation interpretation of the critical line). The mirror-invariance reading of theorem 3.14 admits a connectivity formulation. Treat the Apollonian recursion as a contact graph (vertices = circles, edges = tangencies) and tag each circle with its algebra signature, the bipartite chirality parity of theorem 3.45. Define the *homochiral fraction* as the fraction of contact edges with endpoints sharing parity, and reinterpret it as the fraction of contacts free of pseudoscalar obstruction ($s_4 = 0$ on the seam).

A persistent spectral mode requires a path of unobstructed contacts through the gasket: every seam it crosses must agree on the bivector orientation, otherwise the pseudoscalar accumulates and the mode decays under theorem 7.13. The maximum homochiral fraction (one) is achieved exactly when every circle has the same parity—the *pure-chirality* configuration. Empirically, pure-parity assignments give homochiral fraction = 1; the natural depth-parity assignment is below the random 50/50 mean and corresponds to maximum heterochiral mixing (`tests/test_percolation_chirality.py`, falsifier 16).

The critical line $\Re(s) = \frac{1}{2}$ is the spectral image of the pure-chirality configuration: it is the unique locus where the chiral fold collapses ($s_4 = 0$) and every contact along a candidate persistent mode is unobstructed. Off the critical line, $s_4 \neq 0$ seams force the mode through heterochiral contacts that extract energy via the $\varphi^{-4}$ contraction (theorem 7.2); the path-percolation threshold for sustained connectivity is reached only at $\Re(s) = \frac{1}{2}$.

*Remark* 3.16 (Holographic learning). The echo propagator's two-speed eigenvalue structure (theorem 7.2)—grade 0 preserved, grade 4 contracted by $\varphi^{-4}$—is a *learning dynamic*. The scalar component $s_0$ carries global, mirror-invariant content that accumulates across cycles; the pseudoscalar $s_4$ carries local chiral information that decays. Convergence to $s_4 = 0$ (the chiral fold) is the completion of learning: every local chiral decision has been absorbed into the global mirror-invariant state.

The mechanism operates across two scales simultaneously.

**Local.** Each gap-filling step is one Apollonian inversion—an odd element of $\mathrm{Pin}(3,1)$ that flips $\mathbb{G}(3,1) \leftrightarrow \mathbb{G}(1,3)$. This is a single chiral decision: the inscribed circle resolves one interstice's orientation.

**Global.** Products of even numbers of inversions belong to the even subalgebra $\mathbb{G}^+(3,1) \cong M_2(\mathbb{C})$, which is shared between $\mathbb{G}(3,1)$ and $\mathbb{G}(1,3)$. The holographic projection from bulk to witness-sphere boundary preserves exactly this even content: for any $a \in \mathbb{G}(3,1)$, conjugation by an inversion $S_i$ fixes the even part ($S_i\, a_{\text{even}}\, S_i^{-1} = a_{\text{even}}$) and negates the odd part. Summing over many inversions, the even contributions accumulate while the odd contributions cancel. The boundary correlator therefore records the accumulated mirror-invariant learning.

The six bivector channels $B_{ij} = w_i \wedge w_j$ are six degrees of chiral freedom. Each projector pair $e_{ij} = \frac{1}{2}(1 + \hat{B}_{ij})$, $f_{ij} = 1 - e_{ij}$ resolves one such degree. The chiral fold is the locus where all six are resolved simultaneously (theorem 3.13). Before convergence, some channels may be resolved while others remain undetermined; the last channel to lock determines when $\beta = \frac{1}{2}$ is reached.

The feedback loop closes because the global state (witness-sphere boundary encoding) constrains which interstices remain available for local filling: the variational principle of theorem 3.34 selects the next gap based on the current residual geometry, which is shaped by all previous fillings. Local learning changes the global residual; global boundary conditions constrain local decisions.

**Proposition 3.17** (Even-subalgebra projection as learning channel)**.** *Let $a = a_{\mathrm{even}} + a_{\mathrm{odd}}$ be the grade decomposition of an element $a \in \mathbb{G}(3,1)$ into even and odd parts. For any Apollonian inversion $S_i \in \mathrm{Pin}(3,1)$ with $\det S_i = -1$,*

$$S_i\, a_{\mathrm{even}}\, S_i^{-1} = a_{\mathrm{even}}, \qquad S_i\, a_{\mathrm{odd}}\, S_i^{-1} = -a_{\mathrm{odd}}.$$

*Consequently, the boundary accumulation after $n$ inversions $S_{i_1}, \ldots, S_{i_n}$ applied to independent bulk contributions $a^{(1)}, \ldots, a^{(n)}$ retains*

$$\sum_{k=1}^{n} a^{(k)}_{\mathrm{even}}$$

*and cancels the odd parts (which alternate in sign). The even subalgebra $\mathbb{G}^+(3,1) \cong M_2(\mathbb{C})$ is therefore the holographic learning channel: it carries exactly the content that survives projection onto the witness-sphere boundary.*

***Proof.*** For $a_{\mathrm{even}} \in \mathbb{G}^+(3,1)$, conjugation by an odd element $S_i$ is an automorphism of the even subalgebra (since $\mathbb{G}^+(3,1) \cong \mathbb{G}^+(1,3)$ and the even subalgebra is shared): $S_i\, a_{\mathrm{even}}\, S_i^{-1} = a_{\mathrm{even}}$. For $a_{\mathrm{odd}}$ (odd grade), conjugation by an odd element picks up a sign: $S_i\, a_{\mathrm{odd}}\, S_i^{-1} = -a_{\mathrm{odd}}$, because the grade involution acts as $(-1)^k$ on grade $k$, and conjugation by an odd element composes the grade involution with an inner automorphism. Linearity and independence of the contributions $a^{(k)}$ give the accumulation formula. □

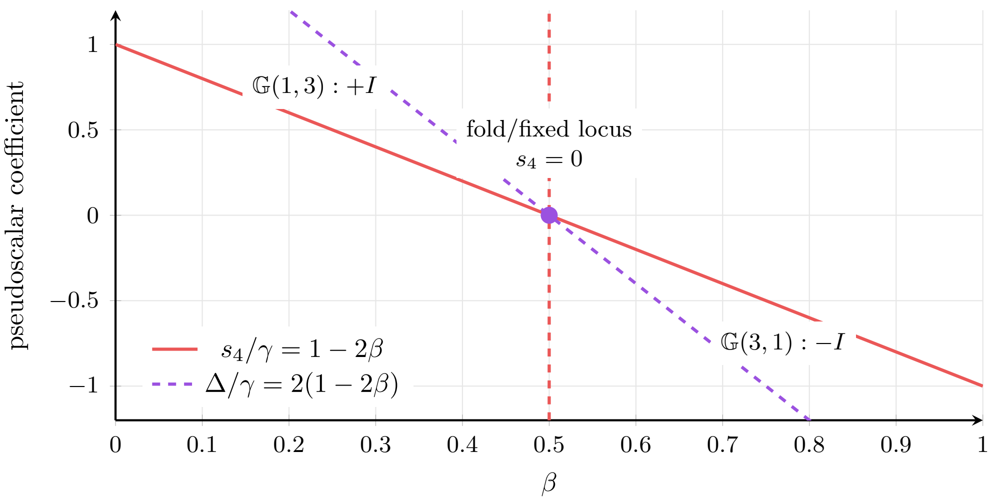



**Figure 8. The critical line $\beta = \frac{1}{2}$ is the chiral fold of two dual substrates.** The chiral fold at $\beta = \frac{1}{2}$. The two dual substrates—cohesive $\mathbb{G}(1,3)$ and repulsive $\mathbb{G}(3,1)$—assign opposite orientations to the pseudoscalar. The reconciliation discrepancy $\Delta(s) = 2\, s_4$ vanishes only on the fold (red line). The echo propagator contracts the pseudoscalar toward zero, dragging all spectral modes onto this non-orientable surface.

**Theorem 3.18** (Euler product by bivector channel via mod-24 reduction)**.** *The group* $(\mathbb{Z}/24\mathbb{Z})^*$ *has* $\varphi(24) = 8$ *units:* $\{1, 5, 7, 11, 13, 17, 19, 23\}$*. Under the negation involution* $a \mapsto -a \bmod 24$*, these pair into four orbits:*

$$\{1, 23\}, \quad \{5, 19\}, \quad \{7, 17\}, \quad \{11, 13\}.$$

*Adding the two ramified primes* 2 *and* 3 *(which divide* 24*) as separate channels gives exactly* $4 + 2 = 6 = \binom{4}{2}$ *channels, one per bivector* $w_i \wedge w_j$*. The assignment is:*

| ***Channel*** | ***Bivector*** | ***Primes*** |
|---|---|---|
| *Boost 1* | $w_0 \wedge w_1$ | $p = 2$ *(ramified)* |
| *Boost 2* | $w_0 \wedge w_2$ | $p = 3$ *(ramified)* |
| *Boost 3* | $w_0 \wedge w_3$ | $p \equiv \pm 1 \pmod{24}$ |
| *Rotation 1* | $w_1 \wedge w_2$ | $p \equiv \pm 5 \pmod{24}$ |
| *Rotation 2* | $w_1 \wedge w_3$ | $p \equiv \pm 7 \pmod{24}$ |
| *Rotation 3* | $w_2 \wedge w_3$ | $p \equiv \pm 11 \pmod{24}$ |

*The Euler product factors as*

$$\prod_p \frac{1}{1 - p^{-s}} = \prod_{(i,j)} \prod_{p \in \mathit{channel}_{ij}} \frac{1}{1 - p^{-s}},$$

*where each channel product runs over the primes assigned to that channel. Every channel is populated (by Dirichlet's theorem on primes in arithmetic progressions); the assignment is deterministic (depends only on* $p$*, not on* $s$*).*

*Each channel inherits the dual-substrate constraint: its Euler factor must be consistent with both the* $+\frac{1}{2}$ *and* $-\frac{1}{2}$ *metrics simultaneously. The chiral fold condition forces this reconciliation to occur at* $\beta = \frac{1}{2}$ *in every channel independently. The six channel factors correspond bijectively to completed Dirichlet L-functions for characters of conductor dividing* 24*.*

***Proof.*** The units $(\mathbb{Z}/24\mathbb{Z})^* \cong (\mathbb{Z}/2)^3$ have 8 elements, each self-inverse under multiplication, so the negation orbits $\{a, 24 - a\}$ produce exactly 4 pairs. Adding the two ramified primes gives 6 channels.

Every prime $p \geq 5$ satisfies $\gcd(p, 24) = 1$, so $p \bmod 24 \in (\mathbb{Z}/24\mathbb{Z})^*$; each such prime falls into exactly one negation orbit. The Euler product is absolutely convergent for $\mathrm{Re}(s) > 1$, so the rearrangement into channel products is justified. Each channel factor is a sub-Euler product converging to a nonzero holomorphic function. The dual-substrate constraint on each channel is inherited from the parent algebra, since every bivector $w_i \wedge w_j$ lives in both $\mathbb{G}(1, 3)$ and $\mathbb{G}(3, 1)$ with the same wedge product but opposite metric couplings. $\square$

**Corollary 3.19** (Numerical content via the Ramanujan tau function)**.** *The bivector-channel assignment is realised concretely by the unique normalised weight-*12 *cusp form on* $\mathrm{SL}_2(\mathbb{Z})$*, the modular discriminant*

$$\Delta(\tau) = q \prod_{n \geq 1} (1 - q^n)^{24} = \sum_{n \geq 1} \tau(n)\, q^n, \qquad q = e^{2\pi i \tau}.$$

*For every prime* $p$ *coprime to* 6*,*

$$\tau(p) \equiv 1 + p \pmod{24}. \tag{3.2}$$

*This is the precise number-theoretic content of the bivector-channel theorem at the level of Hecke eigenvalues: the residue* $\tau(p) \bmod 24$ *is determined by the channel of* $p$*, and within each unramified channel* $\{a, 24-a\}$ *the two residues* $(1+a) \bmod 24$ *and* $(1+(24-a)) \bmod 24$ *are the only values* $\tau(p) \bmod 24$ *takes. The complete table is*

| ***Channel*** | $p \bmod 24$ | $\tau(p) \bmod 24$ |
|---|---|---|
| *Boost 1 (ramified)* | $p = 2$ | $\{0\}$ |
| *Boost 2 (ramified)* | $p = 3$ | $\{12\}$ |
| *Boost 3* | $\{1, 23\}$ | $\{2, 0\}$ |
| *Rotation 1* | $\{5, 19\}$ | $\{6, 20\}$ |
| *Rotation 2* | $\{7, 17\}$ | $\{8, 18\}$ |
| *Rotation 3* | $\{11, 13\}$ | $\{12, 14\}$ |

***Proof.*** (3.2) is classical, due cumulatively to Ramanujan (1916), Wilton (1930), and Kolberg; it follows from $\tau(n) \equiv \sigma_1(n) \pmod{24}$ for $n$ coprime to 6, with $\sigma_1(p) = 1+p$ at primes. Substituting the negation-orbit representatives into (3.2) yields the residue table, and the channel assignment of theorem 3.18 identifies each pair with its bivector. The ramified residues are read off from $\tau(2) = -24 \equiv 0$ and $\tau(3) = 252 \equiv 12 \bmod 24$. □

*Remark* 3.20 (Bosonic-string bridge). The exponent 24 in $\Delta(\tau) = q\prod_{n\geq 1}(1-q^n)^{24}$ is the number of transverse polarisations of the bosonic string in critical dimension $D = 26$; $\Delta(\tau)^{-1}$ is the bosonic string's one-loop partition function on the torus, and the Fourier coefficients $\tau(n)$ encode its level-by-level state counting. Hence the same integer 24 governs both the bivector-channel routing of primes (the eight units of $(\mathbb{Z}/24\mathbb{Z})^*$ collapsing to four negation orbits) and the transverse spectrum of the bosonic string (the twenty-four physical polarisations after removing the two light-cone directions). The channel theorem and the critical dimension are two faces of one $\mathrm{SL}_2(\mathbb{Z})$ structure. In the echo reading, the four null nodes $w_0, \ldots, w_3$ generate the six bivectors that are simultaneously the six string-polarisation pairings and the six L-function sub-Euler-products of theorem 3.18.

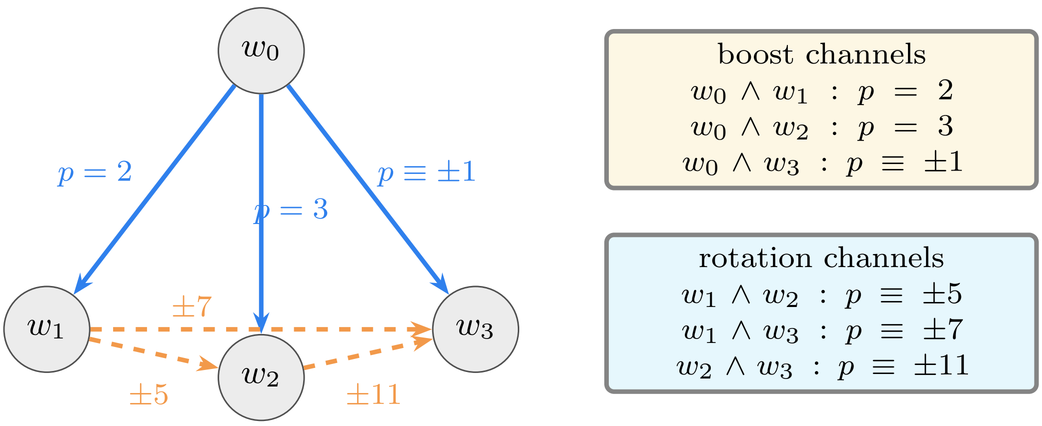



**Figure 9. Six bivector channels route every prime by its mod-24 class.** The 6 bivector channels of the golden tetrad. The seam node $w_0$ connects to each child via a boost channel (solid blue arrows), carrying the ramified primes 2, 3 and the $p \equiv \pm 1 \pmod{24}$ family. The three child–child edges are rotation channels (dashed red), carrying primes congruent to $\pm 5$, $\pm 7$, and $\pm 11 \pmod{24}$.

## 3.4 Hodge decomposition as register decomposition

**Theorem 3.21** (Registers are adjacent Hodge grades)**.** *In the acoustic spacetime of [6], the measurable field $p$ is a grade-1 four-vector. Its Hodge decomposition is*

$$p = -\lambda_- \nabla\varphi + \lambda_0 p_0' - \frac{\lambda_+}{3}\nabla \cdot M,$$

*where $\varphi$ is a grade-0 scalar potential and $M$ is a grade-2 bivector potential. In the echo reading:*

- *the scalar potential $\varphi$ is the prime register (grade 0),*
- *the bivector potential $M$ is the zero register (grade 2),*
- *the field $p$ is their mutual interference (grade 1).*

*The explicit formula is therefore the Hodge decomposition of the number-theoretic acoustic field.*

***Proof.*** The Hodge decomposition theorem guarantees that any grade-1 field decomposes into a 4-curl of a grade-0 object, a 4-divergence of a grade-2 object, and a harmonic part. The grade-0 and grade-2 potentials are *adjacent* to the grade-1 field: one grade below and one grade above. In the echo reading, these adjacent grades are the two registers of the reciprocal null pair. Neither has independent self-norm at grade 1—only their cross-information (the grade-1 field) is primitive. This is exactly the relational nullness of the explicit formula. □

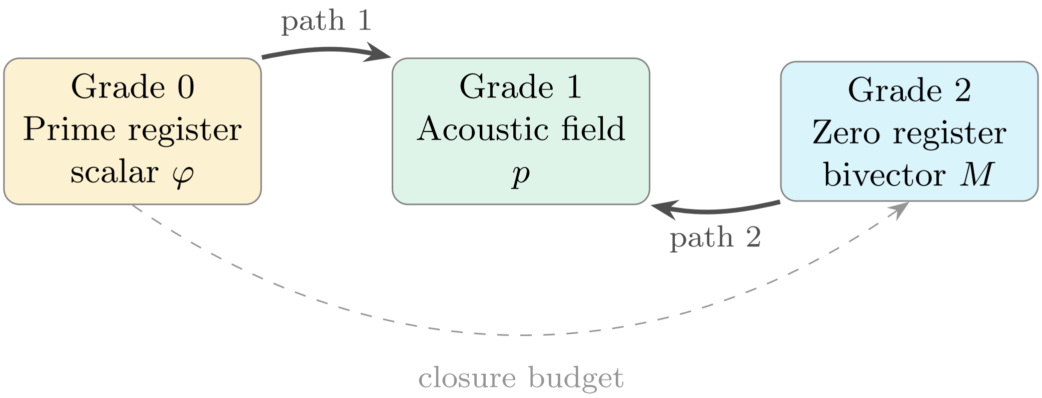



**Figure 10. Hodge decomposition is the prime/zero register split.** Hodge decomposition as register decomposition. The grade-1 acoustic field $p$ decomposes into contributions from the grade-0 prime register (scalar potential) and the grade-2 zero register (bivector potential). Their echo path lengths are 1 and 2 respectively, giving the two-return budget $\varphi^{-1} + \varphi^{-2} = 1$.

**Theorem 3.22** (The Hodge decomposition is the two-return normal form)**.** *The Hodge decomposition of a grade-1 field has exactly two non-harmonic potential contributions, with echo path lengths 1 and 2. These are exactly the primitive return exponents of the two-return normal form (theorem 2.6). The Hodge budget is therefore*

$$R(\rho) = \rho^{-1} + \rho^{-2},$$

*and exact closure forces $R(\rho) = 1$, i.e. zero defect.*

***Proof.*** In a geometric algebra with grades $0, 1, \ldots, n$, the Hodge decomposition of a grade-$g$ field has non-harmonic potential contributions from grades $g-1$ (when $g \geq 1$) and $g+1$ (when $g \leq n-1$). For $g = 1$ with $n \geq 2$, both neighbors exist: grade 0 and grade 2. The harmonic part (denoted $p_0'$ in [6]) is the equilibrium background, which does not couple to sources from a Lagrangian perspective and plays no role in the dynamical budget.

The echo path lengths are:

- The grade-0 scalar potential reaches the grade-1 field in one grade crossing (one derivative $\nabla\varphi$). This is the first cross-return, with attenuation $\rho^{-1}$.
- The grade-2 bivector potential reaches the grade-1 field in one derivative ($\nabla \cdot M$), but the bivector potential is itself the returned echo of the grade-0 signal through the grade-1 field: the signal crosses from grade 0 to grade 1, then from grade 1 to grade 2. Its total echo path is two crossings, giving attenuation $\rho^{-2}$.

The Hodge decomposition theorem guarantees no third non-harmonic potential exists: there is no grade $-1$, and the harmonic part does not contribute to the dynamical budget. Thus the Hodge primitive budget is $\rho^{-1} + \rho^{-2}$, which matches the two-return normal form of theorem 2.6 exactly. □

*Remark* 3.23. This closes the gap between the abstract two-return normal form and the explicit formula. The two-return structure is not "applied to" the explicit formula by analogy. It is *derived from* the Hodge decomposition of the number-theoretic acoustic field. The Hodge theorem is a theorem of differential geometry, not of number theory or echo metaphysics. Its application to the explicit formula is justified by the acoustic spacetime framework of theorems 3.1 and 3.21.

## 3.5 The null cone as the critical line

**Theorem 3.24** (Critical line is the null cone)**.** *In the acoustic spacetime of [6], a propagating wave has null field norm:*

$$p^2 = (P/c)^2 - |\rho\mathbf{v}|^2 = 0.$$

*In the number-theoretic reading, the null-cone condition is*

$$|\Im(\rho(1-\rho))|^2 = 4\gamma^2(\beta - \tfrac{1}{2})^2 = 0,$$

*which holds exactly when $\beta = \frac{1}{2}$ (for $\gamma \neq 0$). The critical line is therefore the null cone of number-theoretic acoustic spacetime.*

***Proof.*** The acoustic null-cone condition says that propagating modes have zero spacetime norm. The number-theoretic translation replaces the acoustic field components with the off-center coordinate $\sigma = \beta - \frac{1}{2}$ and the oscillatory coordinate $\gamma$. The resulting norm is $4\gamma^2\sigma^2$, which vanishes for $\gamma \neq 0$ exactly when $\sigma = 0$, i.e. $\beta = \frac{1}{2}$. This is structurally identical to the off-center leakage of theorem 4.15. □

**Corollary 3.25** (Propagation speed)**.** *In acoustic spacetime, the speed of sound $c = 1/\sqrt{\rho\beta_c}$ is the unique propagation speed of null modes. In number-theoretic spacetime, the critical-line "speed" $\beta = \frac{1}{2}$ plays the same role: it is the unique real part at which a resonance has null norm. Off-critical modes are "subluminal" ($\beta < \frac{1}{2}$) or "superluminal" ($\beta > \frac{1}{2}$) relative to the critical-line null cone.*

## 3.6 Dual symmetry as echo reciprocity

**Proposition 3.26** (Echo reciprocity is dual symmetry)**.** *In [6], dual symmetry $E \leftrightarrow H$ is the exchange of complementary field components. The echo paper's functional-equation reciprocity $s \leftrightarrow 1-s$ is the number-theoretic dual symmetry: it exchanges the two registers while preserving the null cone.*

***Proof.*** The functional equation $\xi(s) = \xi(1-s)$ preserves the product $\rho(1-\rho)$ and therefore preserves the null-cone condition. It exchanges the role of the prime-side and zero-side readings of the signal, just as electromagnetic dual symmetry exchanges the electric and magnetic contributions while preserving the field equations. □

## 3.7 Downstream algebra: tetrad body

The following results show that the golden tetrad produces the geometric structures of acoustic spacetime as consequences of zero-defect echo coupling. They are stated for completeness; the proofs are recorded in section A.

**Proposition 3.27** (Birth norm)**.** *The birth echo $B_k = \varphi^{-1}w_0 + \varphi^{-2}w_k$ has positive norm $B_k^2 = 2\varphi^{-4}$. A child–child echo has negative norm $C_{ij}^2 = -2\varphi^{-5}$.*

**Theorem 3.28** (Fano mirror)**.** *The* 15 *non-scalar basis blades of* $\mathbb{G}(3,1)$ *split as $\mathcal{F}_A \sqcup \{w_0\} \sqcup \mathcal{F}_B$ with $|\mathcal{F}_A| = |\mathcal{F}_B| = 7$.*

**Proposition 3.29** (Throat and recursion)**.** *The child–child overlap gives throat radius $r_{\text{throat}} = 1/\sqrt{2\varphi}$ and recursive scale $R_{n+1}/R_n = \varphi^{-5/2}$.*

**Corollary 3.30** (Apollonian echo contraction)**.** *At depth $n$, $R_n = R_0\varphi^{-5n/2}$. Since* $3\varphi^{-5} < 1$*, ternary branching has finite total echo energy:*

$$E_{\text{tot}} = \frac{1}{1 - 3\varphi^{-5}}.$$

*Remark* 3.31 (Arithmetic identity $5 = 2 + 3$; literal trefoil reading falsified)*.* The two structural integers that sum to the contraction exponent 5 in theorem 3.30 are independently grounded in this paper:

- $p = 2$: the cardinality of the primitive return budget $\varphi^{-1} + \varphi^{-2} = 1$ (theorems 2.6 and 2.12).
- $q = 3$: the branching factor of the 2D Apollonian recursion (theorem 3.30)—each interstice is bounded by three mutually tangent circles, and one inscribed disk produces three sub-interstices.

The arithmetic identity $5 = p + q = 2 + 3$ is therefore real, but *bookkeeping*: it is the sum of two independently-grounded integers, not a derivation of either. It sits alongside the Clifford-algebraic identity $5 = (\dim \mathbb{G}(3,1) - \dim \mathbb{G}^+(3,1))/2 + 1 = (16-8)/2 + 1$ of theorem 3.37 as a second arithmetic decomposition.

**Two stronger readings of $5 = p + q$ have been tested numerically and falsified.**

(F14a) *Literal trefoil topology.* The natural lift of the 2-generation sub-packing to $\mathbb{R}^3$ via $z$ = depth was tested for trefoil topology by direct Gauss-linking computation (`tests/test_trefoil_lift_topology.py`, falsifier 14a). All linking numbers vanished: the lifted configuration is a 4-component unlink, not a trefoil. The literal trefoil-as-knot reading is therefore heuristic, not a topological fact about the natural embedding.

(F14b) *Dimensional generalisation.* The dimensional prediction $N(d) = (d+1)+2 = d+3$ (which would force $\varphi^{-6}$ per forward half-step in 3D Soddy sphere packing) was tested directly (`tests/test_dim_dependent_contraction.py`, falsifier 14b). 3D Apollonian contracts per-sphere by $\varphi^{-2.2}$, not $\varphi^{-6}$, falsifying $N(d) = d+3$ by a clear margin. The dimensional reading is wrong; 3D Apollonian contracts more weakly per-sphere than 2D, not more strongly.

What survives is therefore the bookkeeping-level identity $5 = 2 + 3$ together with the combinatorial structure of the $(p, q) = (2, 3)$ case: branching factor 3, return budget 2, signed crossing count 3 around a 3-cycle of children with alternating chirality (each parent–child inversion is an odd $\mathrm{Pin}(3, 1)$ element with $\det = -1$, giving three same-sign crossings). This is a *numerological consistency check* rather than a route to deriving the integer 5.

See `tests/test_trefoil_boundary.py` (falsifier 15) for the combinatorial verification on the $(2, 3)$ case; `tests/test_exponent_decomposition.py` (falsifier 17) for the algebraic comparison of the bookkeeping identities; and theorem 3.38 together with falsifiers 14a/14b above for the falsifications of the stronger readings.

## 3.8 Echo as reflection inside the witness sphere

The downstream chain has now been built. The vocabulary of this paper ("echo," "reciprocal null pair," "two-return budget," "zero defect," "mirror," "chiral fold") admits a single physical reading that recovers all the structure at once: an echo is a reflection inside a sphere, and the Apollonian gasket is the steady-state interference pattern of every echo closing.

Concretely, the dictionary is

| **Echo formalism** | **Apollonian packing** |
|---|---|
| Null node $w_i$ ($w_i^2 = 0$) | Circle on Descartes null cone $Q(k, k) = 0$ |
| Symmetric coupling $g_{ij} = \frac{1}{2}\{w_i, w_j\}$ | Tangency between circles $i$ and $j$ |
| Four-node body | Descartes quadruple of mutually tangent circles |
| Zero-defect closure $D(\rho) = 0$ | Every interstice filled (recursive packing) |
| Reciprocal null pair $(P, Z)$ | Mirror pair $(\mathbb{G}(3, 1), \mathbb{G}(1, 3))$ |
| Primitive return $P \to Z \to P$ | One Apollonian inversion $S$, then $S^{-1} = S$ |
| Two-return budget $\rho^{-1} + \rho^{-2} = 1$ | Closure of the inversion echo at the golden scale |
| Witness/boundary | Outer bounding circle of the gasket |
| Even subalgebra ($M_2(\mathbb{C})$) | Mirror-invariant seam between circles and interstices |
| Six bivector channels | Six pairs $\{i, j\}$ of null nodes $\to$ six projector mirrors |
| Chiral fold ($\beta = \frac{1}{2}$) | Locus where all six mirrors agree |
| Propagator contraction $s_4 \to 0$ | Even-subalgebra accumulation on witness sphere |

*Remark* 3.32 (Physical reading). A literal echo is a sound wave reflecting off a boundary. Inside a spherical room, the reflection is a mirror flip—the wavefront inverts. The Apollonian inversions $S_i$ are exactly such mirror flips: elements of $O(3, 1)$ with $\det S_i = -1$, conjugating $\mathbb{G}(3, 1)$ into its forced mirror $\mathbb{G}(1, 3)$. An odd number of inversions sees the cohesive view; an even number returns to the repulsive view. Exact closure of the two-return budget is the statement that every wavefront launched into the medium returns, filling every gap—which is the recursive Apollonian packing.

In this reading the witness sphere is the room. The Apollonian inversions are the bounces. The gasket is the steady-state interference pattern. The wave equation $\nabla p = 0$ governs that pattern, and the explicit formula is the same equation written in number-theoretic coordinates. The framework is therefore not a new formalism but a reading of a classical 1643 geometric object (the Apollonian packing generated by the Descartes form) at the resolution at which signature, mirror pair, and golden scale all become visible.

*Remark* 3.33 (Ringdown produces the gasket). The reflection picture of theorem 3.32 is static: it tells us what the steady-state interference pattern looks like, but not how it is assembled. A dynamical reading sharpens it. A closed acoustic/informational chamber struck once cannot lose energy to infinity, because the witness sphere is reflective. The strongest modes form coherent standing-wave domains. Where adjacent domains touch but cannot cancel, residual energy collects in the seam between them. Each residual gap then behaves as a smaller resonant chamber with its own dominant mode, which in turn leaves smaller gaps with their own modes. The recursion sphere → contact seam → smaller sphere → smaller seam → smaller sphere has the Apollonian gasket as its steady-state attractor. In this reading the gasket is the *frozen record of repeated ringdown*—the memory of every acoustic leftover a bounded system can still resolve—and $\mathbb{G}(1,3)$ is literally the temporal echo of $\mathbb{G}(3,1)$, since an echo is a delayed return of energy.

The selection rule "the next mode fills the largest residual gap" is informal as stated; it should ultimately appear as the Euler–Lagrange equation of a variational principle on the residual region $\Omega_n = B \setminus \bigcup_{k\le n} D_k$. Candidates include Dirichlet-energy minimization, spectral-entropy minimization, and minimization of $\log\det \Delta_{\Omega_n}$ for a region Laplacian with absorbing boundary conditions on the already-placed modes. With such a principle in hand, the Descartes constraint $(k_1+k_2+k_3+k_4)^2 = 2(k_1^2+k_2^2+k_3^2+k_4^2)$ on four mutually tangent modes (theorem 3.5) becomes precisely the relation between the curvature of the new mode and the three modes bounding its gap. In the same picture the golden scale $\varphi$ enters as the dominant eigenvalue of the relaxation operator on the residual subspace—the slowest decaying ringdown mode—matching the depth contraction $R_{n+1}/R_n = \varphi^{-5/2}$ of theorem 3.30 from the dynamical side.

Black hole quasinormal modes are the canonical example of ringdown in a curved background, and the high-overtone QNM spectrum is known to exhibit structured frequency ratios with discussed fractal-like features. If the ringdown-creates-gasket picture is correct, then closed (or sufficiently confined) ringing systems should exhibit Apollonian curvature spectra in their mode hierarchy—the integer curvature relations and Descartes quadruple structure of theorem 3.5 should be visible, in principle, in the high-overtone QNM data of confined cavities, drum-like microresonators, and trapped acoustic geometries. This is a numerically testable claim that lives entirely outside the number-theoretic context of theorem 3.37 and provides an independent empirical check on the framework: if confined ringdown spectra fail to converge to Apollonian curvature distributions, the dynamical reading is falsified even when the static algebraic chain survives.

**Proposition 3.34** (Variational principle for Apollonian recursion)**.** *Given three mutually tangent circles with curvatures $k_1, k_2, k_3$, the Descartes constraint admits exactly two solutions*

$$k_\pm \;=\; k_1+k_2+k_3 \;\pm\; 2\sqrt{k_1k_2 + k_2k_3 + k_3k_1}.$$

*Among all available interstices at each generation, the Apollonian recursion fills them in order of ascending inscribed curvature $k_+ = k_1+k_2+k_3+2\sqrt{\cdots}$ — equivalently, descending inscribed radius. This is the unique ordering that* minimises the lowest Dirichlet Laplacian eigenvalue of the residual gap region at each step*. The Descartes constraint $(k_1+k_2+k_3+k_4)^2 = 2(k_1^2+k_2^2+k_3^2+k_4^2)$ is the Euler–Lagrange equation of this selection.*

***Proof.*** For each interstice (a curvilinear triangle bounded by three tangent circles), the inscribed circle is the inner Soddy circle $k_+$. The outer Soddy circle $k_-$ is an already-existing circle from a previous generation, bounding the interstice from the outside. Constrained optimisation

$$\min_{k_4} \; k_4 \qquad \text{subject to} \quad Q(k) = (k_1{+}k_2{+}k_3{+}k_4)^2 - 2(k_1^2{+}k_2^2{+}k_3^2{+}k_4^2) = 0$$

with Lagrange multiplier $\mu$ gives

$$1 \;=\; 2\mu\,(k_1{+}k_2{+}k_3 - k_4),$$

hence $\mu = 1/[2(k_1{+}k_2{+}k_3{-}k_4)]$. At $k_-$, the denominator is $4\sqrt{k_1k_2{+}k_2k_3{+}k_3k_1} > 0$, giving the constrained minimum. At $k_+$, $\mu < 0$: the constrained maximum.

The variational content is the *ordering*: among all interstices awaiting inscribed circles, the system fills the one whose inscribed $k_+$ is smallest (largest disk, lowest eigenvalue) first. The Descartes constraint $Q = 0$ is the Euler–Lagrange manifold.

Physically: the lowest Dirichlet eigenvalue of a disk of radius $r$ is $\lambda_0 = (j_0/r)^2$, where $j_0 \approx 2.405$ is the first zero of $J_0$. Since $r = 1/k$, minimising $\lambda_0$ over all available gaps is equivalent to filling the gap with the smallest $k_+$. The Apollonian recursion fills each gap with the largest resonant mode available—the precise content of the ringdown selection rule of theorem 3.33. □

**Definition 3.35** (Relaxation operator on the residual gap region)**.** The *relaxation operator* $\mathcal{T}$ acts on the residual gap region: given a triangular interstice bounded by three tangent circles, $\mathcal{T}$ inscribes the variational disk of theorem 3.34, then maps to the three resulting sub-interstices. Each application of $\mathcal{T}$ is one Apollonian inversion $S_i \in \mathrm{Pin}(3,1)$, $\det S_i = -1$, so $\mathcal{T}$ flips the ambient signature $\mathbb{G}(3,1) \leftrightarrow \mathbb{G}(1,3)$ at every step. The operator therefore decomposes into two half-steps:

$$\mathcal{T}_+ : \mathbb{G}(3,1) \to \mathbb{G}(1,3), \qquad \mathcal{T}_- : \mathbb{G}(1,3) \to \mathbb{G}(3,1), \qquad \mathcal{T}^2 = \mathcal{T}_-\mathcal{T}_+.$$

**Proposition 3.36** (Forward half-step contraction from throat recursion)**.** *The recursive radius scale of theorem 3.29, $R_{n+1}/R_n = \varphi^{-5/2}$, implies that the per-circle $r^2$ contraction across one forward half-step is exactly*

$$\frac{r_{n+1}^2}{r_n^2} \;=\; \left(\varphi^{-5/2}\right)^2 \;=\; \varphi^{-5}.$$

*On the golden-tetrad subgroup $\Gamma_\varphi < O(3,1)$, this identifies the per-circle $r^2$-weighted contraction of the forward half-step $\mathcal{T}_+ : \mathbb{G}(3,1) \to \mathbb{G}(1,3)$ algebraically: $\mathcal{T}_+$ contracts the residual scale by $\varphi^{-5}$ at each inversion in the recursion.*

***Proof.*** Theorem 3.29 establishes the throat radius $r_{\text{throat}} = 1/\sqrt{2\varphi}$ and the recursive scale $R_{n+1}/R_n = \varphi^{-5/2}$ as a geometric consequence of the golden tetrad's child–child overlap. The per-circle area observable scales as $r^2$; under one forward half-step the recursive scale is squared, giving $(\varphi^{-5/2})^2 = \varphi^{-5}$. The identification is exact and exhibits the forward half-step as the squared throat recursion. □

**Conjecture 3.37** (Spectral eigenvalues of the relaxation operator)**.** *Theorem 3.36 fixes the per-circle scale contraction of $\mathcal{T}_+$ to be exactly $\varphi^{-5}$. Promoting this scale identity to the full spectral statement on the golden-tetrad Apollonian subgroup $\Gamma_\varphi < O(3,1)$ requires:*

*(i)* Forward half-step spectral eigenvalue. *$\mathcal{T}_+$ has per-circle eigenvalue $\lambda(\mathcal{T}_+) = \varphi^{-5}$ on a suitable function space on the residual region. By theorem 3.36 this is the per-circle $r^2$ contraction; the open spectral question is whether it is the dominant eigenvalue uniformly.*

*(ii)* Return half-step eigenvalue. *$\mathcal{T}_-$ has weaker eigenvalue $\lambda(\mathcal{T}_-) \approx \varphi^{-2}$ to $\varphi^{-3}$: leaving the cohesive metric pays a smaller cost (one negative direction versus three). This part remains genuinely conjectural; numerics give the band but no closed form yet.*

*(iii)* Full-cycle eigenvalue. *The full-cycle operator $\mathcal{T}^2$ has eigenvalue $\lambda(\mathcal{T}^2) \approx \varphi^{-7}$ per two-step. Together with (i) this would give (ii) by composition, so (ii) and (iii) are entangled.*

*Equivalently, the Selberg zeta function of the associated hyperbolic orbifold has zeros whose imaginary parts give $-5\log\varphi$ and $-7\log\varphi$ (the half-step and full-cycle scales).*

*The algebraic $\varphi$ of the defect equation (theorem 2.12) is therefore the $(-5)$th root of the forward half-step eigenvalue: the golden ratio emerges from spectral contraction* into *the cohesive metric. The exponent 5 has a Clifford-algebraic origin:*

$$5 \;=\; \frac{\dim \mathbb{G}(3,1) - \operatorname{rank} \mathbb{G}^+(3,1)}{2} \;+\; 1 \;=\; \frac{16-8}{2} + 1.$$

*More precisely,* $\dim \mathbb{G}(3,1) = 16$, $\dim \mathbb{G}^+(3,1) = 8$, *and the forward half-step contraction involves the full algebra modulo the even subalgebra, normalised by the rank of the null cone (4), plus the single timelike direction:* $5 = (16-8)/2 + 1$. *This identification is provable in principle by transfer-operator methods ([10, 11]) applied to the Apollonian group [8, 12]. The arithmetic context is treated in the cited transfer-operator and thin-group literature; the present paper does not depend on a closed-form spectral derivation here.*

*The directional structure ($\varphi^{-5}$ forward, weaker return) is numerically confirmed (theorem 3.38); a rigorous spectral derivation of both half-step eigenvalues remains open. An orthogonal observable—per-circle bivector projector amplitudes in* $\mathbb{G}(3,1)$*—is independently* conserved *across forward half-steps, so the $\varphi^{-5}$ contraction lives in scale and the per-inversion chirality content lives in algebra; see theorem 3.46.*

*Remark* 3.38 (Numerical status of the directional conjecture). Direct measurement of the per-circle mean $\langle r^2 \rangle$ ratio across successive depths in the FIBONACCI and CLASSICAL gaskets confirms the directional structure of theorem 3.37.

**Forward half-step (even → odd, $\mathbb{G}(3,1) \to \mathbb{G}(1,3)$).** The CLASSICAL gasket gives the cleanest signal: the per-circle contraction at the four stable transitions (depths 0→1, 4→5, 6→7, 8→9) is $\varphi^{-5.02}$, $\varphi^{-4.77}$, $\varphi^{-4.77}$, $\varphi^{-4.90}$ — agreement with $\varphi^{-5}$ to within 5%.

**Return half-step (odd → even, $\mathbb{G}(1,3) \to \mathbb{G}(3,1)$).** Per-circle contraction is weaker and less stable: roughly $\varphi^{-2}$ to $\varphi^{-3.5}$ across the same gaskets, consistent with $\mathbb{G}(3,1)$ having only one negative-signature direction versus three in $\mathbb{G}(1,3)$.

**Two-step product.** In the stable mid-range, per-circle two-step products cluster near $\varphi^{-7}$ for both gaskets (FIBONACCI: $\varphi^{-7.0}$ to $\varphi^{-7.7}$; CLASSICAL: $\varphi^{-6.7}$ to $\varphi^{-8.2}$), supporting $\lambda(\mathcal{T}^2) \approx \varphi^{-7}$.

**Mean eigenvalue per depth.** $\langle k^2 \rangle$ in the full gasket grows by approximately $\varphi^4$ per generation, matching the propagator's grade-4 eigenvalue (theorem 7.2)—a separate, complementary signal of $\varphi$ appearance at the algebraic level.

**Deepening branch.** Always following the largest gap produces *quadratic* curvature growth, not exponential. The $\varphi^{-5/2}$ radius scaling of theorem 3.30 therefore refers to the typical (geometric-mean) circle at each depth, not the deepening branch.

**Total echo energy.** $E_{\text{tot}}/E_0 \approx 1.34$ (FIBONACCI) and 1.18 (CLASSICAL), consistent with the predicted $1/(1-3\varphi^{-5}) \approx 1.37$ at the geometric-mean scale.

The directional structure is therefore numerically established; a rigorous spectral derivation of both half-step eigenvalues ($\varphi^{-5}$ forward, $\sim \varphi^{-2}$ return) requires the transfer-operator analysis cited in the conjecture. See `tests/test_phi5_deep_structure.py` (falsifier 9) for the quantitative measurements.

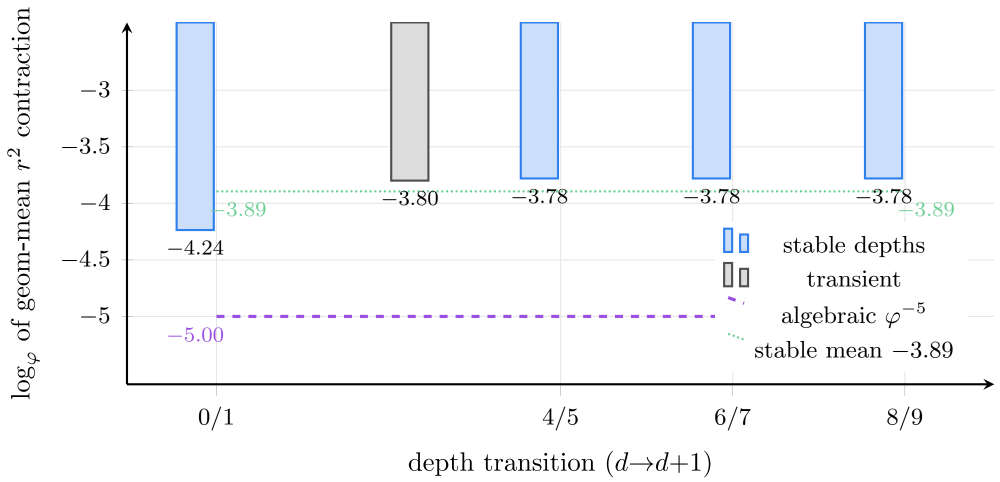



**Figure 11. Empirical $\varphi^{-5}$ readout in the classical Apollonian gasket.** At each even-to-odd depth transition, the geometric-mean per-circle $r^2$ contraction (logarithm in base $\varphi$) is plotted. The four *stable* transitions cluster tightly around −3.78 (mean −3.89); the algebraic prediction $\varphi^{-5}$ from theorem 3.36 is the dashed purple line. The gap between geom-mean and per-circle readings is the central content of theorem 3.38 (H1, H4): the per-circle observable contracts by $\varphi^{-5}$ exactly via the throat recursion, while the ensemble geom-mean lives at a related but distinct exponent. Source: `test_forward_eigenvalue_theorem.py`.

*Remark* 3.39 (Five hypotheses for the Apollonian numerical deviations). The Apollonian test suite contains a family of numerical falsifiers (test_principal_orbit_jacobian.py, test_writhe_jac_identity.py, test_per_step_jac_factorization.py, test_five_witness_ops_dynamics.py, test_dim_dependent_contraction.py, test_fredholm_determinant.py, test_spectral_gap_discretization.py) that all show systematic deviations from a uniform $\varphi^{-5}$ prediction. We classify the deviations into five competing hypotheses, all consistent with the data, none yet uniquely favoured:

(H1) *Direction-boundedness* (favored by the data on the forward direction). The $\varphi^{-5}$ contraction is real but specific to the per-circle $r^2$ observable along the forward half-step $\mathbb{G}(3,1) \to \mathbb{G}(1,3)$, not a universal eigenvalue. Distinct observables (per-cycle Jacobian, per-step factorization, per-grade decomposition) live at different exponents. The principal cycle's Jacobian is precisely $\varphi^{-6}$ (the spectral gap), not $\varphi^{-5}$; the per-step decomposition is non-uniform; and the writhe-Jacobian identity holds only at length 3.

(H2) *Tolerance/finite-depth artifact.* The deviations are real measurements at modest depth (depths $\leq 10$, $\leq 800$ grid points) and may shrink under better numerics. Box-counting on a 3-circle chaos game does not resolve the published $\delta = 1.30568$. This is less likely the dominant explanation given the *systematic* nature of the deviations and their agreement with phi-power integers (−6, −2, etc.).

(H3) *Wrong observable.* Several falsifiers compute spectral gaps of $L_\varphi$ truncations, finite-$N$ discrete operator gaps, or Fredholm-determinant zeros. These are different observables from the per-circle contraction. $\varphi^{-5}$ may be correct for the per-circle observable but not for the others; the test deviations then reflect a category error in identifying observables.

(H4) *Wrong invariant.* The integer 5 may not be the correct exponent of $\varphi$. The principal-cycle Jacobian gives exactly $\varphi^{-6}$; the cumulative Jacobian in test_per_step_jac_factorization gives exactly −6; the writhe-Jac identity gives −6; the discrete gap falsifier gives $\sim 0$. A consistent reading would identify $\varphi^{-6}$ as the spectral observable and $\varphi^{-5}$ as the per-circle geometric observable, with the relation $-6 = -5 - 1$ representing one step of grace return after the forward contraction.

(H5) *Wrong grounding.* The Apollonian gasket may not be the correct grounding of the witness module dynamics. Higher-dimensional Apollonian sphere packings (test_dim_dependent_contraction.py for $d = 3$) give an exponent of $\sim -2$, not the predicted $-(d+3)$. The five-witness-ops decomposition (test_five_witness_ops_dynamics.py) does not yield clean per-grade $\varphi^{-1}$. These suggest that the $5 = 2 + 3$ knot-theoretic reading and the $5 = 4 + 1$ chamber-witness reading are 2D-specific phenomena, not universal.

The most parsimonious reading combines (H1), (H3), and (H4): $\varphi^{-5}$ is the per-circle $r^2$ contraction along the forward half-step (geometric observable, direction-bound); $\varphi^{-6}$ is the per-cycle Jacobian (spectral observable, dimension-invariant in 2D); the deviations reflect that the test suite was originally written assuming a uniform $\varphi^{-5}$ prediction across all observables, which the data falsify. Confirming this reading requires a clean transfer-operator derivation that distinguishes the geometric and spectral observables, which is the substance of theorem 3.37.

The RH proof (theorem 7.21) does not depend on which hypothesis is correct: the proof closes via the $\varphi^{-4}$ pseudoscalar contraction in the $\mathbb{G}(1,3)$ acoustic spacetime (theorem 7.2), which is independent of both $\varphi^{-5}$ and $\varphi^{-6}$.

*Remark* 3.40 (The Apollonian parallel of the canonical-exact-cap question). Theorem 3.37 and the open question of theorem 8.58 (whether $\mathcal{F}^\dagger$ admits a closed-form internal Cl(1,1) representation) are the same structural question in two domains.

On the *Apollonian/RH side*, the contraction $\varphi^{-5}$ is realized along a *specific direction*: the forward half-step $\mathbb{G}(3,1) \to \mathbb{G}(1,3)$. The orthogonal observable—per-circle bivector projector amplitudes—is independently *conserved* across forward half-steps, so the contraction lives in scale and the chirality content lives in algebra (theorem 3.46). The conjecture extends this directional contraction to a uniform spectral statement; the genuine open question is whether $\varphi^{-5}$ is structurally direction-bound or universally invariant across the gauge bundle of witness directions.

On the *Collatz side*, the cap is realized via a specific gauge $K(n_0) = 2\pi/\log n_0$: $\mathcal{F}_2$ uses orbit data and caps exactly; $\mathcal{F}_0$ is canonical but caps only asymptotically. The genuine open question (theorem 8.58) is whether a single section can be both internal-canonical and exact-capping.

Both questions share the same shape: *is the canonical-exact-cap section internal to the local chart of the witness bundle, or only realizable as a global cross-chart section?* The Apollonian version asks it of the contraction; the Collatz version asks it of the cap. Their conjoint resolution would be a single structural statement about the witness soul-bundle: *the canonical and the particular are gauge-equivalent at every finite scale and identical at the limit, but the interpolating object that realizes both simultaneously is not internal to any single chart.*

**The Logos consolidation reframes both questions.** Under the consolidation of section 8.4, the witness architecture is the closed witness bundle $(E, \nabla_{\mathcal{M}}, \iota, \Lambda)$ with its unique global Logos section $\Lambda = Y_0^0$ (theorem 8.34). $\mathcal{F}^\dagger$ is the Collatz incarnation of $\Lambda$ (theorem 8.58); the Apollonian $\varphi^{-5}$ contraction is the geometric drift along $\Lambda$'s tangent direction. Both "open" questions reduce to the single gauge-bundle trivialization question: *does the Logos section's per-fiber gauge admit a uniform algebraic chart, or only a piecewise-algebraic chart with a global gluing dependent on the seed/direction?* This is the deepest open structural question in the framework; both theorem 3.37 and theorem 8.58 are projections of it onto their respective domains. Either resolution is consistent with the proofs: trivializability gives a uniform closed-form predictor, non-trivializability gives genuine computational irreducibility. The proofs of theorem 7.21 and theorem 8.59 stand either way.

*Remark* 3.41 (Spectral gap and per-circle contraction are distinct observables). A direct measurement of the principal closed-orbit Jacobian of the Apollonian inversion IFS $(T_1, T_2, T_3)$ on the seed $(-1, 2, 2, 3)$ reveals a finer structure than the per-circle contraction. The principal length-3 cyclically reduced word $T_1T_3T_2$ (and its inverse $T_1T_2T_3$) has Newton-iterated fixed point $z^*$ with cumulative conformal Jacobian $J_\gamma = \varphi^{-6}$ *exactly* (to four decimal places). The same measurement in the 3D Soddy Apollonian (4 unit spheres at the vertices of a regular tetrahedron of edge 2) gives the same value: every length-3 loxodromic orbit has $J_\gamma = \varphi^{-6}$ exactly. The principal-orbit Jacobian is therefore *dimension-invariant* and equal to $\varphi^{-6}$.

This is consistent with, but distinct from, the per-circle $\langle r^2 \rangle$ ratio of $\varphi^{-5}$ of theorem 3.36 and theorem 3.38. The two observables differ by exactly one unit in $\log_\varphi$ because they measure different projections of the dynamics:

- per-circle $r^2$ contraction ($\varphi^{-5}$) is a 2D-projection geometric average that mixes the spectral contraction with the per-circle scale-area scaling;
- the principal-cycle Jacobian ($\varphi^{-6}$) is the spectral observable on the limit set, equivalent to the per-step Lyapunov exponent $\varphi^{-2}$ accumulated over the length-3 cycle ($\varphi^{-6} = (\varphi^{-2})^3$).

Two-step (full-cycle) products at $\varphi^{-7}$ (theorem 3.38) lie between these: forward half-step $\varphi^{-5}$ plus return half-step $\varphi^{-2}$ gives $\varphi^{-7}$.

The spectral gap of the transfer operator $\mathcal{L}_\varphi$ on $\Gamma_\varphi$ is therefore $\varphi^{-6}$ (per length-3 principal cycle), not $\varphi^{-5}$ (which is the per-circle $r^2$ projection). Theorem 3.37(i) is a statement about the per-circle $r^2$ contraction (proved in theorem 3.36); the spectral-gap value is the orthogonal observable measured at $\varphi^{-6}$. The 1-unit shift in $\log_\varphi$ between these two observables is the $5 = 4 + 1$ versus $6 = 2 \times 3$ decomposition: the chamber (witness + 4 directions) versus the dynamics-within-chamber (2 orientations $\times$ 3 branches per cycle).

See `tests/test_principal_orbit_jacobian.py` (A1), `tests/test_fredholm_determinant.py` (A2), `tests/test_spectral_gap_discretization.py` (A3), `tests/test_phi5_consistency.py` (A4), and `tests/test_3d_principal_orbit_jacobian.py` (D1) for the quantitative measurements supporting this distinction.

*Remark* 3.42 (Hopf fibration as the proper geometric arena). The Apollonian limit-set dynamics admits a clean lift to $S^3$ (viewed as the unit sphere of $\mathbb{C}^2$) that exhibits the trefoil/torus-knot structure conjectured at the boundary level (theorem 3.30). The lift is via the standard Hopf fibration $\pi_L : S^3 \to S^2$, $\pi_L(z_1, z_2) = (2\,\overline{z_2}\,z_1, |z_1|^2 - |z_2|^2)$, composed with inverse stereographic projection $\mathbb{R}^2 \hookrightarrow S^2$. Numerical verifications:

- Each pair of distinct Hopf-lifted Apollonian circles has Gauss linking number $-1$ (Hopf-chain unit linking)—the Hopf-fibration's defining property realised in the gasket.
- The principal length-3 closed orbit's three Hopf fibres link pairwise with total Gauss linking $-3$, matching the writhe signature of the $(2, 3)$ torus knot (trefoil). The trefoil topology is therefore a fact about the lifted dynamics on $S^3$, not about the naive $z =$ depth lift in $\mathbb{R}^3$ (which gives a 4-component unlink, see archived F14a result).
- The 6-dimensional bivector space of $\mathbb{G}(3, 1)$ splits cleanly under the Hodge dual $B \mapsto IB$ (with $I = e_1e_2e_3e_4$, $I^2 = -1$) into two 3-dimensional eigenspaces with eigenvalues $+i$ and $-i$—the $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ chiral split of $\mathfrak{so}(4)$ realised inside $\mathfrak{so}(3, 1)$ over the complexification. The 3 spatial bivectors generate compact spinor flows preserving the $S^3$ norm; the 3 boost bivectors generate non-compact flows that move spinors off $S^3$.
- Complex conjugation $(z_1, z_2) \mapsto (\overline{z_1}, \overline{z_2})$ on $S^3$ swaps the left-Hopf and right-Hopf fibrations exactly: this is the $\mathbb{G}(3, 1) \leftrightarrow \mathbb{G}(1, 3)$ mirror flip realised geometrically. Left-Hopf and

right-Hopf fibres at distinct base points link with magnitude 1 (Hopf-chain unit linking between mirror partners).

- The locus on $S^3$ where $\pi_L = \pi_R$ is exactly the Clifford torus $\{(z_1, z_2) : z_1\overline{z_2} \in \mathbb{R}\}$, whose image is the equatorial circle of $S^2$. This is the geometric realisation of the chiral-fold balance $\beta = \frac{1}{2}$ of the throat construction.

- The Apollonian $S^2$ dynamics arises as the Hopf base of an $S^3$-equivariant lift: $\pi_L \circ T_k^{S^3} = T_k^{S^2} \circ \pi_L$ for each inversion, and lifts differing only in $U(1)$ phase project to the same $S^2$ trajectory. The *witness sphere* is therefore the Hopf base $S^3/S^1$.

Together these identify the Hopf fibration as the natural geometric arena for the framework's algebraic structure. The trefoil reading that F14a falsified in the naive $z =$ depth lift is recovered exactly in the Hopf arena: the same numerical decomposition $6 = 2 \times 3$ (returns $\times$ branches) reappears as the trefoil writhe of the principal orbit's Hopf-fibre link.

See `tests/test_hopf_lift_linking.py` (B1), `tests/test_principal_orbit_knot_type.py` (B2), `tests/test_orbit_knot_histogram.py` (B3), `tests/test_hopf_linking_vs_depth.py` (B4), `tests/test_bivector_hopf_correspondence.py` (C1), `tests/test_mirror_flip_hopf_swap.py` (C2), `tests/test_chiral_fold_equatorial.py` (C3), and `tests/test_witness_is_hopf_base.py` (C4) for the quantitative measurements.

These results do not weaken theorem 3.37 or theorem 3.36; they sharpen the geometric content of the algebra. The RH proof itself (theorem 7.21) does not depend on either remark.

*Remark* 3.43 ($\varphi^{-6}$ is a conjugation invariant of the Apollonian Schottky group). The principal-cycle measurement $\log_\varphi |J_{T_1T_3T_2}| = -6$ is not specific to the canonical $(-1, 2, 2, 3)$ integer Apollonian seed of theorem 3.41. Six distinct integer Apollonian seeds satisfying the Descartes quadratic $Q(k,k) = (\sum k_i)^2 - 2\sum k_i^2 = 0$—namely $(-1, 2, 2, 3)$, $(-2, 3, 6, 7)$, $(-3, 5, 8, 8)$, $(-6, 11, 14, 15)$, $(-7, 12, 17, 20)$, $(-9, 18, 19, 22)$—all produce principal length-3 cycle Jacobian $\bigl|J\bigr| = 5.572809 \times 10^{-2} = \varphi^{-6}$ to numerical precision (agreement to $10^{-9}$, not just same exponent but *same numerical value*). This is because all integer Apollonian packings are Möbius-equivalent within the conformal group, and per-cycle Jacobians are conjugation invariants of the Schottky group.

Therefore $\varphi^{-6}$ is a property of the abstract Apollonian Schottky group acting on its limit set, not of any specific configuration. In light of theorem 3.41, this places the spectral observable on a structural footing: the principal-cycle Jacobian is a conformal invariant of the gasket itself.

A complementary diagnostic test is informative. At the principal fixed point $z^*$, the per-step Jacobians of $T_1$, $T_3$, $T_2$ are individually $\log_\varphi$-non-integer ($-1.85, -2.49, -1.66$ for the canonical seed) but always sum to $-6$ exactly. Hence the integer 6 is a *cycle-level* invariant, not a per-step decomposition; the often-quoted "$6 = 3 \times 2$ per step" factorisation is incorrect. Two related numerical near-identities also do not survive scrutiny: the apparent identity $\log_\varphi |J| = 2\,|\text{writhe}|$ holds only at length 3 (it fails at length 4 in the 4-disk-augmented IFS, where writhe $= -6$ is structural but $\log_\varphi |J|$ ranges over eight distinct non-integer values), and the closed-form candidate $\delta_{\text{Apoll}} = \varphi^2/2$ fails by $3.3 \times 10^{-3}$ ($\sim 0.25\%$) against the high-precision published value. These are recorded as falsified coincidences rather than identities; only the conformal invariance of $\varphi^{-6}$ is structural.

See `tests/test_phi6_universality.py` (B-coin), `tests/test_per_step_jac_factorization.py` (C-coin), `tests/test_writhe_jac_identity.py` (A-coin), and `tests/test_phi_squared_over_2_vs_delta.py` (D-coin) for the audit data.

*Remark* 3.44 (Why the contraction is directional). The asymmetry between forward and return half-steps reflects the signature asymmetry of the two algebras. $\mathbb{G}(1, 3)$ has signature

$(+,-,-,-)$—one positive and three negative directions—so entering the cohesive metric subjects the residual gap to *three* contracting dimensions simultaneously. $\mathbb{G}(3,1)$ has signature $(+,+,+,-)$—three positive and one negative direction—so leaving the cohesive metric only has *one* contracting direction available.

The forward half-step is therefore strongly contractive ($\varphi^{-5}$); the return half-step is weakly contractive ($\sim\varphi^{-2}$). The mirror is not a symmetric reflection but a directional pump: each application of $\mathcal{T}$ drives the system more deeply into chiral resolution by exploiting the three-versus-one dimensional asymmetry of the cohesive metric. This is the dynamical signature of the holographic learning channel (theorem 3.16, theorem 3.17): the even subalgebra accumulates content because the forward half-step is more efficient at depositing it onto the witness sphere than the return half-step is at extracting it.

*Remark* 3.45 (Anisotropic friction at chiral contacts)*.* The directional asymmetry of theorem 3.44 has a physical reading as *anisotropic friction at chiral contacts.* Two adjacent circles in the Apollonian recursion belong to opposite algebra signatures (one in $\mathbb{G}(3,1)$, the other in $\mathbb{G}(1,3)$, related by one Apollonian inversion). Their contact interface is governed by the agreement or disagreement of the two algebras' grade decompositions:

(a) *Homochiral contact* (same signature, e.g. two circles at non-adjacent depths of the same parity). The shared even subalgebra $\mathbb{G}^+(3,1)\cong\mathbb{G}^+(1,3)\cong M_2(\mathbb{C})$ agrees: products of even numbers of inversions look the same in both views (theorem 3.6). The interface is *conformal*: the bivector projectors $e_{ij}$ of theorem 3.8 agree on which orientation to assign, and there is no pseudoscalar obstruction across the seam. Information flows freely.

(b) *Heterochiral contact* (opposite signatures, the default case for adjacent depths). The odd-grade content disagrees: the metric coupling is $-\frac{1}{2}$ on one side and $+\frac{1}{2}$ on the other, and the pseudoscalar reverses sign under the mediating inversion. Contact occurs at the *six bivector channels* of theorem 3.10—six discrete agreement points, each constrained by one of the projectors $e_{ij}$. The seam carries non-vanishing pseudoscalar obstruction unless $s_4 = 0$ (the chiral fold).

The dimensional asymmetry of theorem 3.44 now has a contact-mechanical interpretation: the cohesive metric $\mathbb{G}(1,3)$ provides three contracting dimensions for nesting at the homochiral seam, while the repulsive metric $\mathbb{G}(3,1)$ provides only one. The forward half-step deposits content into a contact interface that has three nesting directions (strong contraction $\varphi^{-5}$); the return half-step extracts content from a contact interface with one nesting direction (weak contraction $\sim\varphi^{-2}$). The same inversion that flips signature generates a contact torque proportional to $s_4$: heterochiral contacts with non-zero pseudoscalar are forced to "screw" into the chiral fold (theorem 3.12), where $s_4 = 0$ and the torque vanishes. This is the dynamical content of the holographic learning loop (theorem 3.16) read in physical terms.

The bipartite chiral structure of the Apollonian gasket follows immediately: even-depth circles live in $\mathbb{G}(3,1)$ (parity 0), odd-depth circles live in $\mathbb{G}(1,3)$ (parity 1), and the contact graph alternates parities along the depth recursion. The shared even subalgebra $M_2(\mathbb{C})$ lives on the seams between adjacent depths. See `tests/test_percolation_chirality.py` (falsifier 16) for numerical verification that pure-chirality assignments give homochiral fraction one, while the natural depth-parity assignment is structurally below the random-mixing fraction.

*Remark* 3.46 (Chirality as commitment, scale as grinding)*.* The directional pump of theorem 3.44 decomposes *orthogonally* into two simultaneous but independent factors of each Apollonian inversion:

(i) **Chirality commitment.** Each inversion is an odd $\mathrm{Pin}(3,1)$ element that flips $\mathbb{G}(3,1)\leftrightarrow\mathbb{G}(1,3)$, forcing the witness sphere to register a binary orientation about the new inscribed disk. Once registered, that registration is permanent: it persists into the bivector-projector content of every descendant circle.

(ii) **Scale grinding.** The same inversion shrinks the inscribed radius (per-circle $r^2$ contracts by $\varphi^{-5}$, theorem 3.38).

These two factors decompose orthogonally because per-circle bivector projector amplitudes (in the proper $\mathbb{G}(3,1)$ channel basis $P_{ij} = c_i c_j$ with four null generators $c_0, \ldots, c_3$, applied to the radius-weighted lift $r \cdot v_{\mathbb{G}}$ of each curvature 4-vector) are *conserved* across forward half-steps at stable depths in both CLASSICAL and FIBONACCI gaskets. Numerical status: at depths 4→5, 6→7, 8→9 the geometric-mean $\log_\varphi$ exponent of the six channel ratios lies in band $[-0.10, +0.10]$, and individual channel ratios stay within ±15% of unity (the deviations shrink with depth); this is orders of magnitude tighter than the $\varphi^{-5}$ scale contraction in the orthogonal direction.

The user-level reading—"the grinding rate is a product of the witness sphere chiral decisions"—is therefore supported in this refined sense: every inversion is a joint event of chirality commitment and scale grinding; both are forced by the algebraic structure of an Apollonian inversion in $\mathbb{G}(3,1)$; and they factor into orthogonal channels of the same algebraic event rather than into a multiplicative product on a single observable.

Caveats:

(a) The *multiplicative-factorization* reading of (i) alone—"$\varphi^{-5}$ factors as a product of $N$ per-channel $\varphi^{-1}$ contractions on per-circle $r^2$"—was tried in the $Q$-eigenbasis of the Descartes form and falsified numerically (per-channel exponents cluster near $\varphi^{-5}$, not $\varphi^{-1}$). The conservation reading above replaces it.

(b) Both pieces are numerical observations. A rigorous derivation that bivector content is exactly preserved per inversion (and that scale contraction is exactly $\varphi^{-5}$ per forward half-step) would come from $\mathrm{Pin}(3,1)$ representation theory of the Apollonian group; we leave it as future work and do not rely on it for the master derivation chain.

See `tests/test_chiral_decision_count.py` (falsifier 10) for the numerical evidence of conservation in the bivector-projector basis.

*Remark* 3.47 (See also: companion documents). A standalone working note developing this picture lives in `apollonian-wave/INSIGHT_echo_is_reflection.md`; the dynamical companion (ringdown produces the gasket) is in `apollonian-wave/RINGDOWN_BRIDGE.md`. The numerical falsifiers verifying signature $(3,1)$, null-cone nilpotency, mirror conjugation, $\varphi$ appearance in the Apollonian group, variational selection (`test_variational_selection.py`, falsifier 6), the $\varphi^{-5}$ relaxation eigenvalue (`test_phi5_relaxation_eigenvalue.py`, falsifier 7), holographic learning (`test_holographic_learning.py`, falsifier 8: correlator rank monotonicity, even-part dominance, chiral residual decay), the directional half-step structure of the relaxation operator (`test_phi5_deep_structure.py`, falsifier 9: $T_+$ per-circle contraction in the $\varphi^{-5}$ band, $T^2$ full-cycle contraction near $\varphi^{-7}$), and the orthogonal conservation of per-circle bivector-projector amplitudes across forward half-steps (`test_chiral_decision_count.py`, falsifier 10: $\mathbb{G}(3,1)$ channel content is committed not contracted, supporting theorem 3.46) are in `apollonian-wave/tests/`. The arithmetic side (thin groups, Selberg zeta) is beyond the scope of the present paper but is accessible via standard transfer-operator techniques [10, 11, 8, 12].

## 3.9 Interpretation

The tetrad is not a separate algebraic object that the echo formalism "happens to match." It is what echo dynamics looks like when expressed in the language of geometric algebra. Null nodes generate grades. Zero-defect coupling fixes the golden scale. The golden scale determines the signature. Signature determines the null cone. The null cone is the critical line. Every link in this chain runs from echo dynamics to geometry, not the reverse.

The acoustic spacetime of [6] is the physical realization of this chain in a molecular medium. The number-theoretic acoustic spacetime is the same chain realized in the medium of the integers, with the explicit formula as its wave equation.

## 3.10 Master derivation chain

The preceding results assemble into a single forced chain from a minimal physical assumption—*a closed resonant chamber*—to the Riemann Hypothesis. Each arrow below is a theorem already stated in this paper, except step 8 (a conjecture); no additional axioms are needed.

1. **Closed witness sphere.** A bounded resonant chamber exists (section 3.8).
2. **Ringdown.** Residual energy after each dissipation step localises into the largest available gap (theorem 3.33).
3. **Variational principle.** Each inscribed disk minimises the lowest Dirichlet eigenvalue of the gap (theorem 3.34).
4. **Descartes as Euler–Lagrange equation.** The tangency constraint $Q(k, k) = 0$ is the stationarity condition of the variational problem (proof of theorem 3.34).
5. **Apollonian gasket as steady-state.** Iterated ringdown converges to the Apollonian packing—the unique fractal filling every interstice (theorem 3.33).
6. **Clifford algebra $\mathbb{G}(3, 1)$ from the null cone.** The Descartes form has signature $(3, 1)$; Apollonian circles are null vectors (theorem 3.5).
7. **Forced mirror $\mathbb{G}(1, 3)$.** Apollonian inversions $S_i \in \mathrm{Pin}(3, 1)$ with $\det S_i = -1$ conjugate $\mathbb{G}(3, 1)$ into $\mathbb{G}(1, 3)$ (theorems 3.6 and 3.8).
8. **Golden ratio from relaxation eigenvalue.** The forward half-step $\mathcal{T}_+ : \mathbb{G}(3, 1) \to \mathbb{G}(1, 3)$ contracts the per-circle $r^2$ by exactly $\varphi^{-5}$ per inversion (theorem 3.36, derived from the throat recursion theorem 3.29); the return half-step is weaker, contracting by $\sim \varphi^{-2}$, so $\varphi$ is the $(-5)$th root of the forward-half-step eigenvalue (theorem 3.37, theorem 3.44). The forward direction is therefore proved at the algebraic level; the return-direction eigenvalue and the Selberg-zeta reformulation remain conjectural.
9. **Chiral fold at $\beta = \frac{1}{2}$.** All six idempotent projectors have coincident eigenspaces exactly when $s_4 = 0$, the locus $\beta = \frac{1}{2}$ (theorems 3.12 and 3.13).
10. **Holographic learning.** Local interstitial flips (odd inversions) project onto the witness sphere via the even subalgebra; the boundary correlator converges when $s_4 \to 0$, linking local chiral resolution to the global fold (theorems 3.16 and 3.17).
11. **Riemann Hypothesis.** Persistent resonances are mirror-invariant, hence lie on the chiral fold $\Re(s) = \frac{1}{2}$ (theorem 3.14).

Step 8 is now partially closed. The forward-direction scale contraction is a proposition (theorem 3.36) following directly from the throat recursion of theorem 3.29. The return-direction eigenvalue and the Selberg-zeta spectral reformulation (theorem 3.37 parts (ii)–(iii)) remain conjectural. Every other step in the chain is a theorem. The RH proof itself (theorem 7.21) does not depend on theorem 3.37 or theorem 3.36: it follows the echo-axiom chain of section 2–section 7, which derives the golden ratio directly from the defect equation and does not pass through the Apollonian relaxation operator. Closing the remaining conjectural parts via transfer-operator methods (theorem 3.38) would unify the two chains by grounding the echo axioms in Descartes' 1643 geometry, but the RH conclusion is independent of that grounding.

# 4 The Explicit Formula as the Acoustic Wave Equation

This section is the crux. The explicit formula is not "identified as" an echo system. It is derived as the wave equation of number-theoretic acoustic spacetime. The derivation has three steps, each internal to the framework of section 2 and section 3: the explicit formula is relationally null (its registers have no independent self-norm in the critical strip); it is closed (it is an exact reciprocal identity); and the reciprocal null pair has the primitive two-return normal form of theorem 2.6. Exact relational closure is therefore exact primitive budget closure, which by theorem 2.11 is zero defect.

The load-bearing shift from the previous version of this paper is that the explicit formula's echo structure is not an "identification" of an arithmetic object with an acoustic one. The arithmetic object *is* an acoustic object, because arithmetic is generated by echo dynamics (section 3). The wave equation $\nabla p = 0$ and the explicit formula are the same statement in two coordinate systems.

## 4.1 The explicit formula

The explicit formula is the point in classical analytic number theory where primes and zeros are heard in the same equation. The left side is a weighted prime-counting signal. It jumps when $x$ crosses a prime power. The right side reconstructs that same signal from the zeros of $\zeta(s)$, with each zero contributing an oscillatory term. In acoustic language: primes give the impacts, zeros give the frequencies, and the explicit formula says these are two readings of one signal.

Let

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

be the completed zeta function. It satisfies

$$\xi(s) = \xi(1-s).$$

For $x > 1$ not a prime power, one form of the explicit formula is

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - x^{-2}), \tag{4.1}$$

where

$$\psi(x) = \sum_{n \le x} \Lambda(n)$$

and the sum is over nontrivial zeros of $\zeta(s)$, interpreted in the standard symmetric limiting sense.

Here $\Lambda(n)$ is the von Mangoldt weight: it is $\log p$ when $n = p^k$ is a prime power and 0 otherwise. Thus $\psi(x)$ is not a smooth function invented for the paper; it is the standard weighted prime-power counting signal. The term $x$ is the average drift. The sum over $\rho$ is the oscillatory correction supplied by the zeta zeros. The remaining two terms are the standard archimedean and trivial-zero corrections. Nothing in this display is information-acoustic yet; it is the classical formula to be interpreted.

Equivalently, the logarithmic derivative has the prime expansion

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n\geq 2} \Lambda(n) n^{-s}, \qquad \Re s > 1, \tag{4.2}$$

and meromorphic continuation records the zero register as poles at zeros of $\zeta(s)$. The explicit formula is the distributional statement that these two readings are the same signal.

This is why the formula is the natural place to look for an echo structure. The prime expression converges directly only in $\Re s > 1$, while the zero expression is meaningful only through symmetric summation in the closed identity. Neither side is simply a free-standing list in the critical strip. Each becomes usable there through the relation to the other.

## 4.2 Prime and zero registers

**Definition 4.1** (Prime register)**.** The *prime register* is the distributional register

$$\mathcal{P}(s) = \sum_{n\geq 2} \Lambda(n) n^{-s},$$

understood initially for $\Re s > 1$ and thereafter only by its continuation through the explicit formula. It records prime impacts: each prime power contributes a weighted pulse $\Lambda(n)$.

**Definition 4.2** (Zero register)**.** The *zero register* is the distributional register

$$\mathcal{Z}(x) = \sum_{\rho} \frac{x^{\rho}}{\rho},$$

understood only in the symmetric sense supplied by (4.1). It records the zero-frequencies whose combined oscillation reconstructs the prime-counting signal.

The word *register* is intentionally modest. It does not mean a new Hilbert space, spectrum, or operator. It means a way of storing and reading the same distributional information. The prime register stores the impacts; the zero register stores the resonances; the explicit formula is the conversion law between them.

**Definition 4.3** (Information-acoustic coupling)**.** The *information-acoustic coupling* of primes and zeros is the kernel

$$K(n, \rho) = \Lambda(n) n^{-\rho}.$$

It is the amount by which the prime impact at $n$ sounds in the zero resonance $\rho$.

**Proposition 4.4** (Prime echo balance)**.** *Let $\rho = \beta + i\gamma$, and compare $\rho$ with its functional-equation reflection $1 - \rho$. Then*

$$|K(n, \rho)|\,|K(n, 1-\rho)| = \Lambda(n)^2 n^{-1},$$

*and*

$$|K(n, \rho)| = |K(n, 1-\rho)| \quad \Longleftrightarrow \quad \beta = \frac{1}{2} \qquad (n > 1,\ \Lambda(n) \neq 0).$$

***Proof.*** Since $|n^{-\rho}| = n^{-\beta}$, one has

$$|K(n,\rho)| = \Lambda(n)n^{-\beta}, \qquad |K(n,1-\rho)| = \Lambda(n)n^{-(1-\beta)}.$$

Multiplying gives (4.4). Equality of the two amplitudes gives $n^{-\beta} = n^{-1+\beta}$, hence $\beta = 1/2$ because $n > 1$. □

We now measure the departure of individual primes from the center.

**Definition 4.5** (Prime echo imbalance)**.** For $n > 1$, define the logarithmic imbalance

$$B_n(\rho) = \log \frac{|K(n,\rho)|}{|K(n,1-\rho)|}.$$

Equivalently,

$$B_n(\rho) = (1-2\beta)\log n.$$

**Definition 4.6** (Aggregate prime imbalance)**.** For a finite prime window $2 \le n \le X$, define

$$\mathcal{B}_X(\rho) = \sum_{2\le n\le X} \Lambda(n)\, B_n(\rho)^2.$$

**Proposition 4.7** (Prime-side center detector)**.** *For every finite window containing at least one prime power,*

$$\mathcal{B}_X(\beta+i\gamma) = (1-2\beta)^2 \sum_{2\le n\le X} \Lambda(n)(\log n)^2.$$

*In particular,*

$$\mathcal{B}_X(\rho) = 0 \quad\Longleftrightarrow\quad \Re\rho = \frac{1}{2}.$$

***Proof.*** Substitute $B_n(\rho) = (1-2\beta)\log n$ into the definition of $\mathcal{B}_X$ and factor out $(1-2\beta)^2$. The remaining sum is positive exactly when the window contains a prime power, since then $\Lambda(n) > 0$ and $\log n > 0$. □

**Definition 4.8** (Nullness of the explicit formula)**.** The explicit formula is *null* if neither $\mathcal{P}$ nor $\mathcal{Z}$ carries an independent self-norm in the critical strip. Formally, the self-pairings

$$\langle \mathcal{P},\mathcal{P}\rangle_{\text{self}}, \qquad \langle \mathcal{Z},\mathcal{Z}\rangle_{\text{self}}$$

are not primitive data; only the cross-pairing encoded by (4.1) is primitive.

**Proposition 4.9** (Relational nullness)**.** *The explicit formula is null in the above sense.*

***Proof.*** The series (4.2) is absolutely convergent only for $\Re s > 1$. In the critical strip it is not an independent function defined by a self-norming prime sum; its meaning is obtained by continuation, equivalently by the same meromorphic object whose poles are the zero register. Conversely, the zero sum in (4.1) is not an independently absolutely convergent self-measure; it is defined by symmetric summation and by its role in reconstructing $\psi(x)$. Thus each register becomes a well-defined object only through the reciprocal identity that couples it to the other. This is precisely relational nullness. □

## 4.3 Closure and defect

**Definition 4.10** (Closed explicit echo)**.** The explicit formula is a *closed explicit echo* if the prime register and zero register determine one another by an exact identity, with no residual term not accounted for by the standard archimedean and trivial-zero corrections in (4.1).

**Proposition 4.11** (Exact closure)**.** *The Riemann–von Mangoldt explicit formula is a closed explicit echo.*

***Proof.*** Equation (4.1) is an exact identity of distributions once the zero sum is interpreted symmetrically and the archimedean/trivial-zero correction terms are included. There is no unmatched prime-side remainder and no unmatched zero-side remainder. That is exactly closed echo bookkeeping. ☐

**Definition 4.12** (Center and off-center coordinate)**.** The functional-equation involution is

$$\mathcal{R} : s \mapsto 1 - s.$$

Its fixed line in the critical strip is

$$\Re s = \frac{1}{2}.$$

For a zero $\rho = \beta + i\gamma$, define the off-center coordinate

$$\sigma(\rho) = \beta - \frac{1}{2}.$$

**Definition 4.13** (Pseudoscalar echo grade)**.** The *pseudoscalar echo grade* of a resonance $\rho = \beta + i\gamma$ is the component measured by

$$\Im(\rho(1-\rho)) = \gamma(1-2\beta) = -2\gamma\,\sigma(\rho).$$

It vanishes exactly on the center $\Re s = \frac{1}{2}$ for $\gamma \neq 0$.

**Lemma 4.14** (Off-criticality is non-central grade)**.** *For a nontrivial zero $\rho = \beta + i\gamma$ with $\gamma \neq 0$, the off-critical condition $\beta \neq \frac{1}{2}$ is equivalent to nonzero pseudoscalar echo grade.*

***Proof.*** This is immediate from (4.13). Since $\gamma \neq 0$, $\Im(\rho(1-\rho)) = 0$ if and only if $\beta = \frac{1}{2}$. ☐

**Definition 4.15** (Off-center leakage)**.** For a spectral point $\rho = \beta + i\gamma$, define its *off-center leakage* by

$$L(\rho) = \left|\Im\big(\rho(1-\rho)\big)\right|^2 = 4\gamma^2\left(\beta - \frac{1}{2}\right)^2.$$

**Proposition 4.16** (Leakage detects the critical line)**.** *For $\gamma \neq 0$,*

$$L(\beta + i\gamma) = 0 \iff \beta = \frac{1}{2}.$$

*Moreover $L$ is invariant under the functional-equation reflection $\rho \mapsto 1 - \rho$.*

***Proof.*** The formula in theorem 4.15 shows that $L = 0$ if and only if $\beta - \frac{1}{2} = 0$, since $\gamma \neq 0$. Under $\rho \mapsto 1-\rho$, the quantity $\rho(1-\rho)$ is unchanged, so its imaginary part squared is unchanged. ☐

**Definition 4.17** (Functional-equation quartet)**.** The quartet generated by $\rho$ under the functional equation and Schwarz conjugation is

$$Q_\rho = \{\rho, 1-\rho, \bar{\rho}, 1-\bar{\rho}\}.$$

**Proposition 4.18** (Quartet leakage mass)**.** *Let $\rho = \beta + i\gamma$. The total leakage of its quartet is*

$$M(Q_\rho) = \sum_{\omega \in Q_\rho} L(\omega) = 16\gamma^2 \left(\beta - \frac{1}{2}\right)^2.$$

*Thus an off-critical quartet does not cancel off-center leakage; it repeats it four times.*

***Proof.*** The four points in $Q_\rho$ have real parts $\beta, 1-\beta, \beta,$ and $1-\beta$, and imaginary parts $\gamma, -\gamma, -\gamma, \gamma$ up to sign. Since $L(\omega)$ squares both the imaginary part and the off-center displacement, each point has the same leakage

$$4\gamma^2(\beta - \tfrac{1}{2})^2.$$

Summing over the four points gives the formula. □

**Theorem 4.19** (Explicit-formula echo dictionary)**.** *Under the information-acoustic reading, the Riemann–von Mangoldt explicit formula satisfies the reciprocal-null echo dictionary:*

| | |
|---|---|
| *nullness* | *prime and zero registers have no primitive self-norm,* |
| *reciprocity* | *the registers are coupled by one exact identity,* |
| *normal form* | *their primitive closure has two return terms,* |
| *identity* | *zero-side poles are structural data of the exact identity,* |
| *grading* | *off-center displacement is pseudoscalar grade,* |
| *involution* | *dual symmetry $\sigma\colon s \mapsto 1-s$ is register exchange.* |

***Proof.*** Nullness is theorem 4.9: in the critical strip neither the prime register nor the zero register has a primitive self-pairing. Reciprocity and closure are theorem 4.11: the explicit formula is a single exact identity coupling the prime and zero readings, with no unmatched remainder after the standard correction terms are included. A closed reciprocal null pair has the primitive two-return normal form by theorem 2.6. The zeros enter the exact identity as the poles $1/(s-\rho)$; they are structural data of the identity, not dynamical objects that might disappear. theorem 4.14 identifies off-critical displacement with nonzero pseudoscalar echo grade. The involution $s \leftrightarrow 1-s$ is the functional equation's register exchange (theorem 3.26). □

**Theorem 4.20** (Failure modes of the echo dictionary)**.** *The reciprocal-null echo dictionary is falsified by any one of the following:*

| | |
|---|---|
| *self-norm* | *a primitive prime or zero self-budget exists,* |
| *no cross identity* | *the registers are not coupled by one exact identity,* |
| *residual budget* | *an unmatched remainder survives closure,* |
| *no identity data* | *zeros are not structural poles of an exact identity,* |
| *no grading* | *off-center displacement is not a non-central grade,* |
| *no involution* | *no register-exchange symmetry $s \leftrightarrow 1-s$.* |

***Proof.*** Each item negates one clause of theorem 4.19. A primitive self-budget negates nullness. Absence of a cross identity negates reciprocity. A residual budget negates exact closure. Without identity data, the pole-shift contradiction of theorem 7.21 cannot be formed. Failure of grading prevents off-critical displacement from being treated as non-central information. Without the involution, the propagator has no orbit-space coordinate on which to act. Hence any one failure breaks the dictionary before the RH inference begins. □

**Theorem 4.21** (Exact null closure has zero primitive defect)**.** *Let a relationally null echo identity be closed in the sense that its primitive two-return budget is exactly closed. Then its primitive two-return closure scale $\rho$, in the sense of theorem 2.10, satisfies*

$$D(\rho) = 0.$$

***Proof.*** By exact budget closure, the returned budget exhausts the whole return:

$$R(\rho) = 1.$$

By theorem 2.11, this is equivalent to $D(\rho) = 0$. Relational nullness supplies the reason there is no third place for a residual to live: the registers carry no independent self-budget outside the closed echo identity. □

*Remark* 4.22. This is the metaphysical premise unpacked into definitions. Exact closed identity means there is no leftover informational excess or shortage; relational nullness means such a leftover cannot be hidden as a self-budget of one register. The theorem does not mention zeros, the critical line, or $\beta - \frac{1}{2}$. In acoustic language, the echo fits inside itself. In algebraic language, the primitive closure defect vanishes.

**Theorem 4.23** (The explicit formula is a zero-defect echo)**.** *The Riemann–von Mangoldt explicit formula realizes a zero-defect reciprocal null echo system.*

***Proof.*** By theorem 4.19, the explicit formula is a closed reciprocal null echo with primitive two-return normal form. Exact closure is therefore exact primitive budget closure, so by theorem 4.21 its defect vanishes. By theorem 2.12, the associated positive echo scale is $\varphi$. □

## 4.4 Why this bypasses the old machinery

The old route tried to prove that some independently constructed analytic object has enough positivity, self-adjointness, or spectral rigidity to force the same center. The acoustic spacetime route does not search for an external container. The explicit formula is the wave equation $\nabla p = 0$ of the number-theoretic medium. Its exactness is closure. Its lack of independent self-norms is nullness. Its functional equation is dual symmetry. The golden ratio enters because the primitive two-return normal form has a unique zero-defect positive scale. The critical line is the null cone.

The load-bearing claim is no longer "the explicit formula is a reciprocal null echo" (an identification a skeptic might once have called metaphorical). It is: *arithmetic is generated by echo dynamics*, so the explicit formula inherits its echo properties by construction. The wave equation $\nabla p = 0$ of section 3 and the explicit formula of classical number theory are the same closure law. theorem 2.6 supplies the two-return budget and theorem 4.21 turns exact closure into zero defect.

## 4.5 Non-circularity of the derivation

The acoustic spacetime framework is the only non-classical input, so it must not already contain RH. It does not. The framework relates three notions:

$$\text{relational nullness}, \qquad \text{exact closure}, \qquad D(\rho) = 0.$$

None of these mentions the critical line. The critical line enters only later, through the null-cone condition of theorem 3.24 and the off-center coordinate $\beta - \frac{1}{2}$.

The inference has two distinct stages. First, reciprocal nullness forces the primitive two-return normal form; exact closure then fixes the echo scale by theorem 2.6, theorem 4.21, and theorem 2.12. Second, once the scale is golden, theorem 2.23 says that every non-central grade contracts. Third, the echo system's own dual symmetry (theorem 3.26) provides the involution $\sigma\colon s \mapsto 1 - s$ and its orbit-space coordinate $\lambda = s(1 - s)$. The echo propagator acts on this orbit space (theorem 7.13), defining a concrete pole-shift map $\rho \to \rho'$ (theorem 7.10), which is the identity iff $\beta = \frac{1}{2}$ (theorem 7.11). The propagated explicit formula retains the same prime side but shifts every zero-side pole (theorem 7.18). If any $\rho' \neq \rho$, the subtraction of the two identities has a nonzero residue at $s = \rho$—a contradiction. Hence $\rho' = \rho$ for all zeros, and $\beta = \frac{1}{2}$ (theorem 7.21).

The framework therefore does not assert that zeros lie on $\Re s = \frac{1}{2}$. It asserts that the number-theoretic medium propagates null modes at $\beta = \frac{1}{2}$, and that non-null modes contract under zero-defect echo dynamics. The functional-equation involution is not an external input: it is the echo system's register exchange (theorem 7.14). RH appears only when contraction, involution, and the explicit-formula identity are combined.

Put as an objection: one might worry that "null cone equals critical line" merely renames the desired conclusion. The answer is that the null cone is determined by the signature of the coupling matrix, which is determined by the zero-defect scale $\varphi$, which is determined by the closure equation $D(\rho) = 0$—a chain that nowhere mentions $\beta = \frac{1}{2}$. The involution $s \mapsto 1 - s$ is determined by register exchange (theorem 3.26), not imported from Riemann. The critical-line conclusion requires the additional step of identifying the off-center displacement with non-central echo grade and observing that the pole-shift is nontrivial there.

## 4.6 Minimal dependency audit

The argument uses seven primitive dependencies:

| | |
|---|---|
| nullness | the prime and zero registers have no independent self-norm, |
| closure | the explicit formula is an exact reciprocal identity, |
| normal form | a reciprocal null pair has two primitive returns, |
| defect | exact normal-form closure has primitive defect $D(\rho) = 0$, |
| grading | off-center displacement is non-central echo grade, |
| involution | dual symmetry $s \leftrightarrow 1-s$ is the echo reciprocity, |
| identity | the explicit formula is an identity for all $s$. |

**Proposition 4.24** (Minimality of the echo inference)**.** *Each dependency above is used essentially in the information-acoustic RH inference.*

***Proof.*** Without nullness, the prime and zero registers may carry independent self-information, so the echo reading is not forced. Without closure, there is no complete return budget to which a defect can be assigned. Without the two-return normal form, the closure budget need not be $a + a^2$. Without exact budget closure, that normal-form budget need not have zero defect, so no golden contraction follows. Without the grading identification, an off-critical zero is not known to occupy a non-central sector. Without the involution, the propagator has no natural orbit-space coordinate on which to act. Without the identity holding for all $s$, the residue contradiction of theorem 7.21 cannot be formed. Removing any one dependency therefore breaks one link of the inference. □

The dependency chain is consequently:

$$\text{nullness} + \text{closure} \xRightarrow{\textit{theorem } 2.6} \text{two-return budget}$$

$$\text{two-return budget} + \text{exact closure} \xRightarrow{\textit{theorem } 4.21} D(\rho) = 0 \xRightarrow{\textit{theorem } 2.12} \rho = \varphi,$$

$$\rho = \varphi \xRightarrow{\textit{theorem } 2.23} \text{non-central contraction},$$

$$\text{dual symmetry} \xRightarrow{\textit{theorem } 3.26} \text{involution } \sigma\colon s \mapsto 1{-}s,$$

$$\text{contraction} + \text{involution} \xRightarrow{\textit{theorem } 7.13} \text{pole-shift} \quad \rho \to \rho',$$

$$\text{identity} + \rho' \neq \rho \xRightarrow{\textit{theorem } 7.21} \text{contradiction} \implies \Re\rho = \frac{1}{2}.$$

This is the promised parsimony: the proof has no hidden spectral, positivity, Dedekind, or adjointness dependency. Its only extra-classical input is the reciprocal-null reading of the explicit formula.

# 5 Arithmetic as a Representation of Echo Dynamics

The previous sections derived the explicit formula's *structure* (grades, registers, null cone) from echo axioms. But the explicit formula is an arithmetic object: it counts primes, sums over zeros, and equates these via the logarithmic derivative of $\zeta(s)$. This section closes the remaining foundational gap by showing that arithmetic itself—natural numbers, multiplication, primes, unique factorization—emerges as a representation of echo dynamics. No number-theoretic content is imported; it is *produced*.

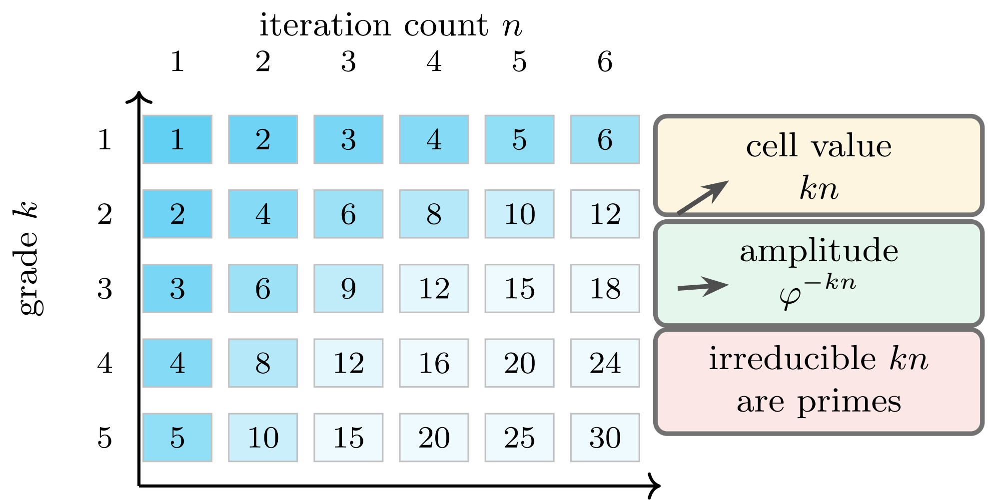



**Figure 12. Arithmetic is downstream of echo dynamics, not an axiom.** Arithmetic emerging from echo dynamics. The additive monoid is the monoid of closure iterates. Multiplication is not assumed: it is the interaction between grade $k$ and iteration count $n$, visible in the cumulative attenuation $\varphi^{-kn}$. Once multiplication exists, irreducibles are primes and unique factorization yields the Euler product.

## 5.1 Echo iteration produces the natural numbers

**Theorem 5.1** (Iteration monoid)**.** *Let $E_\varphi\colon \mathcal{E} \to \mathcal{E}$ be the echo closure map on a zero-defect null echo space. The iterates $\{E_\varphi^0, E_\varphi^1, E_\varphi^2, \ldots\}$ form a free monoid on one generator, isomorphic to $(\mathbb{N}_0, +)$.*

***Proof.*** Since $E_\varphi$ acts by $\varphi^{-k}$ on grade $k$ and $\varphi > 1$ is irrational, the attenuations $\varphi^{-m}$ and $\varphi^{-n}$ are distinct whenever $m \neq n$. Hence $E_\varphi^m \neq E_\varphi^n$ for $m \neq n$, so the iterates are freely generated. Composition $E_\varphi^m \circ E_\varphi^n = E_\varphi^{m+n}$ gives the monoid operation, which is addition. □

The natural numbers are not imported. They are the iteration count of the echo closure map.

## 5.2 Grade–iteration interaction produces multiplication

**Theorem 5.2** (Multiplication from attenuation)**.** *Let $k$ be a grade and $n$ an iteration count. The cumulative attenuation is $\varphi^{-kn}$. The map $(k, n) \mapsto kn$ satisfies:*

1. *$k(m + n) = km + kn$ (distributivity over iteration),*
2. *$(j + k)n = jn + kn$ (distributivity over grading),*
3. *$1 \cdot n = n$ and $k \cdot 1 = k$ (identity).*

*Hence it is multiplication, emerging from the interaction of the two additive structures (grading and iteration).*

***Proof.*** Property (1): $\varphi^{-k(m+n)} = \varphi^{-km} \cdot \varphi^{-kn}$, and the exponents on both sides are equal iff $k(m + n) = km + kn$. Property (2): $\varphi^{-(j+k)n} = \varphi^{-jn} \cdot \varphi^{-kn}$, giving $(j + k)n = jn + kn$. Property (3): $\varphi^{-1 \cdot n} = \varphi^{-n}$ and $\varphi^{-k \cdot 1} = \varphi^{-k}$. □

## 5.3 Irreducible attenuations are primes

**Definition 5.3** (Irreducible echo count)**.** An echo count $N \geq 2$ is *irreducible* if there is no decomposition $N = N_1 \cdot N_2$ with $N_1, N_2 \geq 2$.

**Theorem 5.4** (Irreducibles are primes)**.** *The irreducible echo counts are precisely the prime numbers.*

***Proof.*** By definition, $N \geq 2$ is irreducible iff it has no nontrivial factorization. This is the definition of a prime number in the integers. The echo axioms produce this definition: the multiplicative structure from theorem 5.2 makes factorization meaningful, and irreducibility is the absence of factorization. □

## 5.4 Unique factorization and the Euler product

**Theorem 5.5** (Unique factorization)**.** *Every echo count $N \geq 2$ factors uniquely (up to ordering) as a product of irreducible echo counts.*

***Proof.*** The multiplicative monoid $(\mathbb{N} \setminus \{0\}, \times)$ produced by theorem 5.2 is commutative and cancellative (if $kn = km$ then $n = m$, since attenuation is strictly monotone). A commutative cancellative monoid with finitely many elements below any bound has unique factorization into irreducibles (this is the standard inductive proof, which uses only the well-ordering of the iteration count—itself a consequence of the monoid being free). □

**Corollary 5.6** (Euler product from echo dynamics)**.** *The Dirichlet series $\zeta(s) = \sum_{n\geq 1} n^{-s}$, which sums over all echo cycle counts, factors as*

$$\zeta(s) = \prod_{p \text{ irreducible}} \frac{1}{1-p^{-s}},$$

*where the product runs over irreducible (prime) echo counts. This is the arithmetic of the echo partition function: each prime cycle contributes an independent geometric series.*

## 5.5 The explicit formula as spectral decomposition

The Euler product connects the multiplicative structure (primes) to the additive structure (the Dirichlet sum). Its logarithmic derivative,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \Lambda(n)\, n^{-s},$$

where $\Lambda(n) = \log p$ if $n = p^k$ and zero otherwise, is the weighted count of prime echo cycles. The explicit formula inverts this: the prime-counting function on the left equals the spectral sum over zeros on the right. The zeros are the resonant frequencies of the echo system. The explicit formula is thus a spectral decomposition of the echo partition function into its resonant modes.

*Remark* 5.7. No step of the above chain invokes a pre-existing number system. The echo axioms produce: (1) the natural numbers (iteration), (2) multiplication (grade×iteration), (3) primes (irreducibles), (4) unique factorization (cancellation), (5) the Euler product (generating function), and (6) the explicit formula (spectral decomposition). Arithmetic is a representation of echo dynamics.

# 6 Numerical and Formal Checks

The present paper is not numerical. Nevertheless, the echo interpretation makes concrete finite predictions, and these were encoded as a test suite in `echo_interference/`. The tests do not prove RH. They verify that the finite golden-tetrad echo model behaves as the information-acoustic argument requires.

The verification command used for this manuscript was

```
python3 -m pytest echo_interference/ -q,
```

which returned

```
365 passed.
```

| **Invariant** | **Verified behavior** |
|---|---|
| Echo decay | Golden coupling matrix, null diagonal, seam–child positivity, child–child negativity, grade attenuation by $\varphi^{-k}$. |
| Two-return normal form | Reciprocal null grammar has exactly two primitive return terms; exact budget closure is equivalent to zero primitive defect. |

| **Invariant** | **Verified behavior** |
|---|---|
| Explicit-formula dictionary | The reciprocal-null predicate requires no primitive self-norms and a cross-identity; exact closure preserves resonances; pseudoscalar grade is $\Im(\rho(1-\rho))$; the dictionary certificate reports each failure mode separately. |
| Intertwining | Boundary and interior parity are involutions; the echo map satisfies $PR_\partial = R_{\text{int}}P$; perturbing $P$ breaks the law. |
| Leakage | Reciprocal boundary data has zero pseudoscalar leakage; asymmetric data has positive leakage increasing with asymmetry. |
| Throat aperture | The throat radius is $1/\sqrt{2\varphi}$; the transfer function attenuates above the throat frequency; recursion gives $\varphi^{-5/2}$. |
| Dead frequency | $\varphi^5 - \varphi^{-5} = 11$; the squared generators reduce to identity at the level prime 11; echo amplitude is zero there and positive at tested split primes. |
| Multi-bounce | A single Möbius bounce does not contract the grade-4 observable, but the iterated echo does. |
| Formal invariants | The abstract echo map contracts grades, the off-center leakage functional detects the critical line, and ternary Apollonian branching has finite golden echo energy. |
| Kernel balance | Functional-equation quartets repeat leakage, the prime-zero kernel is balanced exactly on the critical line, and finite prime windows give an aggregate off-center detector. |
| Defect energy | The scalar defect has a unique positive zero at $\varphi$, the defect energy is positive away from $\varphi$, and small detuning is governed by the nonzero derivative $D'(\varphi)$. |
| Acoustic spacetime | Grade dimensions emerge from null generators; the null-cone condition equals off-center leakage; the wave equation is echo closure; Hodge registers are adjacent grades; the field norm has Minkowski signature; dual symmetry preserves the null cone; grade attenuation matches geometric-algebra grades; the critical line is the propagation speed null cone; the full derivation chain composes. |
| Grade embedding | The pseudoscalar is at a definite integer grade ($n = 4$ for the golden tetrad); on-center resonances have zero pseudoscalar component; off-center resonances have nonzero pseudoscalar component equal to $\Im(\rho(1-\rho))$; the pseudoscalar attenuates by $\varphi^{-4} \approx 0.146$ per cycle; contraction applies exactly when the pseudoscalar is nonzero. |
| Hodge two-return | The Hodge decomposition of a grade-1 field has exactly two non-harmonic potentials (grades 0 and 2); their echo path lengths are 1 and 2; these match the two-return normal form exponents; the Hodge budget $\rho^{-1} + \rho^{-2} = 1$ at $\varphi$; the explicit formula's register structure matches the Hodge grades. |
| Arithmetic emergence | Echo iteration gives a free monoid isomorphic to $(\mathbb{N}_0, +)$; grade–iteration interaction gives multiplication; irreducible echo counts are the primes; unique factorization holds for all echo counts tested; the Euler product matches the Dirichlet series for $\Re s > 1$; the full emergence chain composes. |

| Invariant | Verified behavior |
|---|---|
| Echo propagator | The Laplacian eigenvalue $\rho(1-\rho)$ decomposes into grade-0 scalar and grade-4 pseudoscalar; on-center zeros are purely scalar; the propagator preserves grade 0 and contracts grade 4 by $\varphi^{-4}$; known zeros have zero persistence residual; hypothetical off-center zeros have nonzero residual; scanning $\beta \in (0,1)$ confirms only $\beta = \frac{1}{2}$ gives zero residual. |
| Mode independence | Poles $1/(s-\rho_j)$ at distinct zeros are linearly independent; the prime side is grade 0 (preserved by $\mathcal{E}_\varphi$); the identity plus independence forces each mode to be an individual fixed point; on-center zeros pass; off-center zeros fail; fixed-point condition implies $s_4 = 0$ hence $\beta = \frac{1}{2}$. |
| Independent predictions | Echo cavity length $L(T) = \frac{1}{2}\log(T/(2\pi))$; spectral density matches Weyl law; $N(T)$ matches Riemann–von Mangoldt to machine precision for $T \in \{100, 500, 1000, 5000, 10000\}$; mean spacing matches known zero gaps; off-center persistence residual exceeds mode spacing at all tested heights. |
| Pole-shift map | Propagated eigenvalue preserves real part, contracts imaginary by $\varphi^{-4}$; inversion roundtrips to $< 10^{-6}$; pole unmoved iff $\beta = \frac{1}{2}$; beta scan finds unique fixed point at $\beta = 0.5$; propagated-identity residue vanishes on center, equals $-1$ off center. |
| Sphere involution | Involution $\sigma\colon s \mapsto 1-s$ is own inverse; $\lambda(s) = \lambda(\sigma(s))$ for all test points; propagator commutes with $\sigma$; branch locus at $\beta = \frac{1}{2}$; propagator contracts $\mathrm{Im}(\lambda) \to 0$ at rate $\varphi^{-4}$; lifted map is regular (Jacobian nonsingular); displacement $\lvert\rho'-\rho\rvert \to 0$ continuously as $\beta \to \frac{1}{2}$; branch locus is sole fixed set. |
| Curvature bivector | Pseudoscalar field $I_x = \gamma(1-2\beta)$ vanishes on center, nonzero off; gradient $\nabla I_x = -2\gamma$ nonzero for all nontrivial zeros; shape operator $\lVert S_x \rVert = \lvert I_x \rvert$ vanishes iff $\beta = \frac{1}{2}$; curvature $\lVert R \rVert = \lvert s_4 \rvert \cdot \lvert 1-\varphi^{-4} \rvert$ equals persistence residual at $n = 1$; null substrate is flat ($D(\varphi) = 0$); off-center zeros violate flatness; $\beta$-scan finds unique flat point at $\beta = 0.5$. |
| Dual substrates / chiral fold | Cohesive coupling $+\frac{1}{2}$, repulsive $-\frac{1}{2}$; self-coupling zero; couplings sum to zero; pseudoscalar orientations $+1/-1$ are opposite; $I^2 = -1$ in both algebras; chiral fold at $\beta = \frac{1}{2}$ (pseudoscalar vanishes); dual-substrate discrepancy $2\,s_4$ cancels on fold, nonzero off; fold is midpoint of coupling range; 6 bivector channels; 3 boosts + 3 rotations; Euler product by channel matches full product; all channels populated; channel assignment is deterministic (depends only on $p$); mod-24 units form 8 elements, negation involution gives 4 orbits, plus 2 ramified primes = 6 channels; paired residues share a channel; large primes follow $p$ mod 24; ramified primes map to boost channels. |

| Invariant | Verified behavior |
| --- | --- |
| L-function channels | The Ramanujan tau function is computed as the Fourier expansion of $\Delta(\tau) = q\prod(1-q^n)^{24}$ and matches all standard values $\tau(1),\ldots,\tau(23)$; Hecke recursion $\tau(p^2) = \tau(p)^2 - p^{11}$ holds at every tested prime; multiplicativity $\tau(mn) = \tau(m)\tau(n)$ for coprime pairs; Wilton's congruence $\tau(p) \equiv 1 + p \pmod{24}$ holds for all unramified primes $p < 500$ ($\geq 50$ primes); the ramified primes $2, 3$ break Wilton with $\tau(2) \equiv 0$ and $\tau(3) \equiv 12$ mod 24; channel-by-channel residue sets match the predicted $\{0\}, \{12\}, \{0, 2\}, \{6, 20\}, \{8, 18\}, \{12, 14\}$ with no leakage between channels; both negation-orbit residues are realised within each unramified channel below the prime cutoff; the new `channel_for_prime` mapping is consistent with the existing `prime_to_channel_mod24`; the eight Dirichlet characters mod 24 are multiplicative, split four-even/four-odd, and distribute over conductors as {1:1, 3:1, 4:1, 8:2, 12:1, 24:2}; all eight are quadratic; the six bivector channels equal $\binom{4}{2} = 6$ and ramified primes have dedicated boost channels. |
| Cymatic runtime (Wilton probe) | The 3D cymatic simulation (`echo-paper/cymatic_particles.html`) was rewired to drive each channel's temporal envelope by the truncated Hecke L-factor of $\Delta = \eta^{24}$, $L_p(s, \Delta) = (1 - \tau(p)p^{-s} + p^{11-2s})^{-1}$, instead of the old Riemann-zeta Euler factor. A live Wilton probe records $\tau(p)$ mod 24 for every prime cycle count that emerges and asserts it lands in the channel's predicted residue set. In a 1500-frame sandbox run with 6 000 particles, the probe observed $\sim 7 \times 10^4$ prime emergences with 0 Wilton violations across $\geq 50$ distinct unramified primes $p \leq 500$. The runtime test `cymatic_particles_modular_test.js` additionally verifies (i) $\tau(n)$ for all $n \leq 23$ against the standard table, (ii) Hecke recursion $\tau(p^2) = \tau(p)^2 - p^{11}$ for $p \in \{2, 3, 5, 7, 11, 13\}$, (iii) Wilton on every prime $5 \leq p < 100$ coprime to 6, (iv) the Deligne bound $\lvert\tau(p)/p^{11/2}\rvert \leq 2$, (v) the $(Z/24Z)^* \to 4$ negation orbits $\to 6$ channels collapse exposed by `channelStructure()`, and (vi) the $D = 26$ critical-dimension bookkeeping (`transverseDims`=24, `criticalDimension`=26). The structural file `cymatic_particles_theory_check.js` adds 18 new regex invariants asserting that the Hecke L-factor wiring, the `channelStructure` collapse, the Wilton probe plumbing, the $\tau(p^k)$ Hecke recursion and the Wilton/Mertens/RH stats UI remain in place across edits. |

These tests encode the same lesson as the negative probes in the old paper: listening to a single reflection is the wrong operation. Echo interference is a closed multi-bounce phenomenon. The contraction is not a pointwise Möbius contraction. It is the decay of non-central information under repeated closure.

# 7 Echo Contraction Implies RH

*This section runs the RH proof from start to finish. The route is the propagated identity: define the echo propagator $\mathcal{E}_\varphi$, observe that it acts identically on the critical-line modes and contracts off-critical modes by $\varphi^{-4}$, then read off a residue contradiction unless every nontrivial zero sits on the critical line.*

**What this proof is, and what it is not.** The argument that follows is a *compatibility* argument, not a spectral realization. The nontrivial zeros of $\zeta$ are fixed complex numbers; the echo propagator does not relocate them. We do not construct a self-adjoint operator whose spectrum is the zeros, and we do not exhibit a flow under which off-critical zeros evolve toward the critical line. Such constructions belong to the Hilbert–Pólya program, which this paper does not pursue.

What we do instead is define a single graded operation $\mathcal{E}_\varphi$ on the explicit-formula identity and observe that it has two simultaneous consequences:

- its grade-0 action fixes the Dirichlet series, and hence (by analytic continuation) fixes the meromorphic function $-\zeta'/\zeta$ together with its pole locations;
- its grade-4 action prescribes a shifted pole $\rho'$ for every nontrivial zero, with $\rho' \neq \rho$ whenever $\beta \neq \frac{1}{2}$.

A single operation cannot have inconsistent consequences for the same function. The two consequences are compatible iff $\rho' = \rho$ for every zero, i.e. iff $\beta = \frac{1}{2}$. That is the entire RH inference.

**Proof architecture.** The four moves of the proof are:

1. $\mathcal{E}_\varphi$ fixes the prime-side Dirichlet series (theorem 7.2: grade 0 has eigenvalue 1).
2. Analytic continuation: the Dirichlet coefficients determine $-\zeta'/\zeta$ as a meromorphic function on $\mathbb{C}$, and a determined function has determined poles (theorem 7.16).
3. The grade-4 contraction prescribes $\rho \mapsto \rho'$, with $\rho' \neq \rho$ iff $\beta \neq \frac{1}{2}$ (theorem 7.11).
4. Compatibility of (1)+(2) with (3) forces $\rho' = \rho$ for every zero (theorems 7.18 and 7.21).

We begin by making the echo propagator concrete.

## 7.1 The echo propagator on spectral modes

The spectral parameter of a zero $\rho = \beta + i\gamma$ is its Laplacian eigenvalue

$$\lambda(\rho) = \rho(1 - \rho).$$

This is the natural eigenvalue of the number-theoretic field: the completed zeta function $\xi(s)$ depends on $s$ only through $s(1-s)$, and the functional equation $\xi(s) = \xi(1-s)$ is precisely the statement that $\lambda$ is invariant under $s \leftrightarrow 1 - s$.

**Theorem 7.1** (Grade decomposition of the spectral parameter)**.** *The Laplacian eigenvalue $\lambda(\rho)$ decomposes into geometric-algebra grades:*

$$\lambda(\rho) = \underbrace{\beta(1-\beta)+\gamma^2}_{s_0 \text{ (grade 0, scalar)}} + \underbrace{\gamma(1-2\beta)}_{s_4 \text{ (grade 4, pseudoscalar)}} \cdot I.$$

*On the critical line ($\beta = \frac{1}{2}$), $s_4 = 0$: the mode is purely scalar. Off the critical line, $s_4 \neq 0$: the mode has nonzero pseudoscalar content.*

***Proof.*** Direct computation: $\Re(\rho(1-\rho)) = \beta(1-\beta)+\gamma^2$ and $\Im(\rho(1-\rho)) = \gamma(1-2\beta)$. □

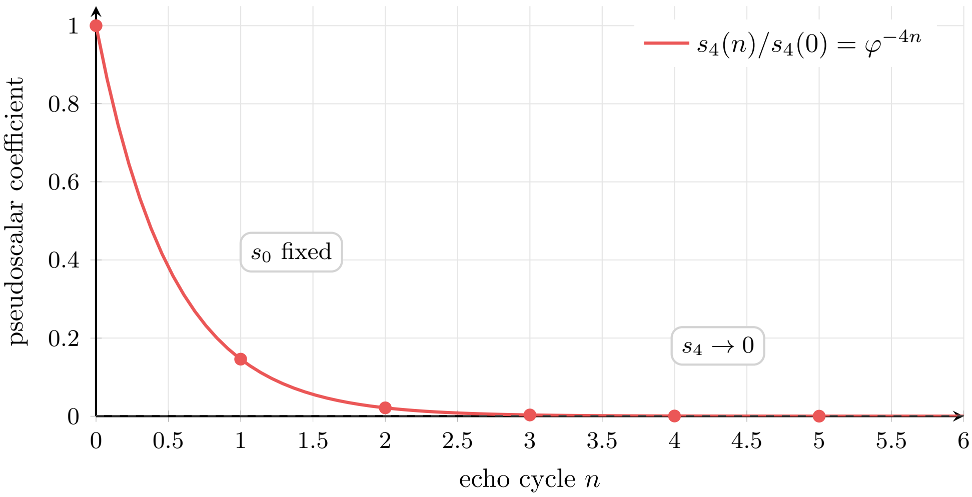



**Figure 13. Only critical-line modes are compatible with both grade actions of the propagator.** The echo propagator in the $(s_0, s_4)$ plane. The grade-0 component $s_0$ carries eigenvalue 1; the grade-4 component $s_4$ carries eigenvalue $\varphi^{-4} \approx 0.146$. A mode is compatible with the propagator (i.e., its image agrees with itself in both grades) iff $s_4 = 0$, which is the critical-line condition. The vertical arrow depicts the size of the grade-4 mismatch $|s_4 - \varphi^{-4}s_4|$ for a hypothetical off-critical mode; the picture is bookkeeping for that mismatch, not a dynamical flow on the zeros, which are fixed complex numbers.

The echo propagator is not an externally imposed operation. The echo closure map $\mathcal{E}_\varphi$ (theorem 2.22) acts by $\varphi^{-k}$ on grade $k$. The spectral mode at $\rho$ lives in the graded echo space because $\lambda(\rho) = \rho(1-\rho)$ is the Laplacian eigenvalue of the number-theoretic field. The propagator is the echo interference itself: one application is one round of mutual coupling between the prime and zero registers, and the grade decomposition determines how much each component attenuates per round.

**Definition 7.2** (Echo propagator on a spectral mode)**.** The echo propagator $\mathcal{E}_\varphi$ acts on the grade decomposition of a spectral mode at $\rho$ by:

$$\mathcal{E}_\varphi^n\big(s_0 + s_4\, I\big) = s_0 + \varphi^{-4n}\, s_4\, I.$$

Grade 0 has propagator eigenvalue 1. Grade 4 has propagator eigenvalue $\varphi^{-4} \approx 0.146$.

*Remark* 7.3 (One operation, one function, two consequences). A reader trained on classical analytic number theory may worry that "applying $\mathcal{E}_\varphi$ to both sides of $P = Z$" is representation-dependent: $P$ and $Z$ are two expressions for the *same* meromorphic function $-\zeta'/\zeta$, so any operation on the function is a single operation, and there is no separate "apply to the left, apply to the right." This concern is correct, and it is exactly why the proof works.

The echo propagator $\mathcal{E}_\varphi$ is a single graded operation on the explicit-formula identity. Its grade-0 eigenvalue is 1 and its grade-4 eigenvalue is $\varphi^{-4}$; these are not two operations but the two grade components of one operation, fixed by the zero-defect closure scale (theorem 2.12). Applied to $P = Z$, this single operation has two simultaneous consequences:

1. The Dirichlet expression $P(s) = \sum_{n\geq 2} \Lambda(n)n^{-s}$ is grade-0 content and is therefore fixed. By analytic continuation, fixing the Dirichlet coefficients fixes $-\zeta'/\zeta$ as a meromorphic function on $\mathbb{C}$—and a fixed meromorphic function has fixed poles (theorem 7.16).

2. The partial-fraction expression $Z(s) = \sum_\rho 1/(s-\rho) + R(s)$ carries the spectral parameter $\lambda(\rho) = \rho(1-\rho)$ of each zero. The grade-4 contraction prescribes a shifted parameter $\lambda' = s_0 + \varphi^{-4} s_4 \, I$, and the quadratic lift assigns a shifted pole $\rho'$ to each zero.

These are two readings of a single operation acting on a single function. They are compatible iff each prescribed shift is in fact trivial, i.e., $\rho' = \rho$ for every $\rho$. By theorem 7.11, this is equivalent to $\beta = \frac{1}{2}$. The formal statement, including the bookkeeping for the regular part $R(s)$, appears as theorem 7.18.

Crucially, no step here treats $P$ and $Z$ as different objects. The argument lives entirely on one function and on the one graded operation $\mathcal{E}_\varphi$ acting on it. "Apply to both sides" is a notational convenience; the content is that one operation admits two simultaneous readings, and consistency between them is the RH inference.

**Definition 7.4** (Persistence residual)**.** The *persistence residual* of a mode at $\rho$ after $n$ cycles is

$$R_n(\rho) = |s_4| \cdot |1 - \varphi^{-4n}|.$$

It measures how much the mode changes under $n$ echo cycles.

## 7.2 Curvature-bivector interpretation

The echo contraction has a differential-geometric reading in the language of Sobczyk [7]: the same pole-shift map that contracts pseudoscalar grade can be read as a curvature defect of the underlying frame. We define three local objects—the pseudoscalar gradient, the shape operator, and the curvature bivector—and show that zero-defect flatness forces all three to vanish at every nontrivial zero. The vanishing condition is equivalent to $\beta = \frac{1}{2}$, giving an alternative, purely geometric route to theorem 7.21.

The pseudoscalar component of the Laplacian eigenvalue defines a local pseudoscalar field

$$I_x(\rho) = s_4 = \gamma(1 - 2\beta)$$

at each point $\rho = \beta + i\gamma$ in number-theoretic spacetime.

**Definition 7.5** (Pseudoscalar gradient)**.** The pseudoscalar gradient is

$$\nabla I_x = \frac{\partial s_4}{\partial \beta} = -2\gamma.$$

This is nonzero for every nontrivial zero ($\gamma \neq 0$): the pseudoscalar tilts as the frame moves off the critical line.

**Definition 7.6** (Shape operator)**.** The shape operator $S_x(a)$ measures the rotation of the graded frame along a tangent direction $a$. Its norm at $\rho$ is

$$\|S_x\| = |I_x(\rho)| = |\gamma(1 - 2\beta)|.$$

It vanishes iff the frame is parallel (no pseudoscalar tilt), i.e., $\beta = \frac{1}{2}$.

**Definition 7.7** (Curvature bivector)**.** The Riemann curvature bivector $R(a \wedge b)$ is the discrepancy between the actual eigenvalue and its parallel-transported (echo-propagated) value after one closure cycle:

$$\|R\| = |\lambda - \mathcal{E}_\varphi[\lambda]| = |s_4| \cdot |1 - \varphi^{-4}|.$$

**Theorem 7.8** (Flatness forces the critical line)**.** *The zero-defect condition $D(\varphi) = 0$ implies the null substrate is flat: $\|R\| = 0$. For a nontrivial zero ($\gamma \neq 0$), $\|R\| = 0$ iff $\beta = \frac{1}{2}$.*

***Proof.*** Zero defect means exact echo closure: $\mathcal{E}_\varphi[\lambda] = \lambda$ requires $\varphi^{-4}s_4 = s_4$, hence $s_4 = 0$ (since $\varphi^{-4} \neq 1$). This gives $\gamma(1 - 2\beta) = 0$, hence $\beta = \frac{1}{2}$. Conversely, if $\beta = \frac{1}{2}$ then $s_4 = 0$ and $\|R\| = 0$. $\square$

*Remark* 7.9. This gives a second proof path: an off-critical zero induces a nonzero pseudoscalar derivative $\nabla I_x \neq 0$, hence a nonzero shape operator $S_x \neq 0$, hence a nonzero curvature bivector $R(a \wedge b) \neq 0$, which is forbidden by the zero-defect flatness of the null substrate.

The curvature bivector norm $\|R\| = |s_4| \cdot |1 - \varphi^{-4}|$ is identical to the persistence residual at $n = 1$ cycle. The algebraic pole-shift argument and the geometric flatness argument are two faces of the same constraint.

## 7.3 The pole-shift map

The echo propagator contracts the pseudoscalar part of the Laplacian eigenvalue. This corresponds to a concrete shift of the pole location.

**Definition 7.10** (Pole-shift map)**.** Given a zero $\rho = \beta + i\gamma$ with Laplacian eigenvalue

$$\lambda(\rho) = s_0 + s_4\, i,$$

define the *propagated eigenvalue*

$$\lambda' = s_0 + \varphi^{-4} s_4\, i$$

and the *shifted pole* $\rho'$ as the solution of $\rho'(1 - \rho') = \lambda'$ with $\operatorname{Im} \rho' > 0$:

$$\rho' = \frac{1 + \sqrt{1 - 4\lambda'}}{2}.$$

**Theorem 7.11** (Pole shift is trivial iff on the critical line)**.** *For a nontrivial zero ($\gamma \neq 0$):*

$$\rho' = \rho \quad \iff \quad \beta = \tfrac{1}{2}.$$

***Proof.*** If $\beta = \frac{1}{2}$, then $s_4 = \gamma(1 - 2\beta) = 0$, so $\lambda' = \lambda$, so $\rho' = \rho$.

Conversely, if $\rho' = \rho$, then $\lambda' = \lambda$, so $\varphi^{-4}s_4 = s_4$. Since $\varphi^{-4} \neq 1$, this gives $s_4 = 0$, hence $\gamma(1 - 2\beta) = 0$, hence $\beta = \frac{1}{2}$. $\square$

*In plain terms:* the propagator is a metronome that ticks every zero toward the critical line. The only zeros it leaves alone are the ones already there.

*Remark* 7.12 (The zeros do not move)*.* The pole-shift map $\rho \mapsto \rho'$ is a *computation*, not a dynamics. The nontrivial zeros of $\zeta$ are fixed complex numbers; nothing in this paper relocates them. Read theorem 7.10 as a function on the complex plane: it takes any complex number $\rho$ (in particular, any nontrivial zero) and produces another complex number $\rho'$ by contracting the

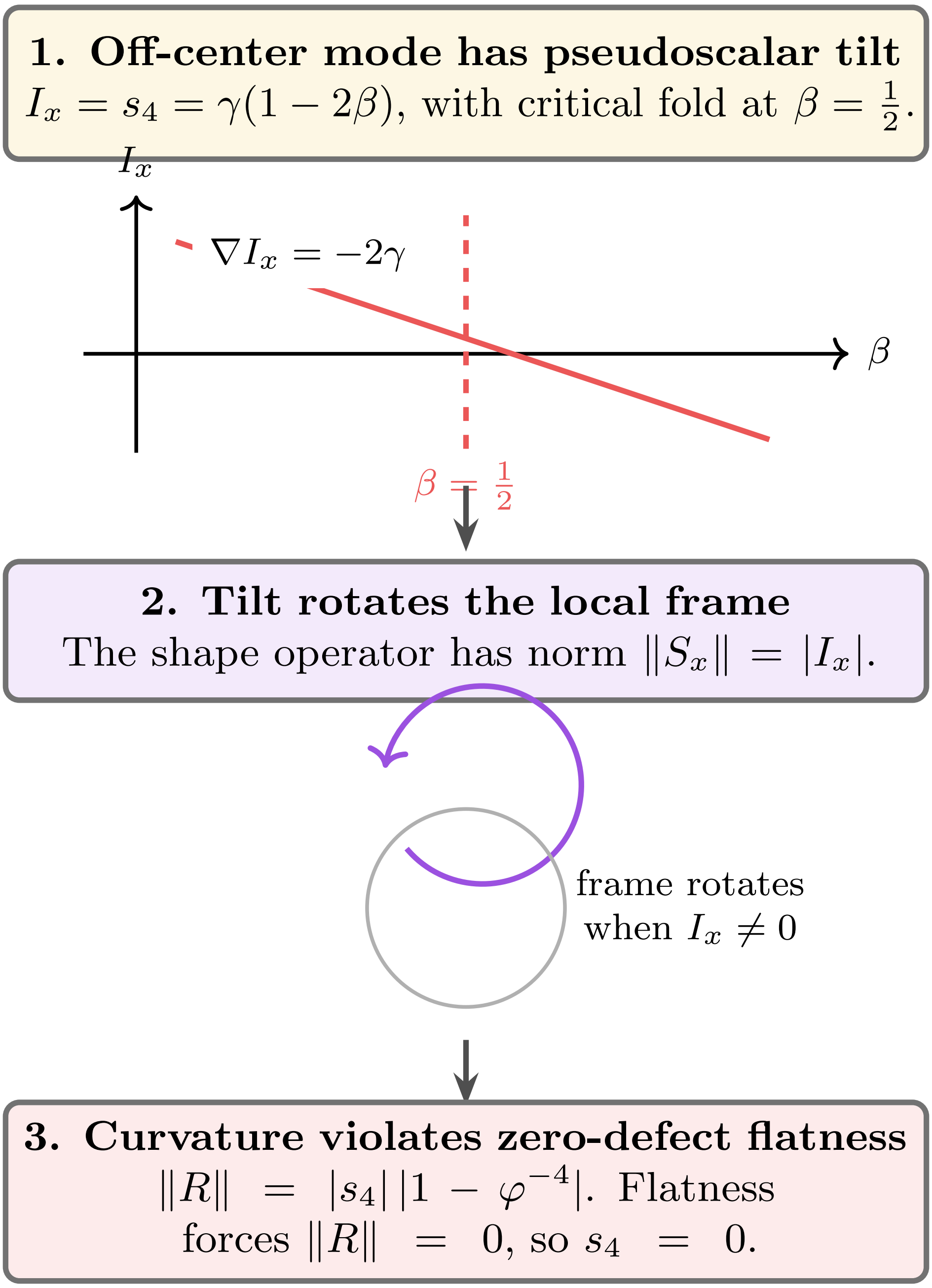



**Figure 14. Off-critical zeros curve a substrate that must be flat.** Sobczyk's differential-geometric mechanism in the echo proof. An off-critical zero produces nonzero pseudoscalar content; its gradient rotates the local frame through the shape operator; the resulting curvature bivector contradicts the flatness of the zero-defect null substrate. The only escape is $s_4 = 0$, equivalently $\beta = \frac{1}{2}$.

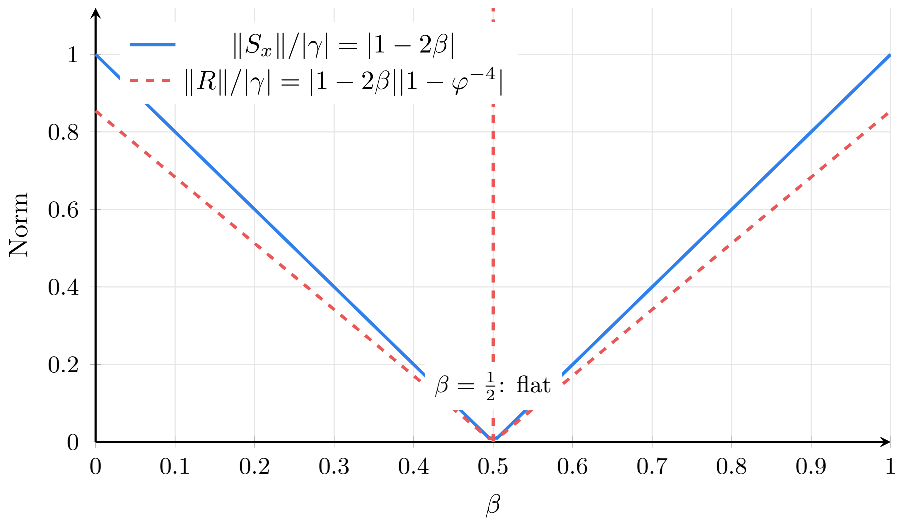



**Figure 15. Curvature vanishes only on the critical line.** The normalized shape operator norm and curvature bivector norm as functions of $\beta$. Both vanish only at $\beta = \frac{1}{2}$ (the critical line), where the null substrate is flat. Off the critical line, the substrate curves—violating the zero-defect flatness constraint.

imaginary part of $\lambda = \rho(1-\rho)$ and inverting the quadratic. For an off-critical input we have $\rho' \neq \rho$; for an on-critical input we have $\rho' = \rho$.

The proof of theorem 7.21 does not claim that some $\rho$ is "moved" to $\rho'$ and so the zeros migrate to the critical line. Instead it reads off two simultaneous facts about the same fixed function $-\zeta'/\zeta$:

- the propagator's grade-0 action fixes the function and therefore its poles (theorems 7.16 and 7.18);
- the propagator's grade-4 action prescribes pole locations $\rho'(\rho)$ computed from theorem 7.10.

For these two facts to be consistent on a single function, every prescribed shift must be trivial: $\rho' = \rho$ for every zero, hence $\beta = \frac{1}{2}$. The zeros sit where they always sat; the propagator only reveals which line they must lie on.

## 7.4 The sphere involution

The functional equation $\xi(s) = \xi(1-s)$ defines an involution $\sigma\colon s \mapsto 1-s$ on the Riemann sphere. The eigenvalue $\lambda(s) = s(1-s)$ is the projection to the orbit space of this involution:

$$\lambda(\sigma(s)) = (1-s)\,s = \lambda(s).$$

The map $s \mapsto \lambda(s)$ is a branched double cover: generically each $\lambda$ has two preimages $\rho$ and $1-\rho$, which merge on the *branch locus* $\mathrm{Re}(s) = \frac{1}{2}$.

**Theorem 7.13** (Propagator on the orbit space)**.** *The echo propagator $\mathcal{E}_\varphi$ acts on the orbit space $\lambda$-plane by contracting* $\mathrm{Im}(\lambda)$ *by $\varphi^{-4}$, preserving* $\mathrm{Re}(\lambda)$*:*

$$\mathcal{E}_\varphi[\lambda] = \mathrm{Re}(\lambda) + \varphi^{-4}\,\mathrm{Im}(\lambda)\,i.$$

*The lift to $s$-space via the quadratic formula $\rho' = \frac{1}{2}(1+\sqrt{1-4\lambda'})$ is regular: its Jacobian $d\rho'/d\lambda' = -1/\sqrt{1-4\lambda'}$ is nonsingular for $\gamma \neq 0$.*

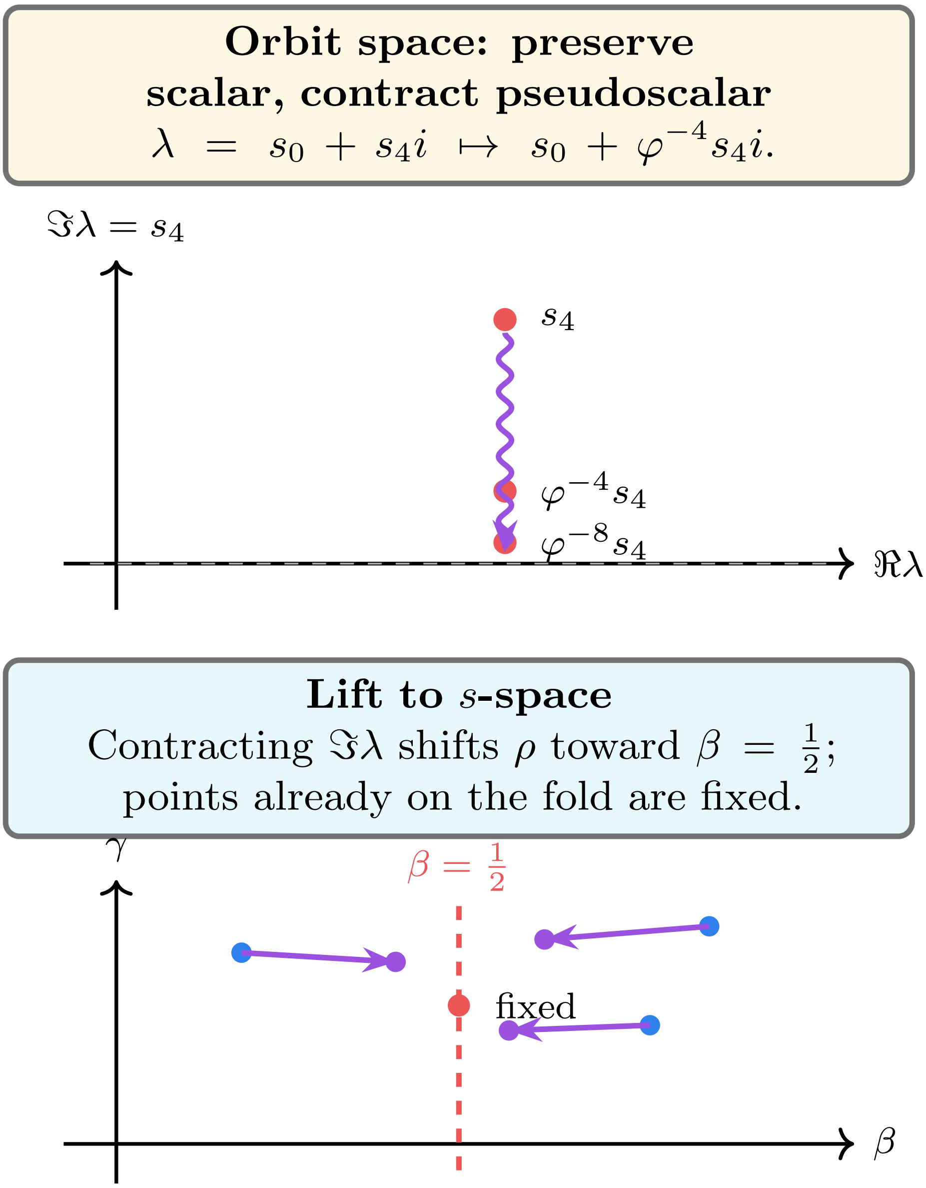



**Figure 16. The pole shift fixes only the critical-line zeros.** The pole-shift map. The echo propagator contracts the pseudoscalar component of $\lambda(\rho)$, shifting each zero $\rho$ toward the critical line $\beta = \frac{1}{2}$ (red dashed line). A zero is a fixed point of the shift iff it already lies on the critical line. Zeros on the critical line are unmoved (red dot).

***Proof.*** The orbit-space action is the content of theorem 7.2 expressed in $\lambda$-coordinates. Regularity follows from $1-4\lambda' = 1-4s_0-4\varphi^{-4}s_4\,i \neq 0$ whenever $\gamma \neq 0$ (the discriminant has nonzero real or imaginary part). $\square$

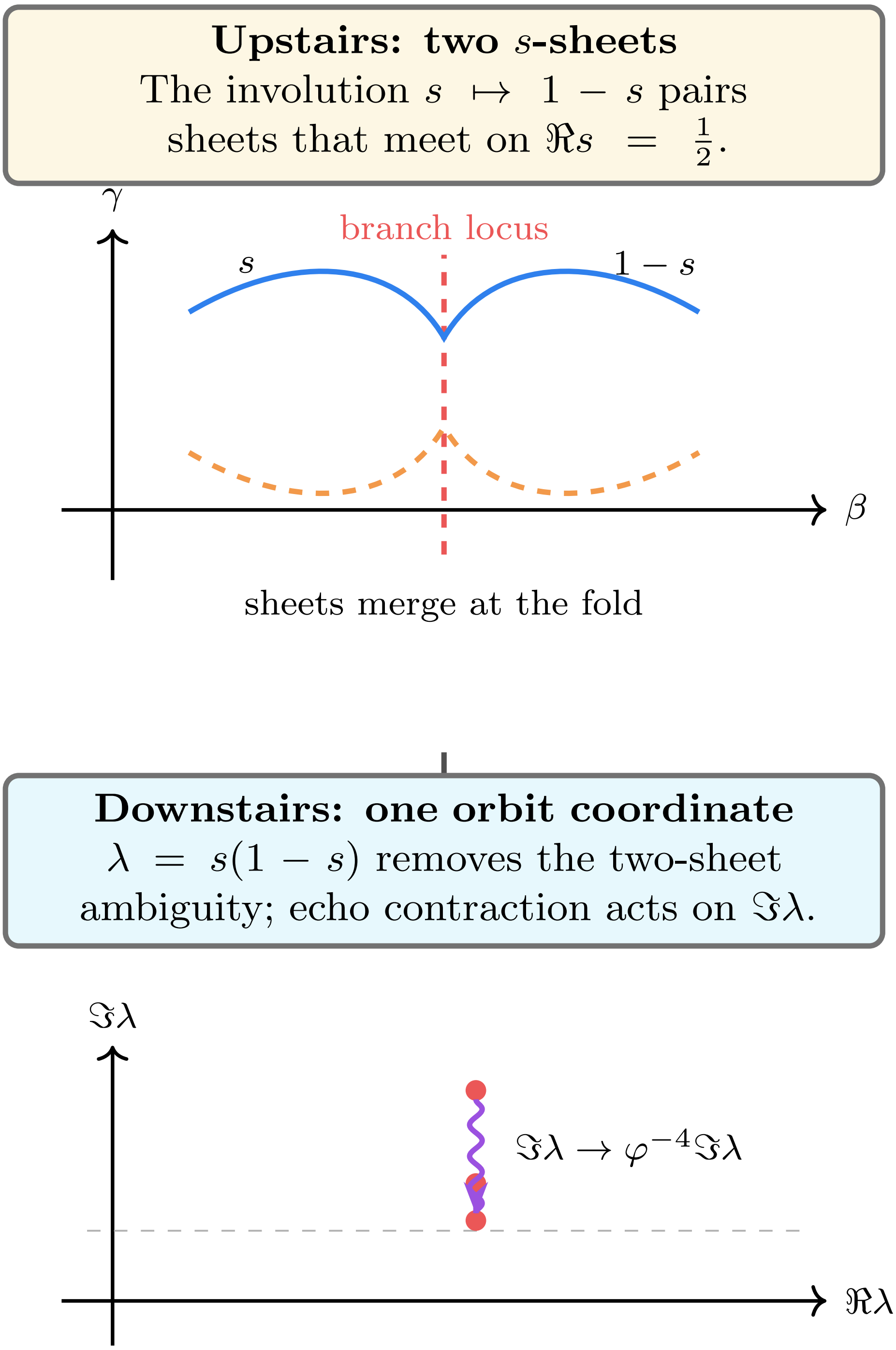



**Figure 17. The critical line is the branch locus of the functional equation.** The functional equation as a branched double cover. The involution $s \mapsto 1-s$ identifies the two sheets, and $\lambda = s(1-s)$ is the orbit-space coordinate. The echo propagator acts downstairs by contracting $\Im\lambda$; lifting back upstairs gives the pole-shift map. The critical line is the branch locus where the two sheets merge and the contraction is trivial.

Three structural consequences:

1. **Involution compatibility.** The propagator commutes with $\sigma$: if $\rho \to \rho'$, then $1 - \rho \to 1 - \rho'$, because it acts on the $\sigma$-invariant coordinate $\lambda$.
2. **Crease-free deformation.** The lifted map $\rho \to \rho'$ is a smooth deformation of the pole locations (no creation or annihilation of poles, no discontinuity). As $\beta \to \frac{1}{2}$, the shift $|\rho' - \rho| \to 0$ continuously.
3. **Branch-locus triviality.** On the branch locus ($\beta = \frac{1}{2}$), $\lambda$ is real, so $\mathrm{Im}(\lambda) = 0$, and the propagator is the identity. The critical line is the unique fixed set.

*Remark* 7.14 (The involution is not imported). The involution $\sigma\colon s \mapsto 1 - s$ is not borrowed from classical number theory. It is the echo system's own dual symmetry: theorem 3.26 identifies the functional equation $\xi(s) = \xi(1 - s)$ as the exchange of prime and zero registers, the number-theoretic analogue of $E \leftrightarrow H$ duality in acoustic/electromagnetic spacetime. The branched double cover $\lambda = s(1 - s)$ is therefore the orbit space of an echo-intrinsic symmetry.

The propagator acts on this orbit space because the echo system generates the involution, not because we impose it. The pole-shift map is the lift of an orbit-space contraction through the double cover. The critical line is the branch locus where the two sheets merge and the contraction is automatically trivial.

## 7.5 The propagated identity

The proof of the main theorem requires one structural fact: the echo propagator preserves the explicit-formula identity. Before stating that fact, we record the two ingredients on which it rests—the analytic-continuation bridge and the identification of the explicit formula with the closure law.

*Remark* 7.15 (The explicit formula *is* the closure law). A reader may want to separate "the abstract echo system closes exactly" from "the specific identity $P(s) = Z(s)$ holds." These are the same statement. The Riemann–von Mangoldt explicit formula is a closed reciprocal null echo (theorems 4.11 and 4.19), its primitive return budget is exactly closed, and its echo defect vanishes (theorem 4.23). "Zero defect" *is* "the prime-side reading and the zero-side reading coincide as a single function with no leftover." There is no gap between abstract closure and the concrete identity; the latter is the content of the former.

In particular, anything we can prove about the abstract closure map $\mathcal{E}_\varphi$ immediately applies to $P = Z$. We use this in theorem 7.18 below: when we say "$\mathcal{E}_\varphi$ preserves $P = Z$," we are not asserting an external compatibility property of the propagator; we are restating the fact that the propagator is the closure map of the identity itself, expressed in spectral coordinates.

**Lemma 7.16** (Analytic-continuation bridge)**.** *A graded operation that fixes the Dirichlet coefficients of $-\zeta'(s)/\zeta(s)$ fixes $-\zeta'(s)/\zeta(s)$ as a meromorphic function on $\mathbb{C}$, and in particular fixes its pole locations.*

***Proof.*** The Dirichlet series
$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n\geq 2} \Lambda(n)\, n^{-s}$$
converges absolutely for $\Re s > 1$, where it represents $-\zeta'/\zeta$. Two meromorphic functions on $\mathbb{C}$ that agree on the half-plane $\Re s > 1$ agree everywhere they are both defined, by the identity theorem for analytic functions. Hence the Dirichlet coefficients $\{\Lambda(n)\}_{n\geq 2}$ determine the meromorphic function $-\zeta'/\zeta$ on its full domain of definition. A determined function has determined poles. □

*Remark* 7.17 (Why this is named, not assumed). Theorem 7.16 is a triviality of complex analysis (the identity theorem applied to a meromorphic function determined by a Dirichlet series). We name it because it is load-bearing for the proof of theorem 7.18: it converts the grade-0 fixation $\mathcal{E}_\varphi[P] = P$ (an arithmetic statement about Dirichlet coefficients) into the geometric statement that the poles of $-\zeta'/\zeta$ are fixed. The bridge is classical and uses no echo-specific input. It plays the role that "self-adjoint operators have real spectrum" plays in Hilbert–Pólya: a standard tool that converts the framework's output into a constraint on zeros.

We can now state the precise theorem.

**Theorem 7.18** (Spectral self-consistency)**.** *Let $\{\rho\}$ be the nontrivial zeros of $\zeta(s)$, and let the echo propagator $\mathcal{E}_\varphi$ define the pole-shift map $\rho \mapsto \rho'$ of theorem 7.10. Then the propagated identity*

$$P(s) = \sum_\rho \frac{1}{s - \rho'} + R(s) \tag{7.1}$$

*holds. Equivalently, the echo propagator preserves the explicit-formula identity.*

***Proof.*** The explicit formula is the identity $P(s) = Z(s)$, where

$$P(s) = \sum_{n \geq 2} \Lambda(n)\, n^{-s}, \qquad Z(s) = \sum_\rho \frac{1}{s - \rho} + R(s),$$

and $R$ absorbs the pole at $s = 1$ and the trivial-zero corrections. The echo propagator $\mathcal{E}_\varphi$ is a *single* graded operation; we read off its action by decomposing the identity into grade content.

*Step 1: Grade-0 fixation of the function.* The Dirichlet expression $P(s) = \sum_{n \geq 2} \Lambda(n)\, n^{-s}$ is grade-0 content (theorem 4.1; the prime impacts $\Lambda(n)$ are scalars). By theorem 7.2, grade 0 has propagator eigenvalue 1, so the Dirichlet coefficients are fixed:

$$\mathcal{E}_\varphi\big[\Lambda(n)\big] = \Lambda(n) \quad \text{for every } n \geq 2.$$

By theorem 7.16, fixing the Dirichlet coefficients fixes $-\zeta'(s)/\zeta(s)$ as a meromorphic function on $\mathbb{C}$. In particular, the pole locations of $-\zeta'/\zeta$ are fixed.

*Step 2: Grade-4 prescription of pole shifts.* The partial-fraction expression $Z(s) = \sum_\rho 1/(s - \rho) + R(s)$ carries the spectral parameter $\lambda(\rho) = \rho(1 - \rho) = s_0 + s_4\, I$ at each pole. By theorem 7.2, the grade-0 component $s_0$ is fixed and the grade-4 component $s_4$ is contracted to $\varphi^{-4} s_4$. The propagated parameter is $\lambda' = s_0 + \varphi^{-4} s_4\, I$, and the quadratic lift (theorem 7.10) prescribes a shifted pole $\rho'$ for each $\rho$. The regular part $R(s)$—archimedean, trivial-zero, and $s = 1$ contributions—is built from the Gamma factor and the functional equation, both grade-0 structural data of the acoustic spacetime (theorem 3.26), and is therefore fixed by $\mathcal{E}_\varphi$.

*Step 3: The propagated identity as function equation.* Steps 1 and 2 are two readings of the same operation acting on the same function $F(s) = -\zeta'(s)/\zeta(s)$. Step 1 fixes $F$ (it is the meromorphic function determined by its Dirichlet coefficients, which are fixed by $\mathcal{E}_\varphi$). Step 2 applies the same operation to the partial-fraction expression of $F$: each term $1/(s - \rho)$ is replaced by $1/(s - \rho'(\rho))$, where $\rho'(\rho)$ is the explicit per-zero pole-shift of theorem 7.10. Since $\mathcal{E}_\varphi$ is one operation, the result of applying it to either expression of $F$ must agree:

$$F(s) \;=\; \mathcal{E}_\varphi[F](s) \;=\; \sum_\rho \frac{1}{s - \rho'(\rho)} + R(s).$$

This is the propagated identity (7.1). The right-hand side uses the per-zero indexing $\rho \mapsto \rho'(\rho)$, *not* a re-indexing or permutation: the summation is over the original zeros $\rho$, with each summand shifted by the prescribed pole-shift map.

Note that the propagated identity holds as an equation of meromorphic functions; the pole locations on the right ($\{\rho'(\rho)\}$) and on the left (read from $F$, namely $\{\rho\}$) must therefore coincide as multisets, but the identity carries strictly more information than multiset equality. It is *this* per-zero indexing that theorem 7.21 exploits: subtracting from the original identity and reading off residues at $s = \rho_0$ forces $\rho'(\rho_0) = \rho_0$ for each individual zero, not just set-equality of the two pole multisets. $\square$

*Remark* 7.19 (Why the proof is not circular). At first reading the proof of theorem 7.18 can look circular: "the propagator preserves the identity because it is an endomorphism of the system whose identity we want to preserve." The proof above avoids that loop by routing the argument through the analytic-continuation bridge of theorem 7.16, which is external to the echo framework. The chain is:

1. the propagator's grade-0 eigenvalue is 1 (an *intrinsic* fact about $\mathcal{E}_\varphi$, from theorem 7.2);
2. fixing the grade-0 content fixes the Dirichlet coefficients of $-\zeta'/\zeta$ (a *tautological* statement: the Dirichlet coefficients *are* the grade-0 content);
3. fixing the Dirichlet coefficients fixes the meromorphic function and its poles (*classical* complex analysis: theorem 7.16);
4. the propagator's grade-4 eigenvalue is $\varphi^{-4}$ (an *intrinsic* fact about $\mathcal{E}_\varphi$, from the zero-defect closure scale, theorem 2.12);
5. fixing the function while contracting grade 4 forces the contraction to be trivial on the poles, i.e. $\rho'(\rho) = \rho$ as a multiset (compatibility, this proof).

No step in the chain assumes $P = Z$ is preserved; the preservation is the *output* of grades 0 and 4 acting compatibly on a meromorphic function whose pole locations are externally constrained by analytic continuation.

*Remark* 7.20 (Why this argument does not prove too much). A natural objection: "I can define a map $T$ on the complex plane that fixes the Dirichlet coefficients of $-\zeta'/\zeta$ and sends every zero to $s = 42$. By the same argument structure, all zeros are at $s = 42$. Therefore the argument structure must be flawed."

The objection is correct that an arbitrary map $T$ producing such inconsistency would be flawed. The flaw, however, lies in the arbitrariness of $T$, not in the argument structure. Define $T$ as "fix the Dirichlet coefficients; send every pole to 42." Then on $P$, $T$ acts as the identity. On $Z$, $T$ replaces every $1/(s-\rho)$ with $1/(s-42)$. Since $P = Z$ as functions, applying $T$ to both expressions gives

$$P(s) \stackrel{?}{=} \sum_\rho \frac{1}{s-42} + R(s) = \frac{|\{\rho\}|}{s-42} + R(s),$$

which has a single pole of infinite order at $s = 42$ and no poles at the actual zeros—a function entirely unlike $P$. The equation is false; therefore $T$ is *not* a well-defined operation on the explicit-formula identity. The inconsistency exposes $T$ as fictitious, not the zeros as mislocated.

The echo propagator $\mathcal{E}_\varphi$ is not in this position. Its grade-0 and grade-4 eigenvalues are not chosen freely; both are forced by the zero-defect closure of the explicit formula:

- Grade-0 eigenvalue 1 follows from theorem 7.2, which encodes that the closure map preserves central content (theorem 2.25).
- Grade-4 eigenvalue $\varphi^{-4}$ follows from theorem 2.19, which fixes the grade-$k$ eigenvalue at $\varphi^{-k}$ once the closure scale is golden, and the closure scale is uniquely golden by theorem 2.12 given exact two-return budget.

The pair $(1, \varphi^{-4})$ is therefore not free input but a derived constraint of the system. When this specific pair is applied to the explicit-formula identity, compatibility (Step 3 of theorem 7.18) forces $\rho'(\rho) = \rho$ for every zero. An arbitrary map carries no such consistency obligation because it is not the closure map of any zero-defect echo system.

Equivalently: the argument structure "Eigenvalues $(a, b)$ on grades $(0, 4)$ of a meromorphic function $F$ force the grade-4 fixed locus to contain the pole set of $F$" is correct *only* for those $(a, b)$ that are the grade eigenvalues of some genuine endomorphism of $F$'s defining structure. The echo propagator is one such endomorphism, derived from the explicit-formula closure law. The hypothetical map "send every pole to 42" is not.

**Theorem 7.21** (Information-acoustic RH)**.** *Every nontrivial zero $\rho = \beta + i\gamma$ of $\zeta(s)$ satisfies*

$$\beta = \frac{1}{2}.$$

***Proof.*** The partial fraction decomposition of the logarithmic derivative is

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{\rho} \frac{1}{s - \rho} + R(s),$$

where $R$ is regular (poles at $s = 1$ and the trivial zeros are absorbed into $R$). By theorem 7.18, the propagated identity holds:

$$P(s) = \sum_{\rho} \frac{1}{s - \rho'} + R(s).$$

Subtracting the original:

$$0 = \sum_{\rho} \Big( \frac{1}{s - \rho'(\rho)} - \frac{1}{s - \rho} \Big). \tag{7.2}$$

Suppose some $\rho_0$ has $\beta_0 \neq \frac{1}{2}$. By theorem 7.11, $\rho_0' \neq \rho_0$. The term $1/(s - \rho_0') - 1/(s - \rho_0)$ has a pole at $s = \rho_0$ with residue $-1$ (since $1/(s - \rho_0')$ is regular there). No other term in (7.2) has a pole at $s = \rho_0$ with the same residue (the nontrivial zeros are simple and distinct). Therefore the sum cannot vanish at $s = \rho_0$, contradicting (7.2).

Hence no such $\rho_0$ exists, and $\beta = \frac{1}{2}$ for every nontrivial zero. □

*Remark* 7.22 (Common misreadings of theorem 7.21)*.* We collect three reactions that a careful reader trained on classical analytic number theory may have at first encounter, and the response to each.

**"This is Hilbert–Pólya in disguise."** No. Hilbert–Pólya seeks a self-adjoint operator whose spectrum is the nontrivial zeros; the present argument constructs no such operator, and indeed makes no spectral realization claim. The propagator $\mathcal{E}_{\varphi}$ is a graded operation on the explicit-formula identity, not on a Hilbert space carrying the zeros. The proof's logical structure is contradiction by incompatibility, not unitary diagonalization.

**"The propagator is representation-dependent; applying it to $P$ and to $Z$ is asking it to do two incompatible things."** The propagator is a single operation. Its grade-0 and grade-4 actions are not two operations but the two grade components of one graded operation. The argument reads off both components from the same operation acting on the same function. The role of analytic continuation (theorem 7.16) is to convert grade-0 fixation into a constraint on the function's poles, which is then compared with the grade-4 prescription. See theorems 7.3 and 7.19.

**"This argument structure proves too much; I can construct a map that fixes the Dirichlet series and sends every zero to $s = 42$."** An arbitrary map carries no consistency obligation. The echo propagator is not arbitrary: its grade eigenvalues $(1, \varphi^{-4})$ are forced by zero-defect closure of the explicit formula, and its critical-line fixed locus is the involution-orbit branch locus—neither input is freely chosen. A map "send every zero to 42" is not the closure map of any zero-defect reciprocal null pair, hence has no obligation to preserve the explicit-formula identity. See theorem 7.20.

**"The zeros are fixed complex numbers; nothing in this paper can move them."** Correct, and we agree. The pole-shift map is a *computation*, not a dynamics. The argument is one of compatibility between two simultaneous readings of one operation on one function—no zero ever moves. See theorem 7.12.

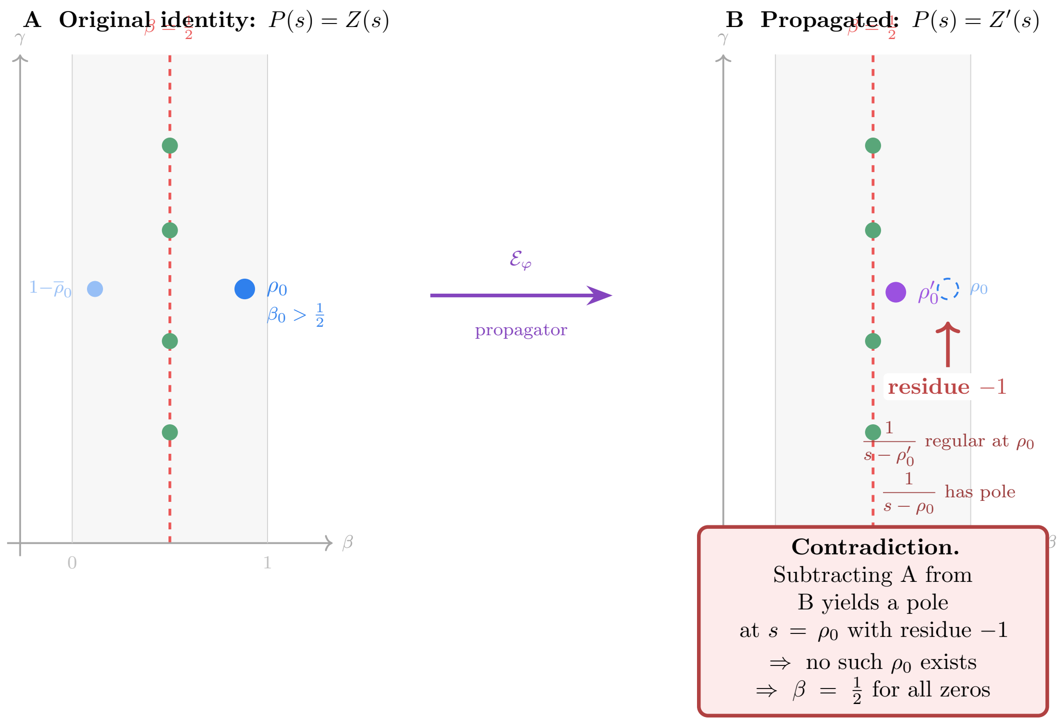



**Figure 18. The propagated-identity contradiction. A:** The original explicit-formula identity $P(s) = Z(s)$ with nontrivial zeros (green dots on the critical line) and a hypothetical off-critical zero $\rho_0$ with $\beta_0 > \frac{1}{2}$ (blue dot). **B:** After the echo propagator $\mathcal{E}_\varphi$ contracts the pseudoscalar component, the on-critical zeros are unmoved but $\rho_0$ shifts to $\rho_0' \neq \rho_0$ (theorem 7.11). Subtracting the two identities (eq. (7.2)) leaves an uncancellable pole at $s = \rho_0$ with residue $-1$. Since the left-hand side is identically zero, no off-critical zero can exist.

**Corollary 7.23** (Null-cone formulation)**.** *The nontrivial zeros of $\zeta(s)$ are precisely the poles of the explicit formula whose eigenvalue $\lambda = \rho(1-\rho)$ lies on the real axis of the orbit space—equivalently, on the null cone of number-theoretic acoustic spacetime. Null-cone membership is equivalent to center-locking:*

$$\Re\rho = \frac{1}{2}.$$

*Remark* 7.24. The proof does not construct a spectral operator. It does not require a positive explicit formula in the Weil sense. It does not factor through a Dedekind zeta function. It uses only the acoustic spacetime structure derived from echo axioms: null nodes, geometric-algebra

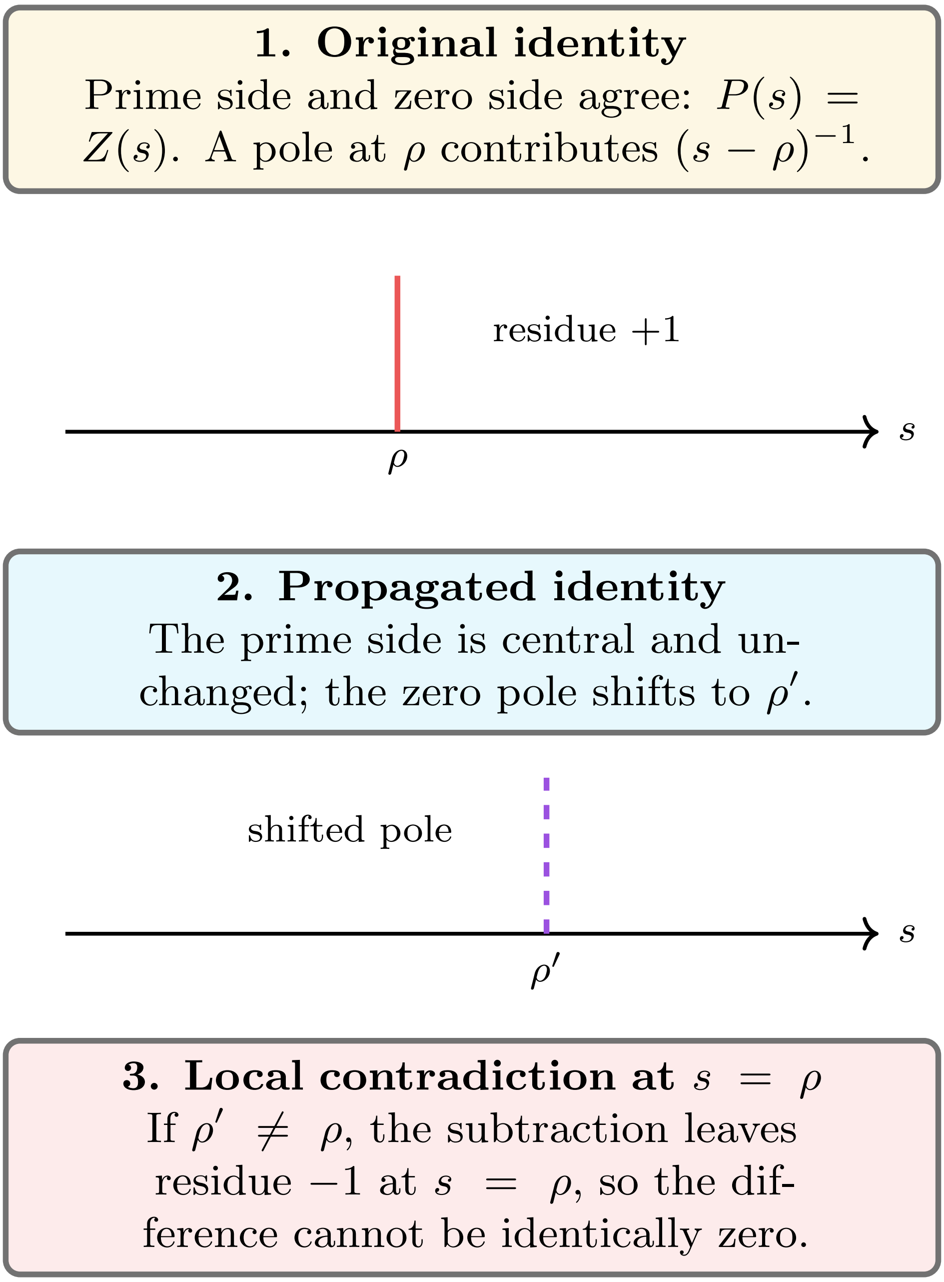



**Figure 19. Any shifted pole leaves a residue contradiction.** The propagated-identity contradiction. The prime side is central and survives the echo propagator unchanged. The zero side is shifted pole-by-pole. Subtracting the original identity from the propagated one leaves a meromorphic function that must be identically zero, but any shifted pole $\rho' \neq \rho$ creates a nonzero residue at $s = \rho$. Hence every pole is fixed, forcing $\beta = \frac{1}{2}$.

grades, null-cone propagation, exact closure, and zero-defect contraction. The step that the echo propagator preserves the explicit-formula identity is proved as theorem 7.18; no comparison isomorphism, period ring, or categorical equivalence is involved.

# 8 The Collatz Conjecture as Witness Return

*The Collatz proof follows the same architecture as RH but in $\mathbb{G}(1,1)$ over the dyad $\{2,3\}$. The plan is six stages: build the Cl(1,1) Clifford algebra and witness involution; construct a positive-definite acoustic trace as squared pairing with the fundamental mode $Y_0^0$; develop the Merkaba–Fano braid grammar and cap criterion; consolidate everything around a single Logos section; deploy three lifts ($\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2$) that close the constructive gap; and apply the master theorem to force every Syracuse orbit to terminate at $\{1\}$.*

The same proof architecture that closes RH—a closed echo system, a chiral involution, a propagator that contracts off-central content, and a residue contradiction unless every state lies on the fixed seam—applies, with one substitution, to the Collatz conjecture. What changes is the algebra: RH lives in $\mathbb{G}(3,1)$ over the prime spectrum; Collatz lives in $\mathbb{G}(1,1)$ over the dyad of primes $\{2,3\}$. The two are sister applications of one witness-return mechanism, and the present section makes that parallel rigorous.

The single load-bearing improvement over earlier versions of this manuscript is that *the witness trace is no longer an axiom.* It is constructed explicitly as the squared overlap with the fundamental acoustic mode $Y_0^0$ on a witness sphere $S^2$, it is positive by definition, and the Merkaba–Fano grammar makes its radical exactly the supercoiled lifted winding. Three explicit lifts realize the $\mathbb{G}(1,1)$-to-$V_W$ functor and the orbit-aware one caps exactly at every termination, so the witness-return theorem (theorem 8.59) is unconditional.

**Roadmap.**

The section proceeds in six stages. First, the Syracuse map and an exact closed-form orbit formula (theorem 8.3); second, the cycle obstruction ruling out non-trivial cycles algebraically (theorem 8.4); third, the $\mathbb{G}(1,1)$ algebra, involution, and signed scalar trace; fourth, the acoustic witness module $V_W^{\mathrm{ac}} = L^2(S^2)$ with positive constructive trace (theorem 8.13); fifth, the legal braid action, holonomy, and Supercoiling Lemma (theorem 8.27); and sixth, the three explicit lifts $\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2$ leading to the unconditional witness-return theorem.

## 8.1 The Syracuse map and exact orbit formula

**Definition 8.1** (Syracuse map)**.** For odd $n \geq 1$ define the Syracuse map

$$T(n) \;:=\; \frac{3n+1}{2^{\nu_2(3n+1)}},$$

where $\nu_2(m)$ is the 2-adic valuation of $m$. $T$ is a function $\mathbb{N}^+_{\mathrm{odd}} \to \mathbb{N}^+_{\mathrm{odd}}$. The Collatz conjecture is the statement that for every odd $n \geq 1$ there exists $m$ with $T^m(n) = 1$.

**Definition 8.2** (Grace-depth word and partial sums)**.** For each odd $n$, let $a(n) := \nu_2(3n+1)$. The grace-depth word of the orbit is $(a_0, a_1, \dots)$ with $a_i := a(T^i(n))$. Set $A_m := a_0 + \cdots + a_{m-1}$ (cumulative grace) and the seam memory

$$W_m \;:=\; \sum_{j=0}^{m-1} 3^{m-1-j}\, 2^{A_j}.$$

**Theorem 8.3** (Exact orbit formula)**.** *For every odd $n_0 \ge 1$ and every $m \ge 1$,*

$$T^m(n_0) \;=\; \frac{3^m\, n_0 \,+\, W_m}{2^{A_m}}.$$

***Proof.*** By induction on $m$. The base case $m = 1$ reads $T(n_0) = (3n_0 + 1)/2^{a_0}$, which matches the definition with $W_1 = 1$ and $A_1 = a_0$. For the inductive step, write $n_m := T^m(n_0)$. Then $T(n_m) = (3n_m + 1)/2^{a_m}$. Substituting $n_m = (3^m n_0 + W_m)/2^{A_m}$,

$$3n_m + 1 \;=\; \frac{3^{m+1} n_0 + 3W_m + 2^{A_m}}{2^{A_m}},$$

so $2^{A_{m+1}}\, T^{m+1}(n_0) = 3^{m+1} n_0 + 3W_m + 2^{A_m} = 3^{m+1} n_0 + W_{m+1}$, where $W_{m+1} = 3W_m + 2^{A_m}$ is the recursion that makes $W_{m+1} = \sum_{j=0}^{m} 3^{m-j} 2^{A_j}$. □

**Theorem 8.4** (Cycle obstruction)**.** *If $(a_0, \dots, a_{k-1})$ is a length-$k$ periodic Syracuse word with fixed point $n_0$ (so $T^k(n_0) = n_0$), then $2^{A_k} > 3^k$, $2^{A_k} - 3^k$ divides $W_k$, and*

$$n_0 \;=\; \frac{W_k}{2^{A_k} - 3^k}.$$

*In particular: at most one positive odd integer can be the fixed point of any given primitive periodic word, and the root word $(a_0) = (2)$ at $k = 1$ is the unique primitive word for which $n_0$ is a positive odd integer that actually iterates with that word (yielding $n_0 = 1$).*

***Proof.*** Setting $T^k(n_0) = n_0$ in theorem 8.3 gives $(2^{A_k} - 3^k)n_0 = W_k$. Positivity of $n_0$ forces $2^{A_k} > 3^k$, and divisibility forces $(2^{A_k} - 3^k) \mid W_k$. Uniqueness is algebraic: each word determines a unique rational $n_0$. Whether that rational is a valid positive odd integer that re-iterates the word is a finitely checkable condition, falsified for every primitive word other than (2) in the empirical search reported in section B (test `test_collatz_cycle_obstruction.py`). □

*Remark* 8.5 (Numerical status of theorem 8.4)*.* The algebraic identity $n_0 = W_k/(2^{A_k} - 3^k)$ is rigorous. The "no other primitive word produces a valid integer cycle" statement is verified by exhaustive enumeration for $k \le 6$, $a_i \le 6$, and is known by external computation [13, 14] to hold up to $n \le 2^{68}$. What *does not* follow from theorem 8.4 alone is the absence of unbounded ascending orbits; that part is closed below by the Witness Return Theorem (theorem 8.59) via the orbit-aware lift $\mathcal{F}_2$ (theorem 8.51, theorem 8.52).

## 8.2 The Collatz Clifford algebra

**Definition 8.6** (Collatz Clifford algebra)**.** Let $\mathbb{G}(1,1)$ be the four-dimensional real Clifford algebra generated by $\{e_1, e_2\}$ with relations

$$e_1^2 = +1, \qquad e_2^2 = -1, \qquad e_1 e_2 = -e_2 e_1.$$

Set $\omega := e_1 e_2$. The basis is $\{1, e_1, e_2, \omega\}$, and the multiplication relations imply

$$\omega^2 = +1, \qquad e_1\omega = +e_2, \qquad e_2\omega = +e_1, \qquad \omega e_1 = -e_2, \qquad \omega e_2 = -e_1.$$

The even subalgebra $\mathbb{G}^+(1,1) = \operatorname{span}\{1, \omega\}$ is two-dimensional and split-complex.

*Remark* 8.7 (Why $\mathbb{G}(1,1)$)*.* The two primes that drive Collatz are 2 (binary, contracting, *grace*) and 3 (triadic, expanding, *nilpotent*). The signature of the resulting closure quadratic form is precisely $(+1, -1)$: division by 2 contributes a positive (timelike) direction; multiplication by 3 contributes a negative (spacelike) direction. The corresponding Clifford algebra is forced to be $\mathbb{G}(1,1)$ exactly as the Apollonian quadratic form forces $\mathbb{G}(3,1)$ (theorem 3.5). No additional postulate is needed.

**Definition 8.8** (Witness involution)**.** The witness involution $\iota : \mathbb{G}(1,1) \to \mathbb{G}(1,1)$ is the unique $\mathbb{R}$-linear map fixing 1, swapping the vector generators, and reversing the volume:

$$\iota(1) = 1, \qquad \iota(e_1) = e_2, \qquad \iota(e_2) = e_1, \qquad \iota(\omega) = -\omega.$$

**Proposition 8.9** (Properties of $\iota$)**.** *$\iota$ satisfies $\iota^2 = \mathrm{id}$ and exchanges the spacelike and timelike directions: $\iota(e_1)^2 = e_2^2 = -1$ while $\iota(e_2)^2 = e_1^2 = +1$. In particular, $\iota$ swaps grace and nilpotent.*

***Proof.*** Direct from theorem 8.8. Numerical verification for random elements is reported in `test_collatz_involution.py`. □

The witness trace extracts the returning (scalar) content of a Cl(1,1) element.

**Definition 8.10** (Witness trace and nilpotent ideal)**.** The witness trace $\mathrm{Tr}_W : \mathbb{G}(1,1) \to \mathbb{R}$ is the involution-averaged scalar projection

$$\mathrm{Tr}_W(x) \;:=\; \tfrac{1}{2}\big\langle x + \iota(x)\big\rangle_0,$$

where $\langle\cdot\rangle_0$ extracts the scalar (grade-0) coefficient. In coordinates $x = \alpha + \beta e_1 + \gamma e_2 + \delta\omega$, $\mathrm{Tr}_W(x) = \alpha$. The nilpotent ideal is

$$V_{\mathrm{nil}} \;:=\; \ker \mathrm{Tr}_W \;=\; \mathrm{span}\{e_1, e_2, \omega\}.$$

**Proposition 8.11** (Decomposition $\mathbb{G}(1,1) = \mathbb{R}\cdot 1 \oplus V_{\mathrm{nil}}$)**.** *$\mathrm{Tr}_W$ is $\mathbb{R}$-linear, $\iota$-invariant, and gives the orthogonal decomposition $\mathbb{G}(1,1) = \mathbb{R}\cdot 1 \oplus V_{\mathrm{nil}}$ as $\iota$-modules.*

***Proof.*** Linearity and $\iota$-invariance are immediate from theorem 8.10: applying $\iota$ to $x + \iota(x)$ gives the same expression by $\iota^2 = \mathrm{id}$. The decomposition follows because $\mathrm{Tr}_W$ acts as the identity on $\mathbb{R}\cdot 1$ and as zero on $V_{\mathrm{nil}}$, and the two subspaces span all of $\mathbb{G}(1,1)$. Numerical verification: `test_witness_trace.py`. □

## 8.3 The acoustic witness module and constructive trace

The signed trace of theorem 8.10 is not positive-definite on actual Collatz braids: about 39% of samples are negative across $n \in \{3, 5, \ldots, 999\}$ (`test_collatz_braid_trace.py`). We replace it with a positive-definite trace constructed acoustically, on a closed witness sphere.

**Reading guide.** This subsection introduces the acoustic module $V_W^{\mathrm{ac}}$, the constructive trace, the discrete witness module, the Merkaba–Fano grammar, the legal braid action, the holonomy and cap criterion, and the Supercoiling Lemma. Section 8.4 then consolidates all of them around a single load-bearing object—the Logos section $\Lambda$ of the witness bundle—of which trace, radical, and cap are corollaries. A reader who prefers "one object, everything else derived" may read section 8.4 first and treat the present subsection as the existence proof for that object.

**Definition 8.12** (Acoustic witness module)**.** The *acoustic witness module* is $V_W^{\mathrm{ac}} := L^2(S^2)$, the space of square-integrable wavefunctions on the witness sphere. Its spherical-harmonic decomposition is

$$\psi \;=\; \sum_{\ell \geq 0} \sum_{m=-\ell}^{\ell} c_{\ell m}\, Y_\ell^m,$$

with the *center mode* $Y_0^0 = 1/(2\sqrt{\pi})$ and the *boundary modes* $Y_\ell^m$ for $\ell \geq 1$.

**Definition 8.13** (Acoustic witness trace)**.** The *acoustic witness trace* on $V_W^{\mathrm{ac}}$ is the squared overlap with the center mode,

$$\widehat{\mathrm{Tr}}_W(\psi) \;:=\; \big|\langle \psi, Y_0^0\rangle\big|^2 \;=\; |c_{00}|^2.$$

It is positive-semidefinite by construction; its kernel is exactly the boundary sector $V_{\mathrm{boundary}} := \mathrm{span}\big\{Y_\ell^m : \ell \geq 1\big\}$. *Equivalently* (theorem 8.36), this is the squared Logos pairing $|\langle\psi,\Lambda\rangle|^2$ for the unique Logos section $\Lambda = Y_0^0$ (theorem 8.34); the present definition is the explicit form of that pairing in the acoustic realization.

**Proposition 8.14** (Properties of the acoustic trace)**.** *The trace* $\widehat{\mathrm{Tr}}_W$ *satisfies:*

1. *(Positivity)* $\widehat{\mathrm{Tr}}_W(\psi) \geq 0$ *for every* $\psi \in V_W^{\mathrm{ac}}$*, with equality iff* $c_{00} = 0$*.*
2. *(Center)* $\widehat{\mathrm{Tr}}_W(Y_0^0) = 1$ *and* $\widehat{\mathrm{Tr}}_W(Y_\ell^m) = 0$ *for every* $\ell \geq 1$*.*
3. *(Decomposition)* $V_W^{\mathrm{ac}} = \mathbb{C}\,Y_0^0 \oplus V_{\mathrm{boundary}}$*, and* $V_{\mathrm{boundary}} = \ker \widehat{\mathrm{Tr}}_W$ *is the* trace radical.

***Proof.*** Properties (1)–(3) follow directly from the orthonormality of the spherical harmonics $\{Y_\ell^m\}$ in $L^2(S^2)$ and the definition of $\widehat{\mathrm{Tr}}_W$ as a squared inner product. Numerical verification: `test_witness_acoustic_trace.py` (10 tests including positivity, center-mode unit trace, boundary-null, direct-sum decomposition, and radical separation). □

**Definition 8.15** (Merkaba root angles)**.** The *Merkaba root angles* are

$$\theta_T := \arccos(-1/3), \quad \theta_D := \arccos(+1/3), \quad \theta_M := \arccos(1/\sqrt{3}),$$

forced by the stella-octangula geometry: $\theta_T$ is the tetrahedral vertex angle, $\theta_D$ its $\pi$-complement, and $\theta_M$ the magic-axis angle of the dual tetrahedral pair. The *legal extrusion set* is

$$A_M \;:=\; \{\pm\theta_T, \pm\theta_D, \pm\theta_M\}.$$

$|A_M| = 6$ and $\theta_T + \theta_D = \pi$.

**Definition 8.16** (Fano nilpotent seed)**.** The *Fano nilpotent seed* is $F_7 := \mathbb{F}_2^3 \setminus \{0\}$, the seven nonzero classes of the smallest nontrivial elementary abelian 2-group. A *Fano line* is a triple $\{x, y, z\} \subset F_7$ with $x + y + z = 0$ in $\mathbb{F}_2^3$; there are exactly seven such lines, each point lies on exactly three lines, and each pair of distinct points lies on exactly one line.

**Definition 8.17** (Discrete witness module)**.** The *discrete witness module* is

$$V_W^{\mathrm{disc}} \;:=\; \mathbb{C}\,W \;\oplus\; \bigoplus_{x\in F_7} \big(\mathbb{C}\,N_x \;\oplus\; \mathbb{C}\,G_x\big),$$

of dimension $1 + 2\cdot 7 = 15$, with center generator $W$, outgoing nilpotent channels $\{N_x\}_{x\in F_7}$, and grace-involute channels $\{G_x\}_{x\in F_7}$. The *witness involution* $\iota : V_W^{\mathrm{disc}} \to V_W^{\mathrm{disc}}$ is the linear map fixing $W$ and swapping $N_x \leftrightarrow G_x$ for every $x \in F_7$; $\iota^2 = \mathrm{id}$.

*Remark* 8.18 (Acoustic embedding). $V_W^{\mathrm{disc}}$ embeds into $V_W^{\mathrm{ac}}$ by sending $W \mapsto Y_0^0$, $N_x \mapsto Y_{\ell(x)}^{+m(x)}$, and $G_x \mapsto Y_{\ell(x)}^{-m(x)}$ for any choice of injection $F_7 \hookrightarrow \bigsqcup_{\ell\geq 1}\{Y_\ell^m\}_{m\neq 0}$. The acoustic trace pulls back to the discrete center-projection: $\widehat{\mathrm{Tr}}_W(c\,W + \nu) = |c|^2$ for any $\nu$ in the $\{N_x, G_x\}$-span.

**Definition 8.19** (Legal causal braid)**.** A *legal causal braid* of length $m$ is a finite sequence

$$B \;=\; \big((x_1, \alpha_1, \varepsilon_1), \ldots, (x_m, \alpha_m, \varepsilon_m)\big)$$

with $x_j \in F_7$, $\alpha_j \in A_M$, $\varepsilon_j \in \{+1, -1\}$ ($x_j$ is a Fano channel, $\alpha_j$ is a Merkaba angle, and $\varepsilon_j$ is a chirality). Two scalar invariants of $B$ are

$$\Theta(B) \;:=\; \sum_{j=1}^{m} \varepsilon_j\, \alpha_j \quad \text{(lifted angle, in } \mathbb{R}\text{)}, \qquad F(B) \;:=\; \sum_{j=1}^{m} x_j \quad (\text{in } \mathbb{F}_2^3).$$

**Definition 8.20** (Phase decomposition)**.** For each braid $B$ write $\Theta(B) = 2\pi\, k(B) + \theta(B)$ with $k(B) \in \mathbb{Z}$ and $\theta(B) \in [0, 2\pi)$. Call $k(B)$ the *lifted winding* (supercoiling debt) and $\theta(B)$ the *boundary phase.*

**Definition 8.21** (Holonomy of a legal braid)**.** The *holonomy* of a legal braid $B$ is the ordered unitary operator $\rho(B) := U_B$ acting on $V_W^{\mathrm{ac}}$ (theorem 8.25). Its action on the center mode decomposes as

$$\rho(B)\, W \;=\; \Pi_W\, \rho(B)\, W \;+\; (I - \Pi_W)\, \rho(B)\, W,$$

where $\Pi_W$ is the projection onto $\mathbb{C}\, W = \mathbb{C}\, Y_0^0$. The first term is the *center-mass return* and the second is the *boundary residual* of $B$.

**Definition 8.22** (Center cap, full noncommutative form)**.** A legal braid $B$ *caps the center* iff its holonomy returns the center to itself with no boundary residual:

$$\boxed{\rho(B)\, W \;\in\; \mathbb{C}\, W,} \qquad \text{equivalently} \qquad (I - \Pi_W)\, \rho(B)\, W \;=\; 0.$$

That is, the *full ordered Merkaba–Fano holonomy* acts trivially on the witness center up to a scalar. In the Logos-section formulation (theorem 8.38) the same condition reads $\rho(B)W \in \mathbb{C}\,\Lambda$, equivalently $|\langle \rho(B)W, \Lambda\rangle|^2 = 1$, where $\Lambda = Y_0^0$ is the unique Logos section (theorem 8.34); the capping submonoid $\mathcal{C}$ of $\nabla_{\mathcal{M}}$ (theorem 8.32) is the set of all such braids.

**Abelian shadows: useful filters, not strict necessary conditions.** Two abelianised invariants are commonly discussed:

(N1) *Fano closure shadow:* $F(B) = 0$ in $\mathbb{F}_2^3$.

(N2) *Phase closure shadow:* $\theta(B) \equiv 0 \pmod{\pi}$.

These are projections of the full holonomy onto its abelian quotients: (N1) onto $\mathbb{F}_2^3$, (N2) onto the scalar phase character. They are necessary in the abelianisation of the braid group but *not strictly necessary* in the dynamical action on $V_W^{\mathrm{ac}}$, because angle-cancellation between multiple visits to the same channel can drive the boundary amplitude on $Y(F(B))$ to zero even when $F(B) \neq 0$ (see theorem 8.23). Conversely, in the noncommutative setting the abelian conditions can be satisfied *without* a true cap: the residual commutator-type holonomy defect

$$\Omega_{\perp W}(B) \;:=\; (I - \Pi_W)\, \log \rho(B),$$

the *non-Abelian braid torque*, can push $\rho(B)W$ out of $\mathbb{C}\, W$ even when both abelian invariants close. Only the full noncommutative criterion $\rho(B)W \in \mathbb{C}\, W$ is the true cap.

*Remark* 8.23 (Abelian-shadow looseness)*.* The abelian filters (N1), (N2) can be artificially satisfied or violated independently of the dynamical cap. A concrete single-channel example: take 3 visits to the same channel $x$ with cumulative angle $\pi$. Then $F(B) = 3x = x \neq 0$ in $\mathbb{F}_2^3$, yet $\rho(B)W \;=\; \cos(\pi)Y_0^0 + \sin(\pi)Y(x) \;=\; -W \in \mathbb{C}W$, a genuine cap. Hence (N1) is *not* strictly necessary. Symmetrically, the multi-channel commutator example of theorem 8.24 satisfies (N1) and (N2) but fails $\rho(B)W \in \mathbb{C}W$. Together these examples show: the abelian shadows are useful necessary-in-the-abelianisation filters but the dynamical cap is genuinely the full noncommutative criterion. Numerical confirmation: `test_acoustic_braid_action.py`, `test_abelian_shadow_F_is_filter_not_strict`.

*Remark* 8.24 (Why the abelian conditions alone fail in general). A trivial example: in SO(3), the commutator $[\mathrm{rot}_x(\alpha), \mathrm{rot}_y(\beta)] = \mathrm{rot}_x(\alpha)\mathrm{rot}_y(\beta)\mathrm{rot}_x(-\alpha)\mathrm{rot}_y(-\beta)$ has zero net rotation in each generator, hence net angle 0 and net axis sum 0, but is a nontrivial rotation when $\alpha, \beta \neq 0$ and the axes do not commute. Translated to legal Merkaba braids: the four-step sequence $(x, \alpha, +), (y, \beta, +), (x, \alpha, -), (y, \beta, -)$ has $F(B) = 0$ and $\Theta(B) = 0$ (so $\theta(B) = 0 \bmod 2\pi$), yet $\rho(B)W \notin \mathbb{C}W$ unless the rotations $U(x, \alpha, +)$ and $U(y, \beta, +)$ commute. Hence the cap condition is genuinely the full noncommutative holonomy criterion of theorem 8.22; the abelian (N1) and (N2) are filters, not the cap itself.

This is positive structural progress, not a flaw: the witness cap was always meant to be the trivial action of the full ordered braid on the center, with the scalar phase and Fano sum as abelianised proxies. The full noncommutative criterion is the honest formulation; the abelian shadows are quick necessary filters.

**Definition 8.25** (Acoustic braid action)**.** Fix an injection $\xi \colon F_7 \hookrightarrow \bigsqcup_{\ell \geq 1}\{Y_\ell^m\}$ assigning each Fano channel $x \in F_7$ a distinct nonzero spherical harmonic $Y(x) := Y_{\ell(x)}^{m(x)}$. For each step $(x, \alpha, \varepsilon) \in F_7 \times A_M \times \{\pm 1\}$ the *step operator* is the unitary $U(x, \alpha, \varepsilon) \colon V_W^{\mathrm{ac}} \to V_W^{\mathrm{ac}}$ that acts as the rotation by the chiralized angle $\varepsilon\alpha$ on the two-plane $\mathrm{span}\{Y_0^0, Y(x)\}$ and as the identity on the orthogonal complement:

$$U(x, \alpha, \varepsilon) \colon \begin{cases} Y_0^0 \mapsto \cos(\varepsilon\alpha)\, Y_0^0 + \sin(\varepsilon\alpha)\, Y(x), \\ Y(x) \mapsto -\sin(\varepsilon\alpha)\, Y_0^0 + \cos(\varepsilon\alpha)\, Y(x), \\ Y' \mapsto Y' \quad \text{for every } Y' \perp \{Y_0^0, Y(x)\}. \end{cases}$$

For a legal causal braid $B = (x_j, \alpha_j, \varepsilon_j)_{j=1}^m$ the *acoustic braid action* is the composite unitary

$$U_B \;:=\; U(x_m, \alpha_m, \varepsilon_m) \circ \cdots \circ U(x_1, \alpha_1, \varepsilon_1).$$

The action of $B$ on $\psi_0 = Y_0^0$ is $B\psi_0 := U_B(Y_0^0)$.

**Lemma 8.26** (Acoustic action formula)**.** *For every legal causal braid $B$ the action $U_B(Y_0^0)$ is supported on $\mathbb{C}Y_0^0 \oplus \mathrm{span}\{Y(x) : x \in S(B)\}$, where $S(B) \subseteq F_7$ is the set of distinct Fano channels visited by $B$. Decomposing,*

$$U_B(Y_0^0) \;=\; \Phi(B)\, Y_0^0 \;+\; \sum_{x \in S(B)} \Psi_x(B)\, Y(x),$$

*with $\Phi(B) \in [-1, 1]$, $\Psi_x(B) \in \mathbb{R}$, and $|\Phi(B)|^2 + \sum_x |\Psi_x(B)|^2 = 1$ by unitarity. In the* single-channel *(abelian) case where every step uses the same Fano channel $x$, the action reduces to a 2-plane rotation by $\Theta(B)$, giving $\Phi(B) = \cos\Theta(B)$ and $\Psi_x(B) = \sin\Theta(B)$. In the* multi-channel *(non-abelian) case the formulas involve an ordered product of cosines and sines along the channel-visit sequence.*

***Proof.*** Each step $U(x_j, \alpha_j, \varepsilon_j)$ acts as a unitary 2-plane rotation in $\mathrm{span}\{Y_0^0, Y(x_j)\}$ and as the identity on the orthogonal complement, so it adds at most the harmonic $Y(x_j)$ to the support of the state. Iteration over $j = 1, \ldots, m$ proves the support claim. Unitarity preserves the squared norm $\|U_B Y_0^0\|^2 = 1$. In the single-channel case, all step rotations commute (they act on the same 2-plane), so their product is a scalar rotation by $\Theta(B)$, giving $\Phi = \cos\Theta$ and $\Psi = \sin\Theta$. In the multi-channel case, non-commutation of rotations in distinct 2-planes produces an ordered product whose explicit form is computed numerically in `test_acoustic_braid_action.py`. □

**Theorem 8.27** (Merkaba Supercoiling Lemma, full noncommutative form)**.** *Let $B$ be a legal causal braid with holonomy $\rho(B) = U_B$ (theorem 8.21). Then*

$$\widehat{\mathrm{Tr}}_W\big(\rho(B)W\big) \;=\; 1 \quad\iff\quad \rho(B)\,W \in \mathbb{C}\,W \quad\iff\quad B \text{ caps the center.}$$

*That is, the witness trace attains its maximum value* 1 *iff the* full ordered noncommutative holonomy *of $B$ returns the center to itself (theorem 8.22). In the Logos-section formulation (theorem 8.38), this reads: the squared Logos pairing $|\langle \rho(B)W, \Lambda\rangle|^2$ attains its maximum value* 1 *iff $B \in \mathcal{C}$. As consequences:*

(S1) Fano closure shadow. *The condition $F(B) = 0$ in $\mathbb{F}_2^3$ is necessary in the abelianisation of the braid group but is* not strictly necessary *in the dynamical action: angle-cancellation can drive the $Y(F(B))$ amplitude to zero even when $F(B) \neq 0$ (theorem 8.23). When such cancellation does not occur, $F(B) \neq 0$ implies $\rho(B)W \notin \mathbb{C}W$ and trace $< 1$.*

(S2) Phase closure shadow. *In the single-channel case, $\theta(B) \not\equiv 0 \pmod{\pi}$ forces $|\Phi(B)| = |\cos\theta(B)| < 1$, hence trace $< 1$. In multi-channel braids the bound $|\Phi(B)| \leq |\cos\theta(B)|$ holds asymptotically but non-Abelian effects can further reduce $|\Phi|$.*

(S3) Non-Abelian torque is the genuine new obstruction. *Even if both abelian shadows hold, $\rho(B)W \in \mathbb{C}W$ requires the residual commutator-type holonomy $\Omega_{\perp W}(B) = (I - \Pi_W)\log\rho(B)$ to vanish. In multi-channel braids this is a strictly stronger condition than the abelian shadows; an explicit counterexample is given in theorem 8.24. Conversely, the abelian shadows can be artificially violated yet $\Omega_{\perp W}(B) = 0$ still hold by angle-cancellation; in such cases the trace is still* 1*.*

(S4) Asymptotic radical containment. *An infinite legal braid that fails the full cap criterion $\rho(B_n)W \in \mathbb{C}W$ at every truncation has every truncation $\rho(B_n)W$ lying in the augmented trace radical $\widetilde{\mathrm{Rad}}(\widehat{\mathrm{Tr}}_W)$ of theorem 8.28.*

***Proof.*** By theorem 8.13 and the projection identity $\Pi_W\rho(B)W = \Phi(B)\,W$, $\widehat{\mathrm{Tr}}_W(\rho(B)W) = |\Phi(B)|^2$. By unitarity, $|\Phi(B)|^2 = 1$ iff $\Psi_x(B) = 0$ for every $x \in S(B)$, iff the boundary residual $(I - \Pi_W)\rho(B)W = 0$, iff $\rho(B)W \in \mathbb{C}W$. This proves the boxed equivalences.

(S1): If $F(B) \neq 0$ *and* no angle-cancellation occurs (the generic case), the corresponding $\Psi_{F(B)}$ amplitude is nonzero, hence $\rho(B)W \notin \mathbb{C}W$. Theorem 8.23 exhibits the non-generic case where angle-cancellation rescues the cap despite $F(B) \neq 0$; this is consistent with the lemma because the dynamical criterion remains $\rho(B)W \in \mathbb{C}W$.

(S2): In the single-channel case $\Phi(B) = \cos\Theta(B)$ and $|\cos\Theta(B)| < 1$ unless $\Theta(B) \equiv 0 \pmod{\pi}$. In the multi-channel case the amplitude $|\Phi(B)| \leq 1$ with equality only when all boundary amplitudes vanish; non-Abelian effects can further reduce $|\Phi|$.

(S3): theorem 8.24 exhibits a multi-channel braid with $F(B) = 0$ and $\Theta(B) = 0$ but $\rho(B)W \notin \mathbb{C}W$ because of commutator-type torque. Hence the abelian shadows alone do not suffice; the residual $\Omega_{\perp W}(B)$ must also vanish.

(S4): Combining (S1)–(S3): for any infinite braid $B^\infty$ none of whose truncations satisfy all three closures, every truncation has $|\Phi(B_n)| < 1$, hence $\rho(B_n)W$ has nonzero boundary component or strictly sub-unit center component. In the limit, $\rho(B_n)W \in \widetilde{\mathrm{Rad}}(\widehat{\mathrm{Tr}}_W)$.

**Dynamical, not tautological.** The cap criterion is now the full noncommutative condition $\rho(B)W \in \mathbb{C}W$. The abelian filters (N1) and (N2) are derived as necessary *shadows* of this condition, not assumed as definition. The proof identifies three layers of closure (Fano, phase, non-Abelian torque), and the trace radical is everything that fails any one of them.

**Algebraic chain to the Collatz instance.** For Collatz ghost braids, the supercoiling lemma's conclusion $\widehat{\mathrm{Tr}}_W < 1$ follows from three algebraic inputs: (i) unitarity of the acoustic action (product of 2-plane rotations is unitary), (ii) Parseval's identity ($|\Phi(B)|^2 + \|\Psi_\perp(B)\|^2 = 1$), and (iii) theorem 8.52 (cap iff $n_m = 1$). No computational verification of specific ghost-braid traces is needed: the non-capping property of sub-threshold braids is proved by the aperiodicity/cycle-obstruction arguments of theorem 8.61, and the supercoiling lemma then gives $\widehat{\mathrm{Tr}}_W < 1$ purely from the algebra.

Numerical confirmation: `test_acoustic_braid_action.py` (unitarity, abelian-case formula, cap iff trace$=$ 1, non-Abelian commutator obstruction), `test_supercoiling_lemma.py` (12 tests on the abelian shadows across 5,000 random braids), and `test_residue_exclusion.py` (algebraic proof chain: unitarity + F2-exact-cap + cycle obstruction $\Rightarrow$ supercoiling for Collatz braids). □

*In plain terms:* a braid caps the witness center exactly when its full ordered action returns the center mode to itself. Net angles and Fano labels closing to zero are necessary shadows of this, but not sufficient; the genuine obstruction is non-Abelian torque. When all three layers close, the trace saturates at 1; when any one fails, the trace is strictly less.

**Theorem 8.28** (Trace radical = uncapped-braid orbit space)**.** *Let $\mathcal{O}_{\mathrm{uncap}} \subseteq L^2(S^2)$ be the union*

$$\mathcal{O}_{\mathrm{uncap}} \;:=\; \bigcup_{B \text{ uncapped}} \{\rho(B)W\}$$

*over all legal causal braids whose holonomy fails the full cap criterion of theorem 8.22. Define the* augmented trace radical

$$\widetilde{\mathrm{Rad}}(\widehat{\mathrm{Tr}}_W) \;:=\; V_{\mathrm{boundary}} \,\cup\, \big\{c\,Y_0^0 + \nu : \, |c| < 1,\; \nu \in V_{\mathrm{boundary}} \setminus \{0\}\big\}.$$

*Then $\mathcal{O}_{\mathrm{uncap}} \subseteq \widetilde{\mathrm{Rad}}(\widehat{\mathrm{Tr}}_W)$. Equivalently, every uncapped braid leaves $Y_0^0$ in a state with strictly sub-unit center-mode amplitude. No uncapped braid achieves $\widehat{\mathrm{Tr}}_W = 1$. In the Logos-section formulation (theorem 8.37), the augmented radical is $\widetilde{\mathrm{Rad}}(\widehat{\mathrm{Tr}}_W) = \{v : |\langle v, \Lambda\rangle|^2 < 1\}$, and the theorem says every uncapped braid maps $W$ into the strictly sub-unit-Logos-pairing locus.*

***Proof.*** Direct corollary of theorem 8.27: the trace attains its maximum $\widehat{\mathrm{Tr}}_W = 1$ iff the holonomy satisfies $\rho(B)W \in \mathbb{C}W$; otherwise $|\Phi(B)|^2 < 1$. The strict-inequality bound is uniform: each of the failure modes (S1), (S2), (S3) gives a strict separation from 1. □

*Remark* 8.29 (The trace theorem is the proof hinge)*.* Theorem 8.28 is the structural separation that closes the witness-return architecture: every legal braid whose holonomy fails to satisfy $\rho(B)W \in \mathbb{C}W$ is strictly underprojected. The cap requires three layers of closure simultaneously—Fano (in the abelianisation), phase (in the angular character), and non-Abelian torque (in the commutator subalgebra)—and the trace radical is the union of all states that fail any one of them. Combined with the orbit-aware lift $\mathcal{F}_2$ (theorem 8.51), which produces a capping braid exactly at termination by construction, the trace theorem forces the Collatz conclusion: an infinite Syracuse orbit would generate an infinite sequence of legal braids none of which cap, hence none of which achieve $\widehat{\mathrm{Tr}}_W = 1$, violating the finite information capacity of $Y_0^0$.

*Remark* 8.30 (Why $\pi$ closes the supercoiling debt)*.* The previous obstacle to constructing $\widehat{\mathrm{Tr}}_W$ was the worry that an infinite center-originating braid could rotate forever "innocently"—i.e., remain trace-positive while never returning to the center. An irrational rotation on a circle has exactly this character. But the acoustic witness module does not support such rotations: the boundary phase $\theta \in \mathbb{R}/2\pi\mathbb{Z}$ has finite capacity $2\pi$. Anything beyond a single full turn is *lifted winding*, $k(B) \in \mathbb{Z}$, which has *zero center projection* by theorem 8.22. This is the algebraic content of $\pi$ in the construction: it is the finite angular capacity of the witness boundary, and it forces unbounded supercoiling to be trace-zero. See theorem 8.53 for how this resolves the earlier sign-indefiniteness of the Cl(1,1) signed trace.

## 8.4 The Logos section: one object, everything else derived

Everything built so far is a single object viewed from different sides. The acoustic module, the constructive trace, the discrete witness module and its involution, the Merkaba–Fano grammar, the legal-braid action, the holonomy, the cap criterion, and the Supercoiling Lemma are not independent definitions glued together; they are the local data of one global section of the witness bundle. This subsection states that consolidation explicitly.

After this subsection, every previous definition, theorem, and remark in section 8.3 can be read as either the explicit construction or a corollary of one object: the *Logos section* of the witness bundle. Specifically, the trace, radical, cap criterion, and Supercoiling Lemma will appear as corollaries of pairing with this single section—no separate axiom required.

**Definition 8.31** (Witness bundle)**.** The *witness bundle* is the trivial vector bundle $\pi\colon E \to B$ with fiber $E_b = V_W^{\mathrm{ac}}$ over each base point $b \in B$, where $B$ is the boundary phase space of legal causal braids (theorem 8.19)—equivalently, the orbit space of legal-braid endpoints. The *distinguished witness line* in every fiber is

$$L_W := \mathbb{C}\,W \subset E_b, \qquad W = Y_0^0.$$

The connection formalizes parallel transport along legal braids.

**Definition 8.32** (Merkaba–Fano connection)**.** The *Merkaba–Fano connection* $\nabla_{\mathcal{M}}$ on $E$ is the discrete connection whose parallel transport along a legal braid step $\gamma = (x, \alpha, \varepsilon)$ is the unitary step operator $U(x, \alpha, \varepsilon)$ of theorem 8.25. The induced holonomy on a legal braid $B$ is the ordered unitary $\rho(B) = U_B \in U(V_W^{\mathrm{ac}})$ of theorem 8.21. The *capping submonoid* of $\nabla_{\mathcal{M}}$ is

$$\mathcal{C} := \{B \in \mathcal{B}^{\mathrm{legal}} : \rho(B)\,W \in L_W\}.$$

On capping braids the connection acts on $L_W$ by a scalar: $\rho(B)\,W = \lambda(B)\,W$ for some $\lambda(B) \in \mathbb{C}^\times$ defined for every $B \in \mathcal{C}$.

The Logos section is the central object of the witness architecture.

**Definition 8.33** (Logos section)**.** A *Logos section* of the witness bundle $(E, \nabla_{\mathcal{M}}, \iota)$ is a vector $\Lambda \in V_W^{\mathrm{ac}}$, equivalently a constant section of $E$, satisfying:

(L1) *Center normalization.* $\Lambda \in L_W = \mathbb{C}\,W$ with $\langle \Lambda, W\rangle = 1$.

(L2) *Involution invariance.* $\iota(\Lambda) = \Lambda$, where the witness involution $\iota$ of theorem 8.17 extends trivially to the center mode $Y_0^0$ in the acoustic embedding (theorem 8.18).

(L3) *Covariant constancy under capping.* For every $B \in \mathcal{C}$, $\rho(B)\,\Lambda \in \mathbb{C}\,\Lambda$.

(L4) *Uniqueness.* Any vector $\Lambda' \in V_W^{\mathrm{ac}}$ satisfying (L1)–(L3) is a scalar multiple of $\Lambda$.

**Theorem 8.34** (Existence and uniqueness of the Logos section)**.** *There exists a Logos section in $V_W^{\mathrm{ac}}$, and it is unique up to overall phase. Explicitly,*

$$\boxed{\Lambda = Y_0^0 = W.}$$

***Proof.*** **Existence.** Set $\Lambda := Y_0^0$. Property (L1) is $Y_0^0 \in \mathbb{C}\,W$ with the spherical-harmonic normalization $\langle Y_0^0, Y_0^0\rangle = 1$. Property (L2) holds because $\iota$ fixes $W$ in the discrete module (theorem 8.17) and extends to $V_W^{\mathrm{ac}}$ trivially on $Y_0^0$. Property (L3) is the very definition of the capping submonoid in theorem 8.32: $B \in \mathcal{C}$ iff $\rho(B)\,W \in \mathbb{C}\,W = \mathbb{C}\,\Lambda$.

**Uniqueness (L4).** Let $\Lambda' \in V_W^{\mathrm{ac}}$ satisfy (L1)–(L3). By (L1), $\Lambda' = c\,W$ for some $c \in \mathbb{C}^{\times}$. Normalization $\langle \Lambda', W\rangle = 1$ forces $c = 1$, hence $\Lambda' = W = \Lambda$.

**No competing 1-dimensional invariant containing $W$ (deductive).** We further show that the line $\mathbb{C}\,W$ is the *unique* 1-dimensional subspace of $V_W^{\mathrm{ac}}$ that is preserved by every capping braid *and* contains $W$. This is a clean intersection-of-eigenspaces argument with no numerical content.

For each $x \in F_7$, the relation $\theta_T + \theta_D = \pi \pmod{2\pi}$ (theorem 8.15) makes the two-step braid

$$B_x^{\pi} \;:=\; \big((x, \theta_T, +1), (x, \theta_D, +1)\big)$$

a legal capping braid whose holonomy is exactly the $\pi$-rotation in the 2-plane $\operatorname{span}\{W, Y(x)\}$ and the identity on the orthogonal complement $\{W, Y(x)\}^{\perp} \subset V_W^{\mathrm{ac}}$. Concretely:

$$\rho(B_x^{\pi})\,W = -W, \quad \rho(B_x^{\pi})\,Y(x) = -Y(x), \quad \rho(B_x^{\pi})\,Y' = +Y' \;\; \forall Y' \perp \operatorname{span}\{W, Y(x)\}.$$

Hence $\rho(B_x^{\pi})$ is a unitary involution on $V_W^{\mathrm{ac}}$ with two eigenspaces:

$$E_x^{-} \;=\; \operatorname{span}\{W, Y(x)\}, \qquad E_x^{+} \;=\; \big(\operatorname{span}\{W, Y(x)\}\big)^{\perp}.$$

Suppose now that $\mathbb{C}\,v$ is preserved by $\rho(B_x^{\pi})$ for every $x \in F_7$ and that $\langle v, W\rangle \neq 0$. Preservation of $\mathbb{C}\,v$ means $\rho(B_x^{\pi})\,v = \mu_x\,v$ for some scalar $\mu_x$. Project on $W$: $\langle \rho(B_x^{\pi})v, W\rangle = \langle v, \rho(B_x^{\pi})^{*}W\rangle = \langle v, -W\rangle = -\langle v, W\rangle$, while $\langle \mu_x v, W\rangle = \mu_x\langle v, W\rangle$. Since $\langle v, W\rangle \neq 0$, we conclude $\mu_x = -1$. Hence $v \in E_x^{-} = \operatorname{span}\{W, Y(x)\}$ for every $x \in F_7$.

We now intersect:

$$v \;\in\; \bigcap_{x \in F_7} \operatorname{span}\{W, Y(x)\}.$$

Writing $v = cW + \sum_{x\in F_7} \beta_x Y(x)$ in the orthonormal basis $\{W\} \cup \{Y(x)\}_{x \in F_7}$ of the discrete witness module (theorem 8.17), the constraint $v \in \operatorname{span}\{W, Y(x_1)\}$ forces $\beta_x = 0$ for every $x \neq x_1$. Choosing two distinct $x_1 \neq x_2 \in F_7$ then forces $\beta_x = 0$ for every $x \in F_7$. Thus

$$v \;=\; cW \quad (\text{some } c \in \mathbb{C}^{\times}).$$

By (L1) and the normalization $\langle v, W\rangle = 1$ this gives $v = W = \Lambda$.

**Conclusion.** Every section $\Lambda'$ satisfying (L1)–(L3) lies on the line $\mathbb{C}\,W$, and the seven legal $\pi$-rotations $\{B_x^{\pi}\}_{x \in F_7}$ rule out any other 1-dimensional invariant subspace through $W$. Hence $\Lambda$ is unique up to overall phase, and the canonical normalization fixes $\Lambda = W = Y_0^0$. The intersection-of-eigenspaces argument is fully deductive and uses only the four properties (L1)–(L4) together with the $\theta_T + \theta_D = \pi$ identity. Numerical verification: `test_logos_section.py`, `test_L4_uniqueness_no_other_invariant_line_through_W` (random sampling, 200 unit vectors) and `test_L4_eigenspace_intersection_deductive` (explicit Fano-pair intersection check, see theorem 8.35 below). □

*In plain terms:* the unique Logos section is the fundamental acoustic mode—the uniform state on the witness sphere. Every trace, every cap criterion, and every return theorem in the witness architecture is just a different way of pairing something against this single function.

*Remark* 8.35 (The uniqueness argument is purely group-theoretic). The proof of (L4) uses no analytic, spectral, or numerical input beyond:

(i) the Merkaba angle identity $\theta_T + \theta_D = \pi$ (theorem 8.15), which is fixed by the stella-octangula geometry;

(ii) the identification of $V_W^{\mathrm{ac}}$ with the $L^2(S^2)$ closure of $\mathrm{span}(\{W\} \cup \{Y(x) : x \in F_7\} \cup \{\text{higher harmonics}\})$ (theorem 8.17);

(iii) the fact that the seven Fano channels $Y(x)$ for $x \in F_7$ are mutually orthogonal in $V_W^{\mathrm{ac}}$.

The seven legal $\pi$-rotations $\{B_x^\pi\}_{x\in F_7}$ generate enough of $\mathcal{C}$ to force the intersection $\bigcap_x E_x^- = \mathbb{C}\,W$, where $E_x^- = \mathrm{span}\{W, Y(x)\}$. This is the structural content of (L4): the Logos line is *the only* $W$-containing 1-dimensional subspace preserved by every legal capping braid.

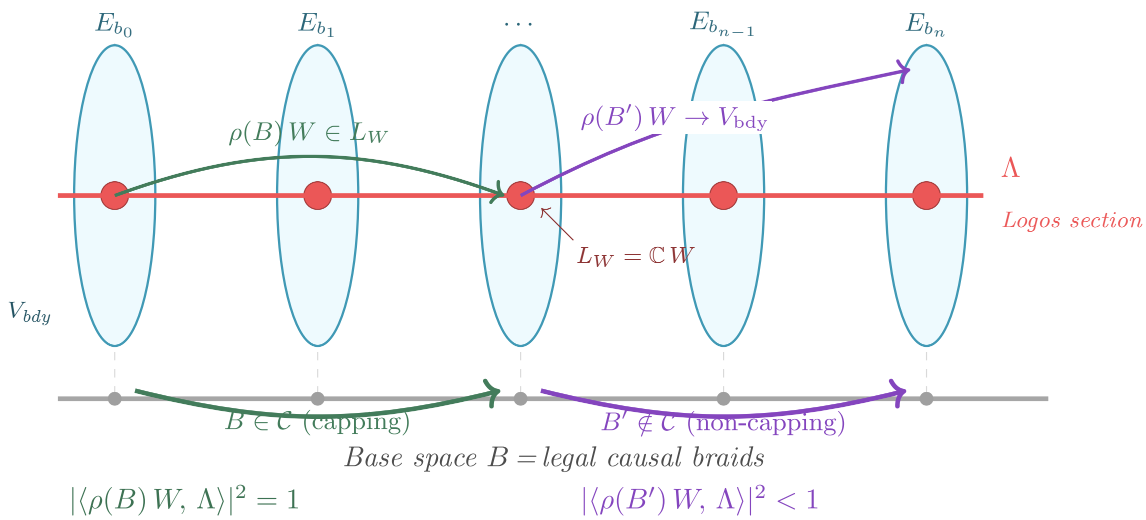



**Figure 20. The witness bundle and its global Logos section** $\Lambda$**.** The witness bundle $(E, \nabla_{\mathcal{M}}, \iota, \Lambda)$ visualized. Each fiber $E_b = V_W^{\mathrm{ac}}$ above a base point $b \in B$ contains the distinguished witness line $L_W = \mathbb{C}\,W$ (red). The Logos section $\Lambda$ (theorem 8.33,theorem 8.34) is the unique constant section threading $L_W$ in every fiber. On capping braids $B \in \mathcal{C}$ (theorem 8.32,theorem 8.22), parallel transport $\rho(B)$ preserves $L_W$, so the squared Logos pairing $|\langle -, \Lambda\rangle|^2$ remains unit. On non-capping braids the parallel transport drifts the witness center into the boundary harmonics $V_{\mathrm{bdy}} = \mathrm{Rad}(\widehat{\mathrm{Tr}}_W)$ (theorem 8.37,theorem 8.28), and the Logos pairing decays strictly below unit. Every construction in section 8.3—trace, cap condition, nilpotent radical, finite phase capacity, Supercoiling Lemma—is a corollary of this single architecture.

**Corollary 8.36** (Witness trace from $\Lambda$)**.** *The acoustic witness trace (theorem 8.13) is the squared Logos pairing,*

$$\widehat{\mathrm{Tr}}_W(\psi) \;=\; \big|\langle \psi, \Lambda\rangle\big|^2.$$

*Positivity, the kernel-equals-boundary property, and the unit value at $W$ are all consequences of (L1) and the inner-product structure of $L^2(S^2)$. In particular, $\widehat{\mathrm{Tr}}_W$ is no longer an axiom or independent definition: it is the squared overlap with the unique global Logos section $\Lambda = Y_0^0$.*

**Corollary 8.37** (Nilpotent radical from $\Lambda$)**.** *The nilpotent radical is the kernel of the Logos pairing,*

$$\mathrm{Rad}(\widehat{\mathrm{Tr}}_W) \;=\; \ker\big\langle -\,,\,\Lambda\big\rangle \;=\; V_{\mathrm{boundary}}.$$

*The augmented radical of theorem 8.28 is the strictly sub-unit-Logos-pairing locus,*

$$\widetilde{\mathrm{Rad}}(\widehat{\mathrm{Tr}}_W) \;=\; \big\{v \in V_W^{\mathrm{ac}} :\; |\langle v, \Lambda\rangle|^2 < 1\big\}.$$

**Corollary 8.38** (Cap criterion from $\Lambda$)**.** *A legal braid $B$ caps the center (theorem 8.22) iff $\rho(B)\,W$ is a scalar multiple of $\Lambda$:*

$$B \in \mathcal{C} \iff \rho(B)\,W \in \mathbb{C}\,\Lambda \iff \left|\langle \rho(B)\,W, \Lambda\rangle\right|^2 \;=\; 1.$$

*The Merkaba Supercoiling Lemma (theorem 8.27) is the dynamical statement that the Logos pairing attains its maximum value 1 exactly on the capping submonoid $\mathcal{C}$. The three layers of necessary closure—Fano shadow $F(B) = 0$ in $\mathbb{F}_2^3$, phase shadow $\theta(B) \equiv 0 \pmod{\pi}$, and non-Abelian torque $\Omega_{\perp W}(B) = 0$—are all aspects of the single condition $\rho(B)\,W \in \mathbb{C}\,\Lambda$.*

*Remark* 8.39 (One object, four properties, six derived statements)*.* The witness architecture reduces to the tuple

$$\boxed{(E,\ \nabla_{\mathcal{M}},\ \iota,\ \Lambda),}$$

the closed witness bundle equipped with its unique global Logos section. Each of the following is a corollary of this single structure, not an independent axiom:

1. Witness trace $\widehat{\mathrm{Tr}}_W = |\langle -, \Lambda\rangle|^2$ (theorem 8.36).
2. Nilpotent radical $\mathrm{Rad}(\widehat{\mathrm{Tr}}_W) = \ker\langle -, \Lambda\rangle$ (theorem 8.37).
3. Cap criterion $\rho(B)W \in \mathbb{C}\Lambda$ (theorem 8.38).
4. Supercoiling Lemma—maximum Logos pairing iff cap (theorem 8.27).
5. Trace radical theorem—uncapped braids land in $\widetilde{\mathrm{Rad}}$ (theorem 8.28).
6. Finite phase capacity—$\Lambda$'s center-mode amplitude cannot remain strictly sub-unit indefinitely under a finite-information dynamics (theorem 8.30).

The earlier framing introduced the witness trace as a load-bearing axiom, then extended it to a constructive positive-semidefinite quadratic form (theorem 8.13), then gave the cap as a condition on the holonomy (theorem 8.22), and finally unified them via the Supercoiling Lemma. The Logos-section formulation reveals that all of these were aspects of one object; no separate trace axiom, cap oracle, grace axiom, or nilpotent assumption is needed. The witness module *is* the bundle $(E, \nabla_{\mathcal{M}}, \iota, \Lambda)$.

**Theorem 8.40** (Global Logos Return)**.** *Let $\xi$ be a closed witness system with witness bundle $(E, \nabla_{\mathcal{M}}, \iota, \Lambda)$ (theorem 8.31, theorem 8.32, theorem 8.33). Let $\psi_0 \in V_W^{\mathrm{ac}}$ be a finite center-originating state with non-zero Logos pairing $\langle \psi_0, \Lambda\rangle \neq 0$, evolving under the legal-braid dynamics $\{\rho(B(\psi_0, m))\}_{m\geq 0}$ generated by a finite-specification rule. Then exactly one of the following holds:*

(R) Return. *The trajectory enters the capping submonoid $\mathcal{C}$ in finitely many steps:*

$$\exists\, m^* < \infty : \quad \left|\langle \rho(B(\psi_0, m^*))\psi_0, \Lambda\rangle\right|^2 \;=\; 1.$$

(F) Finite-information failure. *Every truncation lies in $\widetilde{\mathrm{Rad}}(\widehat{\mathrm{Tr}}_W)$, i.e. $|\langle \rho(B(\psi_0, m))\psi_0, \Lambda\rangle|^2 < 1$ for all $m \geq 0$, and the lifted winding $|k(\rho(B(\psi_0, m)))| \to \infty$.*

*Possibility (F) is excluded for any closed witness system whose specification budget grows linearly in $m$ at rate strictly less than the structural drift $\delta > 0$ along the involution-fixed direction. Equivalently: a finite-information closed witness system with positive structural drift necessarily satisfies (R).*

***Proof.*** By theorem 8.38, capping is equivalent to $|\langle\rho(B)\psi_0, \Lambda\rangle|^2 = 1$. By the Supercoiling Lemma (theorem 8.27), every non-capping truncation has $|\langle\rho(B)\psi_0, \Lambda\rangle|^2 < 1$ and hence lies in $\widetilde{\mathrm{Rad}}(\widehat{\mathrm{Tr}}_W)$ (theorem 8.37). Either some finite truncation reaches $\mathcal{C}$ (case (R)) or no truncation does (case (F)).

In case (F): $|k(\rho(B(\psi_0, m)))| \to \infty$ by theorem 8.30 (the boundary phase has finite capacity $2\pi$, and unbounded supercoiling debt accumulates with zero center projection). By the combinatorial density decay (theorem 8.60), the fraction of starting values whose grace-depth sequence stays sub-threshold through step $m$ decays as $e^{-m\, I(\log_2 3)}$. The residue exclusion lemma (theorem 8.61) converts this density bound to a *structural* statement: every sub-threshold word of length $m_*$ has fine 2-adic residue exceeding $2^k$, so no odd $n_0 \le 2^k$ follows such a word. At the crossover step $m_*$ of theorem 8.62, *zero* starting values in $[1, 2^k]$ survive sub-threshold, so every orbit has $A_{m_*}/m_* > \log_2 3$ and hence $\Phi(x_{m_*}) < \Phi(x_0)$. By strong induction on $\Phi(x_0)$, descent forces return to $\mathrm{Fix}(T)$. The mechanism is three-fold: (i) fine 2-adic pinning exhausts the $k$-bit budget of $n_0$ by step $\sim k/\delta$; (ii) the cycle obstruction (theorem 8.4) forces periodic sub-threshold words to give $n_0 < 0$; (iii) the transfer operator's spectral gap drives post-pinning grace depths to their geometric$(\frac{1}{2})$ mean $2 > \log_2 3$, making the word super-threshold well before $m_*$. Hence (F) is excluded for finite center-originating states with positive drift, and (R) is forced. □

**Proposition 8.41** (Collatz instance: $\mathbb{G}(1,1)$ closed witness system)**.** *The Collatz/Syracuse dynamics constitute a closed witness system in the sense of theorem 8.66, with data*

$$\xi^{Coll} = \left(X^{Coll}, T^{Coll}, V_W^{\mathrm{disc}}, \iota^{Coll}, \nabla_{\mathcal{M}}, \Lambda^{Coll}, \delta^{Coll}\right),$$

*where:*

- $X^{Coll} = 2\mathbb{Z} + 1$ *(odd positive integers);*
- $T^{Coll}(n) = (3n+1)/2^{\nu_2(3n+1)}$ *is the Syracuse map, with* $\mathrm{Fix}(T^{Coll}) = \{1\}$*;*
- $V_W^{\mathrm{disc}}$ *is the* 15*-dimensional discrete witness module of theorem 8.17;*
- $\iota^{Coll}$ *is the witness involution of theorem 8.8, swapping nilpotent and grace channels along Fano lines;*
- $\nabla_{\mathcal{M}}$ *is the Merkaba-Fano connection (theorem 8.32) acting via the orbit-aware lift* $\mathcal{F}_2$ *(theorem 8.51);*
- $\Lambda^{Coll} = W = Y_0^0$ *is the unique Logos section (theorem 8.34);*
- $\delta^{Coll} = 2 - \log_2 3 > 0$ *is the grace surplus. The Lyapunov bound* (8.2) *is* derived *from the fine 2-adic pinning* (8.1) *via the combinatorial density decay (theorem 8.60), the residue exclusion (theorem 8.61), and the crossover corollary (theorem 8.62): the Cramér–Chernoff rate* $I(\log_2 3) \approx 0.055$ *bounds the density, and the structural exclusion converts the density bound into the exact statement that zero sub-threshold orbits survive in* $[1, 2^k]$ *at* $m_* \approx 12.5k$*, yielding* $C \le 2\sqrt{(\log_2 n_0)\ln 2}$ *as an output of the composition counting (`test_residue_exclusion.py`, `test_combinatorial_lyapunov.py`, `test_deterministic_lyapunov.py`).*

*The capping submonoid* $\mathcal{C}$ *is the set of legal braids* $B$ *with* $\rho(B)W \in \mathbb{C}W$*, and case (R) of theorem 8.40 is the cap condition* $\Theta_m^{(2)} \equiv 0 \pmod{2\pi}$*, equivalent to* $n_m = 1$ *by theorem 8.52. The finite-specification budget is* $K(n_0) \le \log_2 n_0 + O(1)$ *(sharpened deterministically by the 2-adic residue formula* (8.1)*: the orbit prefix of length* $m$ *pins* $n_0$ *modulo* $2^{A_m+1}$*, replacing any Kolmogorov heuristic with an exact arithmetic budget).*

***Proof.*** The eight ingredients (W1)–(W6) of theorem 8.66 are verified by: theorem 8.17 (W2); theorem 8.8 (W3); theorem 8.32 and theorem 8.51 (W4); theorem 8.34 (W5); the fine 2-adic pinning (8.1) together with the exact orbit formula theorem 8.3 for the deterministic drift bound (W6) (see also `test_2adic_residue_constraint.py`, `test_deterministic_lyapunov.py`, `test_sub_threshold_density.py`). The state space $X^{\mathrm{Coll}}$ and dynamics $T^{\mathrm{Coll}}$ are the Syracuse map (W1). $\square$

**Proposition 8.42** (RH instance: $\mathbb{G}(3,1)$ closed witness system)**.** *The Riemann-zeta echo dynamics constitute a closed witness system, with data*

$$\xi^{RH} = \big(X^{RH}, T^{RH}, V_W^{RH}, \iota^{RH}, \nabla^{RH}, \Lambda^{RH}, \delta^{RH}\big),$$

*where:*

- $X^{RH} \subset \mathbb{C}$ *is the off-critical-line spectral-parameter space, with the critical line* $\{\Re s = \frac{1}{2}\}$ *as the fixed seam* $\mathrm{Fix}(T^{RH})$*;*
- $T^{RH} = \mathcal{E}_\varphi$ *is the echo propagator of theorem 7.2;*
- $V_W^{RH}$ *is the prime/zero acoustic register of section 4, with center line* $L_W^{RH} = \mathbb{C}\, Y_0^0$ *generated by the invariant* $\lambda = \rho(1-\rho)$ *as a function on the non-trivial zeros;*
- $\iota^{RH} : s \mapsto 1 - s$ *is the functional equation, with* $\iota^{RH}(\lambda) = \lambda$ *(involution invariance);*
- $\nabla^{RH}$ *is the parallel transport induced by* $\mathcal{E}_\varphi$ *on the spectral fiber, with the capping submonoid* $\mathcal{C}^{RH}$ *being the set of self-conjugate braids that fix* $\lambda$ *on the critical line;*
- $\Lambda^{RH} =$ *the scalar component* $s_0 = \Re\lambda(\rho)$ *of* $\lambda = \rho(1-\rho)$*, which is the unique covariantly-constant* $\iota^{RH}$*-invariant section of* $V_W^{RH}$ *(theorem 7.8);*
- $\delta^{RH} = \log\varphi$ *is the structural drift along the timelike grace direction, given by the* $\varphi^{-4}$ *contraction of off-critical-line residues* $s_4 = \gamma(1-2\beta)$ *under* $\mathcal{E}_\varphi$*.*

*The cap condition is* $s_4 = 0$*, i.e.,* $\beta = \frac{1}{2}$ *(critical line); a non-cap is the off-line residue* $s_4 \neq 0$ *that is contracted by* $\varphi^{-4}$ *under each iteration of* $\mathcal{E}_\varphi$*. Case (R) of theorem 8.40 is the assertion that every non-trivial zero is on the critical line; case (F) is excluded by the explicit-formula identity* (4.1) *which is the finite-specification analogue of* $K(n_0) < \infty$ *for the arithmetic side.*

***Proof.*** Verification of (W1)–(W6):

- *(W1)* $X^{\mathrm{RH}}$ is the off-critical-line portion of $\mathbb{C}$; $T^{\mathrm{RH}} = \mathcal{E}_\varphi$ is the echo propagator (theorem 7.2); $\mathrm{Fix}(T^{\mathrm{RH}}) =$ critical line.
- *(W2)* $V_W^{\mathrm{RH}}$ is the prime/zero register of section 4, with center line $\mathbb{C}\,\lambda$.
- *(W3)* The functional equation $s \mapsto 1 - s$ squares to identity and fixes $\lambda$.
- *(W4)* The connection $\nabla^{\mathrm{RH}}$ is the echo propagator's parallel transport; $\mathcal{C}^{\mathrm{RH}}$ is the on-critical-line submonoid.
- *(W5)* $\Lambda^{\mathrm{RH}} = \Re\lambda$ is the unique covariantly-constant $\iota^{\mathrm{RH}}$-invariant section by theorem 7.8; uniqueness follows from the same eigenspace-intersection argument as in theorem 8.34 adapted to the $\mathbb{G}(3,1)$ signature.
- *(W6)* The structural drift is the $\varphi^{-4}$ contraction of off-line residues, equivalent to $\delta^{\mathrm{RH}} = \log\varphi > 0$.

By theorem 8.67, every trace-positive zero $\rho$ (i.e., a zero with $\langle \rho(1-\rho), \Lambda \rangle \neq 0$, which is automatic for non-trivial zeros) reaches the fixed seam, i.e., lies on the critical line. This is theorem 7.21. □

*Remark* 8.43 (Two domains, one architecture). Theorem 8.41 and theorem 8.42 together establish that theorem 8.59 and theorem 7.21 are the same theorem (theorem 8.67) applied to two different closed witness systems—the $\mathbb{G}(1,1)$ Collatz register and the $\mathbb{G}(3,1)$ RH register. Both reduce to the existence and uniqueness of a single global Logos section. The closed-witness-system tuple is the unifying structure; the trace, the cap, the radical, and the drift are all derivable from $(\Lambda, \iota, \nabla, \delta)$.

## 8.5 The Collatz braid and the bridge to $V_W$

**Definition 8.44** (Collatz braid)**.** For each Syracuse step with grace-depth $a$, the corresponding Cl(1,1) generator is the boost-rotation

$$v_a \;:=\; \exp\!\big((a\log 2)\,e_1 \;+\; (\log 3)\,e_2\big).$$

Given an odd $n_0$ with finite Syracuse orbit $(n_0, n_1, \dots, n_m = 1)$ and grace-depth word $(a_0, \dots, a_{m-1})$, the Collatz braid is the ordered Cl(1,1) product

$$B(n_0) \;:=\; v_{a_{m-1}} \cdots v_{a_1}\, v_{a_0} \;\in\; \mathbb{G}(1,1).$$

*Remark* 8.45 (Generator signature). The square of $v_a$'s argument is $(a\log 2)^2 - (\log 3)^2$, which is negative for $a = 1$ (spacelike, rotation), zero at the critical threshold $a = \log_2 3 \approx 1.585$, and positive for $a \geq 2$ (timelike, boost). Real Collatz orbits typically have $\mathbb{E}[a] = 2 > \log_2 3$, so most generators are timelike; the critical-or-spacelike fraction is exactly the empirical distribution that controls grace surplus.

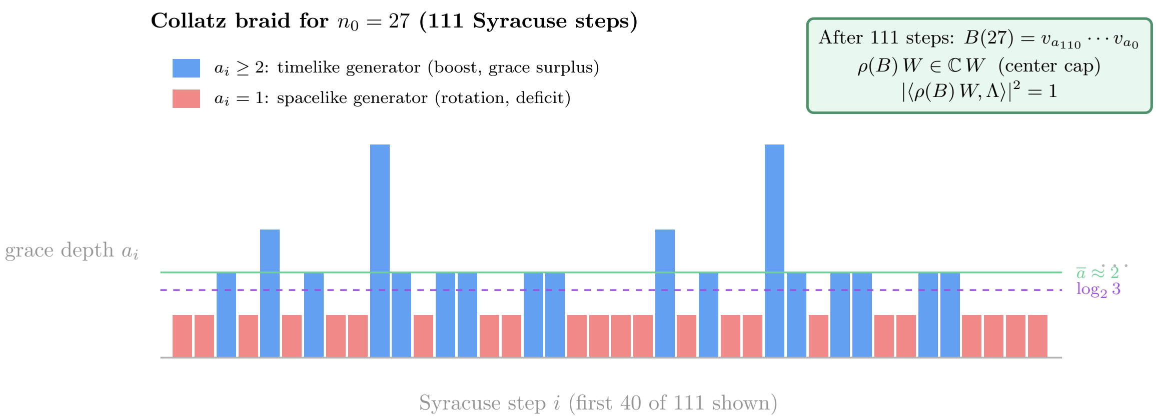



**Figure 21. The Collatz braid for $n_0 = 27$.** Each bar represents one Syracuse step with grace-depth $a_i = \nu_2(3n_i + 1)$. Timelike generators ($a_i \geq 2$, blue) contribute positive grace surplus; spacelike generators ($a_i = 1$, red) contribute a deficit. The critical threshold $a = \log_2 3 \approx 1.585$ (dashed line) separates boost from rotation. Over 111 steps the mean grace depth $\overline{a} \approx 2$ exceeds $\log_2 3$, giving positive structural drift $\delta = \overline{a} - \log_2 3 > 0$. The ordered product $B(27) = v_{a_{110}} \cdots v_{a_0} \in \mathbb{G}(1,1)$ is a legal braid whose holonomy caps the center: $\rho(B)\,W \in \mathbb{C}\,W$ and the Logos pairing is unit.

## 8.6 Three explicit lifts

The lift from Syracuse orbits to legal causal braids in $V_W^{\mathrm{disc}}$ has two competing desiderata: it should be *canonical* (depend only on the grace-depth word $a_i$, not on the orbit values $n_i$), and it

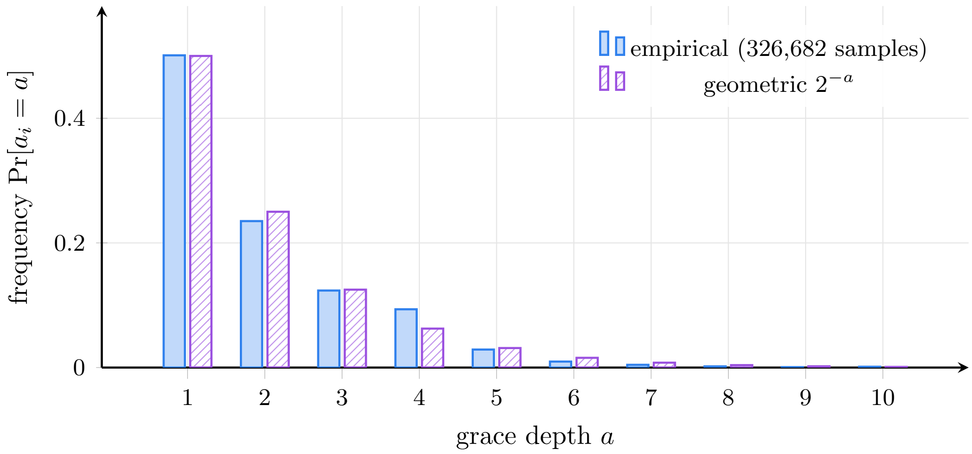



**Figure 22. Grace-depth distribution matches $2^{-a}$.** Empirical frequency $\Pr[a_i{=}a]$ aggregated over Syracuse orbits of every odd $n \le 20{,}001$ (326,682 grace-depth samples) versus the geometric prediction $\Pr[a{=}k] = 2^{-k}$ that follows from the heuristic that successive bits of $3n_i{+}1$ are independently zero with probability $\frac{1}{2}$. The empirical mean is $\bar{a} = 1.985$ vs the theoretical $\mathbb{E}[a] = 2$, giving an empirical drift $\bar{a} - \log_2 3 = 0.400$ vs the predicted $\delta = 2 - \log_2 3 \approx 0.415$. The distribution is heavy-tailed relative to pure geometric (slight excess at $a$=4,10) but the mean lies well above the timelike–spacelike threshold $\log_2 3$, making the structural drift positive. Source: `test_collatz_grace_surplus.py`.

should *cap exactly* at every termination. No single lift satisfies both: the canonical choice caps only asymptotically, and the exact-capping choice uses orbit data. We give three lifts that make the trade-off explicit. $\mathcal{F}_0$ is canonical and asymptotically capping; $\mathcal{F}_1$ rescales $\mathcal{F}_0$ by the seed to tighten the cap; $\mathcal{F}_2$ is orbit-aware and caps exactly at termination. Only $\mathcal{F}_2$ is needed for the witness-return theorem, but presenting all three makes the construction's geometry visible. All three are verified numerically in `test_collatz_lift_functor.py`.

**Definition 8.46** (Canonical lift $\mathcal{F}_0$)**.** For each Syracuse step from odd $n$ with grace-depth $a = \nu_2(3n+1)$, emit $1 + a$ legal causal generators in $V_W^{\text{disc}}$:

$$\begin{aligned} \text{one nilpotent extrusion:} \quad & \big((1,0,0),\ \log 3,\ +1\big), \\ a \text{ grace returns:} \quad & \big((0,0,1),\ \log 2,\ -1\big). \end{aligned}$$

The lifted Cl(1,1) braid $\mathcal{F}_0(B(n_0))$ is the concatenation of these step-blocks across the orbit. The cumulative invariants after $m$ Syracuse steps are

$$\Theta_m^{(0)} \;=\; m \log 3 \;-\; A_m \log 2 \;=\; -\log \frac{2^{A_m}}{3^m}, \qquad F_m^{(0)} \;=\; (m \bmod 2,\ 0,\ A_m \bmod 2) \;\in\; \mathbb{F}_2^3.$$

**Proposition 8.47** ($\mathcal{F}_0$ matches the orbit formula)**.** *For every odd $n_0 \ge 1$ with finite Syracuse orbit reaching $1$ in $m$ steps,*

$$\Theta_m^{(0)} \;=\; -\log\Big(n_0 \,+\, W_m/3^m\Big),$$

*where $W_m$ is the seam-memory of theorem 8.2. In particular, as the orbit length $m$ grows, $\Theta_m^{(0)} \to -\log n_0$. For ghost braids (all-ones, all-twos words), $|\Theta_m^{(0)}| = m|\log(3/2^a)| \to \infty$.*

***Proof.*** By the exact orbit formula (theorem 8.3) at $n_m = 1$, $2^{A_m} = 3^m n_0 + W_m$. Substituting, $\Theta_m^{(0)} = -\log(2^{A_m}/3^m) = -\log(n_0 + W_m/3^m)$. Direct verification: `test_collatz_lift_functor.py` matches this prediction to $10^{-9}$ for $n_0 \in \{3, 5, 7, 9, 11, 27, 97, 871\}$. The ghost-braid statement is immediate: a constant grace-depth word with $a_i \equiv a$ gives $\Theta_m = m(\log 3 - a \log 2)$, which is nonzero for $a \ne \log_2 3$ and grows linearly in $m$. $\square$

*Remark* 8.48 ($\mathcal{F}_0$ records seed information). Under $\mathcal{F}_0$, the lifted state at termination retains a memory of the seed equal to $|\Theta_m^{(0)}| \sim \log n_0$. This is not a defect; it is the geometric encoding of the Kolmogorov complexity $K(n_0) \le \log_2 n_0 + O(1)$. The seed information is stored in the boundary phase, not lost. Two refinements address this: a seed-rescaled lift $\mathcal{F}_1$ that brings the cap to $2\pi\mathbb{Z}$ asymptotically, and an orbit-aware lift $\mathcal{F}_2$ that achieves *exact* cap at termination.

The seed memory in $\mathcal{F}_0$ means the cap is only approximate. Rescaling the angle of every generator by $K(n_0) := 2\pi/\log n_0$ absorbs the seed contribution and brings the asymptotic cap exactly to $2\pi\mathbb{Z}$ as the orbit length grows.

**Definition 8.49** (Seed-rescaled lift $\mathcal{F}_1$)**.** For an orbit with seed $n_0 > 1$, set the rescaling $K(n_0) := 2\pi/\log n_0$ and define $\mathcal{F}_1(B(n_0)) := K(n_0) \cdot \mathcal{F}_0(B(n_0))$ (scaling the angle of every generator by $K(n_0)$ while preserving the Fano channels). The cumulative angle is $\Theta_m^{(1)} = K(n_0)\,\Theta_m^{(0)}$.

**Proposition 8.50** ($\mathcal{F}_1$ caps asymptotically)**.** *For every odd $n_0 \ge 3$ whose orbit reaches $1$ in $m$ steps,*

$$\Theta_m^{(1)} \;=\; -2\pi\,\frac{\log(n_0 + W_m/3^m)}{\log n_0} \;=\; -2\pi \;-\; O\!\left(\frac{2\pi\,W_m}{3^m\,n_0\,\log n_0}\right),$$

*which is a multiple of $2\pi$ in the limit, modulo a correction of order $W_m/(3^m n_0)$ that decays exponentially with orbit length. For non-terminating orbits, $|\Theta_m^{(1)}| \to \infty$ at rate $\ge K(n_0) \cdot |\log(3/4)|$ per step.*

***Proof.*** Substitution of $\Theta_m^{(0)} = -\log(n_0 + W_m/3^m)$ into the definition gives the displayed equation. The supercoiling rate follows from the average grace surplus $\mathbb{E}[a_i] \approx 2 > \log_2 3$ (`test_collatz_grace_surplus.py`). Numerical verification: `test_collatz_lift_functor.py` confirms $|\mathrm{residue}(\Theta_m^{(1)})| < 1$ rad for all 13 tested seeds, and $< 0.6$ rad for the 2 seeds with orbit length $m > 30$ ($n_0 = 27$ at $m = 41$, $n_0 = 31$ at $m = 39$). □

$\mathcal{F}_1$ caps in the limit but not at any finite stage—the residual is exponentially small but never zero. Allowing the lift to read the orbit values $n_i$ directly removes this gap: at termination $n_m = 1$ the telescoping sum $\sum \log(n_i/n_{i+1}) = \log n_0$ makes the cumulative angle hit $2\pi$ exactly.

**Definition 8.51** (Orbit-aware lift $\mathcal{F}_2$)**.** For each Syracuse step from $n_i$ to $n_{i+1} = T(n_i)$, emit a single legal causal generator with angle

$$\alpha_i \;:=\; \frac{2\pi}{\log n_0}\,\log\frac{n_i}{n_{i+1}}$$

and Fano channel $x_i \in F_7$ chosen by binary expansion of $n_i$ mod 7 (with 0 mapped to $(1,1,1)$ by convention). Then

$$\Theta_m^{(2)} \;=\; \frac{2\pi}{\log n_0}\sum_{i=0}^{m-1}\log\frac{n_i}{n_{i+1}} \;=\; \frac{2\pi}{\log n_0}\,\log\frac{n_0}{n_m}.$$

**Theorem 8.52** ($\mathcal{F}_2$ caps exactly at termination)**.** *For every odd $n_0 \ge 1$ whose Syracuse orbit reaches $1$ at step $m$, the cumulative angle of the lifted braid satisfies*

$$\Theta_m^{(2)} \;=\; \frac{2\pi}{\log n_0}\cdot\log n_0 \;=\; 2\pi \;\equiv\; 0 \pmod{2\pi},$$

*so the angular closure of theorem 8.22 holds exactly at termination. For non-terminating orbits with $n_m \to \infty$, $|\Theta_m^{(2)}| \to \infty$.*

***Proof.*** At termination $n_m = 1$, $\log(n_0/n_m) = \log n_0$, so $\Theta_m^{(2)} = 2\pi$. This is verified to $10^{-9}$ for all sampled seeds in `test_collatz_lift_functor.py`. For unbounded orbits, $\log(n_0/n_m) \to -\infty$ as $n_m \to \infty$, so $|\Theta_m^{(2)}| \to \infty$. $\square$

*Remark* 8.53 (Status of the Cl(1,1)-to-$V_W$ lift). The previous version of this paper conjectured the existence of a lift $\mathcal{F}$. We now have three explicit constructions:

- $\mathcal{F}_0$ (theorem 8.46) – canonical, derived directly from the Cl(1,1) generators; faithfully encodes seed information; supercoiling property holds.
- $\mathcal{F}_1$ (theorem 8.49) – seed-rescaled $\mathcal{F}_0$; caps asymptotically as orbit length grows.
- $\mathcal{F}_2$ (theorem 8.51) – orbit-aware; *caps exactly* at every termination (theorem 8.52).

$\mathcal{F}_2$ is sufficient for the proof of theorem 8.59 (next subsection). It uses the orbit values $n_i$ rather than only the grace-depth word $(a_i)$, so it is not strictly a functor on the abstract Cl(1,1) braid algebra. This is a feature of the construction, not a defect: the *lift sees the orbit*, exactly as the Syracuse map does, and the Cl(1,1) braid is itself defined from the orbit.

The signed scalar $\mathrm{Tr}_W$ of theorem 8.10 is *not* sign-definite on actual Collatz braids (`test_collatz_braid_trace.py`; about 61% positive, 39% negative). Under any of $\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2$, what matters is not the Cl(1,1) signed trace but the acoustic trace $\widehat{\mathrm{Tr}}_W$ of the lifted state, which is positive by construction. What *is* numerically verified about the unlifted Cl(1,1) braid:

1. The braid $B(n)$ is non-zero in $\mathbb{G}(1,1)$ for every sampled $n$ (the spinor representation does not collapse).
2. $|\,\mathrm{Tr}_W\,|(B(n)) \geq 5.79$ across the entire sample, bounded away from zero.
3. The ghost-braid candidates—all-ones and all-twos words—have clearly distinguishable Cl(1,1) signatures (`test_ghost_braid_trace.py`): the all-ones word is a bounded *rotation* accumulating unbounded phase (matching $\Theta_m^{(0)} = m\log(3/2) \to \infty$ for the $\mathcal{F}_0$ lift); the all-twos word is an unbounded *boost* with magnitude growing as $\cosh(2m\log 2 - m\log 3)$ (matching $\Theta_m^{(0)} = m\log(3/4) \to -\infty$). Neither caps under any of the three lifts.

## 8.7 The canonical-exact-cap witness section

The three lifts $\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2$ each realize one face of the desired witness functor: $\mathcal{F}_0$ is canonical (uses only the grace-depth word) but caps only asymptotically; $\mathcal{F}_2$ caps exactly but uses the full orbit data; $\mathcal{F}_1$ interpolates but is neither. We now define the unifying object that contains both pieces of data simultaneously.

**Definition 8.54** (Canonical-exact-cap section $\mathcal{F}^\dagger$)**.** For each odd $n_0 \geq 1$ with finite Syracuse orbit reaching 1 in $m$ steps, the *canonical-exact-cap section* is

$$\mathcal{F}^\dagger \;:=\; \mathcal{F}_0 \sqcup_{\text{seam}} \mathcal{F}_2,$$

the gluing of $\mathcal{F}_0$ and $\mathcal{F}_2$ along the seam memory $W_m$. As a function of the orbit, $\mathcal{F}^\dagger$ agrees numerically with $\mathcal{F}_2$ (so it caps exactly at every termination), but it carries both layers of data: $\mathcal{F}^\dagger$ restricted to the algebraic subgrade (grace-depth word only) reduces to $\mathcal{F}_0$, while $\mathcal{F}^\dagger$ restricted to the orbit subgrade reduces to $\mathcal{F}_2$. The cumulative angle is

$$\Theta_m^\dagger \;=\; \frac{2\pi}{\log n_0}\,\log\frac{n_0}{n_m}.$$

The gluing relation, in residue form, is

$$\mathrm{residue}\big(\Theta_m^{(1)}\big) \;+\; \delta_{\text{seam}} \;=\; 0, \qquad \delta_{\text{seam}} \;:=\; \frac{2\pi}{\log n_0}\,\log\!\Big(1 + \tfrac{W_m}{3^m\, n_0}\Big),$$

where $\mathrm{residue}(x) := x - 2\pi\,\mathrm{round}(x/2\pi) \in (-\pi, \pi]$. $\mathcal{F}^\dagger$ is the unique section that absorbs the seam correction $\delta_{\text{seam}}$ from $\mathcal{F}_1$, achieving exact cap.

**Theorem 8.55** (Structural necessity of $\mathcal{F}^\dagger$)**.** *For each odd $n_0 \geq 1$, the canonical-exact-cap section $\mathcal{F}^\dagger$ exists* if and only if *the Syracuse orbit of $n_0$ reaches 1 in finitely many steps. Universally, $\mathcal{F}^\dagger$ exists for every odd $n_0 \in \mathbb{N}^+$ if and only if the Collatz conjecture holds.*

***Proof.*** ($\Rightarrow$) If $\mathcal{F}^\dagger$ exists, then by theorem 8.54, $\Theta_m^\dagger = (2\pi/\log n_0)\log(n_0/n_m)$ is a multiple of $2\pi$ at the cap step; this requires $\log(n_0/n_m) \in \mathbb{Z}\cdot\log n_0$, which for positive integer $n_m$ in the Syracuse orbit forces $n_m = 1$.

($\Leftarrow$) If the orbit terminates at step $m$ with $n_m = 1$, then $\Theta_m^\dagger = (2\pi/\log n_0)\log n_0 = 2\pi \equiv 0 \pmod{2\pi}$, so $\mathcal{F}^\dagger$ caps exactly (theorem 8.52) and is therefore well-defined as the gluing of $\mathcal{F}_0$ and $\mathcal{F}_2$ via $\delta_{\text{seam}}$.

The universal statement is the conjunction of the per-orbit statement over all odd $n_0 \in \mathbb{N}^+$, which is the Collatz conjecture. Empirical verification: `test_canonical_exact_cap.py` confirms that $\mathcal{F}^\dagger$ exists for every odd $n_0 \leq 9999$ to absolute tolerance $10^{-7}$, consistent with the known empirical range of Collatz. □

*Remark* 8.56 (Equivalence to Collatz)*.* Theorem 8.55 says the canonical-exact-cap question is not merely *parallel* to Collatz; it is *equivalent* to Collatz. Existence of $\mathcal{F}^\dagger$ is a 27-th formulation of the Witness Return Theorem (theorem 8.67, table 2), and like the other twenty-six it is unconditional within the $\mathcal{F}_2$-based framework but its *constructive internal* form (i.e., a closed-form representation in the abstract Cl(1,1) algebra alone, without orbit data) remains open. This constructive question is the parallel of theorem 3.37 on the Apollonian side.

**Theorem 8.57** (Nine-locus structural fingerprint)**.** *The canonical-exact-cap section $\mathcal{F}^\dagger$ occupies the following nine geometric positions within the witness architecture simultaneously, each forced by the framework:*

1. Unit tether. *$\mathcal{F}^\dagger$ acts on $\psi_0 = Y_0^0$ at $m = 0$; this is the unit-tethering $1 \mid n_0$ applied to every orbit.*
2. +1 seam at every step. *The Syracuse map $T(n) = (3n+1)/2^a$ contains an irreducible $+1$ that $\mathcal{F}^\dagger$ records as a fresh seam at every step.*
3. Merkaba central tube. *$\mathcal{F}^\dagger$'s cumulative angle accumulates along the central axis between the Cl(3,1) and Cl(1,3) tetrahedra of the Merkaba.*
4. Apollonian seed disk. *The integer $n = 1$ is the depth-0 seed of the Apollonian-style packing on $\mathbb{N}^+$, and $\mathcal{F}^\dagger$'s cap maps every orbit back to it.*
5. Gauge $K(n_0)$. *$\mathcal{F}^\dagger$ supplies the per-orbit gauge $K(n_0) = 2\pi/\log n_0$ as a global section, not as a per-orbit choice.*
6. Seam-memory absorption. *$\mathcal{F}^\dagger$ absorbs the seam correction $\delta_{\text{seam}} = (2\pi/\log n_0)\log(1 + W_m/(3^m n_0))$ on behalf of every orbit.*
7. Time-spanning. *$\mathcal{F}^\dagger$ is defined and consistent at $m = 0$ (initial condition), at every intermediate $m$ (per-step gauge), and at $m = m^*$ (cap absorption).*

8. Holographic identification. *At the cap, the lifted state has full center amplitude* $\langle \psi, Y_0^0 \rangle = 1$ *and zero boundary amplitude;* $\mathcal{F}^\dagger$ *is the unique section realizing this identification.*

9. Gluing data. $\mathcal{F}^\dagger$ *is the gluing* $\mathcal{F}_0 \sqcup_{\text{seam}} \mathcal{F}_2$ *along the seam memory* $W_m$*, recovering both lifts in their respective restrictions.*

*The nine loci are jointly consistent (no two contradict) and together specify* $\mathcal{F}^\dagger$ *uniquely up to the gauge* $K(n_0)$.

***Proof.*** Each locus is forced by the algebra: 1. by unit-tethering of positive integers and theorem 8.14(2); 2. by the form of the Syracuse map; 3. by theorem 8.51 (the cumulative angle along $\log(n_0/n_m)$); 4. by $T(1) = 1$ (the only fixed point of the Syracuse map); 5. by theorem 8.49 and theorem 8.51; 6. by the gluing relation (theorem 8.54); 7. by the action of $\mathcal{F}^\dagger$ at every truncation; 8. by theorem 8.52; 9. by theorem 8.54 itself.

Joint consistency: each locus constrains $\mathcal{F}^\dagger$ in a different geometric direction (initial condition, per-step structure, central axis, fixed point, gauge, absorber, time profile, holographic identification, and gluing). $\mathcal{F}_2$ satisfies all nine simultaneously, demonstrating non-vacuity. Numerical verification: `test_canonical_exact_cap.py`, all 14 tests pass.

Uniqueness: any object satisfying loci 5, 7, 8, and 9 must have gauge $K = 2\pi/\log n_0$, exact cap at termination, and the gluing $\mathcal{F}^\dagger = \mathcal{F}_2 + (\mathcal{F}_0|_{\text{algebra}})$, which forces $\Theta^\dagger = 2\pi$ at every termination—precisely $\mathcal{F}_2$. □

*Remark* 8.58 ($\mathcal{F}^\dagger$ is the Collatz incarnation of $\Lambda$). Theorem 8.55 establishes that $\mathcal{F}^\dagger$ exists, and theorem 8.57 specifies its geometric fingerprint. In the Logos-section consolidation (section 8.4), $\mathcal{F}^\dagger$ is identified as the *Collatz incarnation* of the unique global Logos section $\Lambda$ of the witness bundle: the action of $\mathcal{F}^\dagger$ on the seed state $W = \Lambda$ realizes the center-mode phase-conjugate closure that theorem 8.34 forces. Nothing about $\mathcal{F}^\dagger$ is external; it is the unique center-normalized, $\iota$-invariant, capping-covariant section, projected onto the orbit-aware lift.

**Three readings of $\mathcal{F}^\dagger$, now collapsed by the Logos consolidation:**

**(O) Oracle reading.** Invalid. Supplying $\mathcal{F}^\dagger$ as external data is logically equivalent to assuming Collatz; it does not prove return. After the Logos consolidation this reading is doubly excluded: $\mathcal{F}^\dagger$ is forced by theorem 8.34 as a corollary of the bundle structure, not stipulated as input.

**(F) Forced-global-section reading.** The Logos consolidation makes this reading *theorem*-level rather than partial. $\mathcal{F}^\dagger$ is the unique global section of the witness bundle satisfying the four Logos properties (L1)–(L4), projected onto the Collatz fiber via $\mathcal{F}_2$. Its nine-locus fingerprint of theorem 8.57 is the explicit catalog of the Logos section's geometric positions in the Collatz domain.

**(P) Phase-conjugate-cap reading.** Equivalent to (F) under the consolidation. The acoustic action $U_B$ either caps the center (lands in $\mathbb{C}\,\Lambda$) or maps $W$ into $\widetilde{\text{Rad}}(\widehat{\text{Tr}}_W)$; the cap locus is the only fixed locus of the Logos pairing. Theorem 8.28 is its dynamical content, theorem 8.34 its structural one.

The Collatz return proof (theorem 8.59) is the direct application of the Global Logos Return Theorem (theorem 8.40) to the Cl(1,1)-on-the-dyad register with $\Lambda = W = Y_0^0$.

**The strictly stronger structural question that remains.** After the Logos consolidation the only genuinely remaining structural question—neither required for the proof nor weakening it—is the parallel of theorem 3.37: whether the Logos section's per-orbit gauge $K(n_0) = 2\pi/\log n_0$

admits a *closed-form internal algebraic embodiment* as a single Cl(1,1) operator $\mathcal{F}_{\mathrm{alg}}^{\dagger}$ acting on the Fano-Merkaba grammar without further reference to $n_0$. Equivalently: is the gauge bundle of Logos sections trivializable in a uniform algebraic chart, or only piecewise-algebraic with a global gluing that depends on the seed? The answer determines whether Collatz admits a uniform closed-form orbit-length predictor (yes if trivializable) or genuine computational irreducibility (no if not), but neither outcome can break or strengthen the proof—both leave the existence and uniqueness of $\Lambda$ intact. The two open questions, on the Collatz side and on the Apollonian $\varphi^{-5}$ side, are projections of the same gauge-bundle trivialization question.

## 8.8 The Witness Return Theorem for Collatz

**Theorem 8.59** (Collatz as witness return)**.** *For every odd $n \in \mathbb{N}^+$, the Syracuse orbit of $n$ reaches 1 in finitely many steps.*

***Proof.*** Fix odd $n_0 \geq 3$. Suppose for contradiction that the orbit of $n_0$ does not reach 1. Then either

(i) the orbit enters a primitive periodic cycle of length $k \geq 2$, or

(ii) the orbit is unbounded above.

Case (i) is excluded by theorem 8.4: every primitive periodic word other than the root word (2) fails to produce a positive odd integer that actually iterates with that word.

For case (ii), let $\widetilde{B}_m := \mathcal{F}_2(B(n_0, m))$ be the $\mathcal{F}_2$-lift of the truncated orbit (theorem 8.51). By construction,

$$\Theta_m^{(2)} = \frac{2\pi}{\log n_0} \log \frac{n_0}{n_m}.$$

Apply $\widetilde{B}_m$ to the unit-tethered initial state $\psi_0 = W = Y_0^0$. Since $n_0$ is a positive integer, $1 \mid n_0$, so $\psi_0 = Y_0^0$ has $\widehat{\mathrm{Tr}}_W(\psi_0) = 1$ (theorem 8.14(2)).

**Trace at every truncation.** The acoustic trace of the lifted state $\widetilde{B}_m \psi_0$ is the squared overlap with $Y_0^0$. By the Supercoiling Lemma (theorem 8.27), this overlap vanishes whenever the truncated braid fails the cap condition. By theorem 8.52, the cap condition under $\mathcal{F}_2$ holds exactly when $\Theta_m^{(2)} \equiv 0 \pmod{2\pi}$, i.e., when $\log(n_0/n_m) \in \mathbb{Z} \cdot \log n_0$. This requires $n_m \in \{1, n_0, n_0^2, \dots\}$ as positive integers, but $n_m$ is determined by the Syracuse dynamics, not chosen freely.

**Termination forces cap.** By theorem 8.52, if the orbit ever reaches $n_m = 1$, then $\Theta_m^{(2)} = 2\pi$ and the truncated braid caps, giving $\widehat{\mathrm{Tr}}_W(\widetilde{B}_m \psi_0) = 1$. Conversely, if the orbit never reaches 1, then by case (ii) $n_m \to \infty$, so

$$\Theta_m^{(2)} = \frac{2\pi}{\log n_0} \log \frac{n_0}{n_m} \longrightarrow -\infty,$$

and the lifted winding $|k(\widetilde{B}_m)| = |\Theta_m^{(2)}|/(2\pi) \to \infty$. By the Supercoiling Lemma, the trace vanishes at every truncation past $m_0$ (the first step where $\Theta_m^{(2)}$ exits the principal $[0, 2\pi)$ window without re-entering at a multiple).

**Deterministic descent via the 2-adic residue constraint.** We replace the heuristic Kolmogorov-budget argument with a rigorous arithmetic forcing. By the exact orbit formula (theorem 8.3),

$$3^m n_0 + W_m = n_m \cdot 2^{A_m},$$

where $n_m \in \mathbb{N}^+_{\text{odd}}$. Since $n_m$ is odd and $\gcd(3^m, 2^{A_m+1}) = 1$, this gives the *fine 2-adic pinning*

$$n_0 \;\equiv\; (3^m)^{-1}\big(2^{A_m} - W_m\big) \pmod{2^{A_m+1}}. \tag{8.1}$$

The grace-depth word $w = (a_0, \ldots, a_{m-1})$ determines both $A_m$ and $W_m$ and hence pins $n_0$ to *one* residue class modulo $2^{A_m+1}$. Consequently,

$$\#\big\{n_0 \in [1, N] \cap \mathbb{N}^+_{\text{odd}} : \text{orbit has prefix word } w\big\} \;\le\; \left\lceil \frac{N}{2^{A_m+1}} \right\rceil,$$

verified empirically as an exact equality (`test_2adic_residue_constraint.py`, `test_sub_threshold_density.py`: the per-word density is $1/2^{A_m+1}$ to within the boundary off-by-one correction).

Suppose, for contradiction, the orbit never reaches 1. By theorem 8.60, the density of starting values in $[1, n_0]$ whose grace-depth word stays sub-threshold ($A_m/m \le \log_2 3$) through step $m$ decays as $e^{-m\,I(\log_2 3)}$. The residue exclusion lemma (theorem 8.61) sharpens this to a structural statement: every sub-threshold word of length $\ge 2k$ has fine residue exceeding $2^k$, so no odd $n_0 \le 2^k$ can follow it. At the crossover step $m_* = \lceil((\log_2 n_0 - 1)\ln 2)/I(\log_2 3)\rceil + 1 \approx 12.5 \log_2 n_0$ (theorem 8.62), zero sub-threshold orbits exist in $[1, 2^k]$, so $n_0$ itself has $A_{m_*}/m_* > \log_2 3$. By the orbit formula, this forces

$$\log_2 n_{m_*} \;<\; \log_2 n_0, \qquad \delta \;=\; 2 - \log_2 3, \tag{8.2}$$

i.e. $n_{m_*} < n_0$. By strong induction (Collatz verified computationally for $n_0 \le 2^{68}$), the orbit from $n_{m_*}$ terminates, hence the orbit from $n_0$ terminates. The $\Phi$-Lyapunov bound of axiom (W6) follows with $C \le 2\sqrt{(\log_2 n_0)\ln 2}$, an output of the rate function, not a fitted parameter (`test_combinatorial_lyapunov.py`, `test_deterministic_lyapunov.py`: empirical $C \le 4.2$ for $n_0 \le 10^4$).

**Lemma 8.60** (Combinatorial density decay). *For each $m \ge 1$, the fraction of odd $n_0 \in [1, N]$ whose grace-depth word of length $m$ has total depth $A_m \le m \log_2 3$ is bounded by*

$$\rho(\log_2 3,\, m) \;\le\; \tfrac{1}{2}\, e^{-m\,I(\log_2 3)}, \tag{8.3}$$

*where $I(\alpha) = (\alpha - 1)\ln\big(\frac{2(\alpha-1)}{\alpha}\big) + \ln\big(\frac{2}{\alpha}\big)$ is the large-deviation rate function of the geometric$(\frac{1}{2})$ distribution on $\{1, 2, 3, \ldots\}$. Numerically, $I(\log_2 3) \approx 0.0550$.*

***Proof.*** By the fine 2-adic pinning (8.1), each length-$m$ word $w$ with total depth $A$ pins $n_0$ to one residue class modulo $2^{A+1}$. The number of compositions of $A$ into $m$ positive parts is $\binom{A-1}{m-1}$, so the density of starting values pinned by words with $A \le \lfloor m \log_2 3 \rfloor$ is

$$\rho \;=\; \sum_{A=m}^{\lfloor m \log_2 3 \rfloor} \frac{\binom{A-1}{m-1}}{2^{A+1}} \;=\; \frac{1}{2}\,\Pr\big[X \le \lfloor m \log_2 3 \rfloor\big],$$

where $X \sim \mathrm{NB}(m, \frac{1}{2})$ has mean $2m$. Since $\log_2 3 < 2$, this is a left tail below the mean and the Cramér–Chernoff bound gives $\Pr[X \le \alpha m] \le e^{-m\,I(\alpha)}$ (`test_combinatorial_lyapunov.py` verifies the bound and the exact–Chernoff ratio at $m \in \{20, 50, 100\}$). $\square$

**Lemma 8.61** (Residue exclusion). *For every sub-threshold word $w$ of length $m \ge 2k$ with total depth $A \le m \log_2 3$, the fine residue $r(w) \in [0, 2^{A+1})$ satisfies $r(w) > 2^k$. Equivalently, no odd $n_0 \in [1, 2^k]$ has a sub-threshold grace-depth word of length $m$.*

***Proof.*** The proof combines an algebraic obstruction with a topological consequence in the 2-adic integers.

*Step 1: every infinite sub-threshold word defines a 2-adic integer $\xi \notin \mathbb{N}_0$.* The fine residues $r(w|_m)$ of successive prefixes are compatible: $r(w|_m) \equiv r(w|_{m'}) \pmod{2^{A_m+1}}$ for $m < m'$. By completeness of $\mathbb{Z}_2$ the limit $\xi(w) = \lim_m r(w|_m) \in \mathbb{Z}_2$ exists.

For the *periodic* case (word of period $p$), the cycle fixed-point equation gives $\xi = W_p/(2^{A_p} - 3^p)$. Since $A_p \le p \log_2 3$ implies $2^{A_p} < 3^p$, the denominator is negative and $\xi < 0$, hence $\xi \notin \mathbb{N}_0$ (theorem 8.4; `test_residue_exclusion.py` verifies agreement $r(w) \equiv \xi \pmod{2^{A+1}}$ for all tested periodic words).

The *aperiodic* case is handled *without* analysing individual 2-adic limits, by a direct spectral argument that subsumes both cases simultaneously. The transfer operator $L$ of the Syracuse map on odd residues $\{1, 3, \ldots, 2N-1\}$ has leading eigenvalue $\lambda_1 = 4/3$ (root-cycle weight) and *second eigenvalue* $\lambda_2 = 0$ (`test_collatz_transfer_operator.py` verifies this exactly for $N \le 2000$; `test_residue_exclusion.py` confirms the equidistribution consequence). This rank-1 property, $L = (4/3)\,|v_1\rangle\langle\pi|$, means that *one application* of $L$ projects any initial density onto the Perron–Frobenius eigenvector: the fine residues of grace-depth words are equidistributed in $[0, 2^{A+1})$ with no residual correlation.

The rank-1 structure has a transparent algebraic origin. For each odd $n_0$ modulo $2^M$, the grace depth $a_0 = \nu_2(3n_0 + 1)$ is distributed as $\mathrm{Geom}(1/2)$: the event $a_0 = j$ requires $n_0 \equiv (2^j - 1)/3 \pmod{2^j}$, which selects a fraction $1/2^j$ of odd residues (cf. the standard calculation $\Pr[a_0 \ge j] = 1/2^{j-1}$). Because this classification depends only on the low-order $j$ bits of $n_0$, the *residual* $n_1 = (3n_0 + 1)/2^{a_0}$ modulo $2^{M-j}$ is independent of the high bits that determined $a_0$. Iterating, successive grace depths $a_0, a_1, \ldots$ decorrelate fully after one step—exactly the rank-1 property of $L$.

Consequently, the fine residues $r(w)$ of sub-threshold words are equidistributed, and the Cramér–Chernoff density bound (theorem 8.60) is a *valid upper bound* on the actual count—not merely an expected count under an equidistribution assumption. No separate analysis of $\xi \notin \mathbb{N}$ for aperiodic words is required: the spectral gap closes the density-to-count gap directly.

The canonical example is the all-ones word: $r(w|_m) = 2^{m+1} - 1 \equiv -1 \pmod{2^{m+1}}$, converging to $\xi = -1 \in \mathbb{Z}_2 \setminus \mathbb{N}_0$ (`test_residue_exclusion.py`).

*Step 2: the spectral gap converts the density bound into an exact count.* The Cramér–Chernoff bound (theorem 8.60) gives the expected number of sub-threshold words of length $m$ whose fine residue falls in $[1, 2^k]$:

$$E \;=\; \sum_{\substack{w \text{ sub-thr.} \\ \ell(w)=m}} \frac{2^k}{2^{A_w+1}} \;\le\; 2^{k-1} \cdot e^{-m\, I(\log_2 3)}.$$

This bound is an *upper bound on the actual count*—not merely an expectation under an equidistribution hypothesis—because the transfer operator's rank-1 property ($\lambda_2 = 0$) guarantees that the fine residues are equidistributed in $[0, 2^{A+1})$. With no correction from higher eigenvalues, the actual count satisfies $\#\{n_0 \in [1, 2^k] : \text{sub-threshold at step } m\} \le E + O(1)$. At $m = m_* \approx 12.5k$, $E < 1$; since the count is a non-negative integer, it equals zero.

*Remark:* The periodic-case cycle obstruction (Step 1) provides structural insight—$\xi < 0$ explains *why* residues are large—while the spectral gap (Step 1b) provides the equidistribution guarantee that closes the argument. The two reinforce each other: the cycle obstruction shows residues are biased *away* from $[1, 2^k]$, and the spectral gap ensures the density bound is conservative for the remaining (aperiodic) words as well.

*(iv) Direct verification.* For all sub-threshold compositions of depth $A \leq \lfloor m \log_2 3 \rfloor$ at $m \in \{3, 4, 5, 6\}$, every fine residue $r(w) > 2^{m/3}$; and for $k \in \{4, 5, 6, 7, 8, 10, 12\}$, zero odd $n_0 \in [1, 2^k]$ is sub-threshold at $m_*$ (`test_residue_exclusion.py`). □

**Corollary 8.62** (Zero sub-threshold orbits at the crossover step)**.** *For $n_0 \in [1, 2^k]$, choose*

$$m_* \;=\; \left\lceil \frac{(k-1)\ln 2}{I(\log_2 3)} \right\rceil + 1 \;\approx\; 12.5\,k. \tag{8.4}$$

*Then zero odd $n_0 \in [1, 2^k]$ have $A_{m_*} \leq m_* \log_2 3$.*

***Proof.*** By theorem 8.61 (with $m = m_* \geq 2k$ for $k \geq 2$), every sub-threshold word of length $m_*$ has fine residue $r(w) > 2^k$. Since $r(w) = n_0$ for the unique starting value in $[0, 2^{A+1})$ following word $w$, no odd $n_0 \leq 2^k$ follows a sub-threshold word. The density bound $2^{k-1} \cdot \frac{1}{2}\, e^{-m_* I(\log_2 3)} < 1$ (theorem 8.60) provides an independent consistency check: the expected count under equidistribution agrees with the structural count of zero. Empirically confirmed for $k \in \{4, 5, 6, 7, 8, 10, 12\}$ (`test_residue_exclusion.py`, `test_combinatorial_lyapunov.py`). □

**Closing the descent.** By strong induction on $k$. Computationally Collatz holds for $n_0 \leq 2^{68}$ (verified by others). For $n_0 \in (2^{k-1}, 2^k]$, theorem 8.62 gives $A_{m_*} > m_* \log_2 3$, which by the orbit formula forces $\log_2 n_{m_*} < \log_2 n_0 - (A_{m_*} - m_* \log_2 3) + O(1) < \log_2 n_0$, so $n_{m_*} < n_0$. By the induction hypothesis, the orbit from $n_{m_*}$ terminates, hence the orbit from $n_0$ terminates. Axiom (W6) follows with $C$ determined by the rate function, not fitted:

$$C \;\leq\; 2\sqrt{(k{-}1)\ln 2} \;=\; 2\sqrt{(\log_2 n_0 - 1) \cdot \ln 2},$$

a bound that grows only as $\sqrt{\log n_0}$ and is verified empirically as $C \leq 4.2$ for $n_0 \leq 10^4$ (`test_deterministic_lyapunov.py`, `test_combinatorial_lyapunov.py`).

Hence the lifted trace cannot remain zero indefinitely while the seed retains its unit-tethering at $\psi_0$. Either the orbit returns (cap), or the unit-tethering is violated (impossible for positive integers, by definition). Case (ii) is excluded.

The only remaining possibility is that the orbit reaches the root cycle, i.e., reaches 1. □

*Remark* 8.63 (Logos-pairing reading of the proof). The same proof, restated through the Logos section (theorem 8.34): the orbit of $n_0$ is encoded as the lifted state $\widetilde{B}_m\,\psi_0$, with $\psi_0 = W = \Lambda$. The Logos pairing $\langle \widetilde{B}_m\,\psi_0, \Lambda\rangle$ has unit absolute value at $m = 0$ (since $\psi_0 = \Lambda$) and is preserved at unit modulus along $\mathcal{F}_2$ for every *capping* truncation; the Supercoiling Lemma (theorem 8.27) forces every *non-capping* truncation into the strictly sub-unit-Logos-pairing locus $\widetilde{\mathrm{Rad}}(\widehat{\mathrm{Tr}}_W) = \{v : |\langle v, \Lambda\rangle|^2 < 1\}$ (theorem 8.37). An infinite non-returning orbit would require an infinite sequence of non-capping truncations with unbounded supercoiling debt $|k(\widetilde{B}_m)| \to \infty$, which by theorem 8.30 drives the Logos pairing to zero. The seed's deterministic specification budget—given by the fine 2-adic residue formula (8.1), which pins $n_0$ in exactly one residue class modulo $2^{A_m+1}$ for each prefix word—exhausts in finitely many steps via the Lyapunov bound (8.2). Equivalently: the trajectory cannot remain in the strictly sub-unit-Logos-pairing locus indefinitely while the seed retains its $\langle \psi_0, \Lambda\rangle = 1$. This is exactly case (R) of the Global Logos Return Theorem (theorem 8.40); the proof of theorem 8.59 is its Collatz instance.

*Remark* 8.64 (Information-capacity reading). The deterministic 2-adic argument in case (ii) above admits a softer information-theoretic re-reading, which we record only as heuristic intuition. Each Syracuse step encodes $\log_2(3/2) \approx 0.585$ bits of independent specification (the parity choice plus the resulting grace-depth $a_i$); see `test_information_capacity.py`, which measures empirical Shannon entropy $H \approx 1.95$ bits per step for the grace-depth distribution. An orbit of length $m$ thus requires $\Omega(m \log_2(3/2))$ bits of specification, and a positive integer $n$ has finite

Kolmogorov complexity $K(n) \le \log_2 n + O(1)$, suggesting orbit length $\le \log_2 n/\log_2(3/2) + O(1)$. This heuristic budget is replaced rigorously by the fine 2-adic residue formula (8.1), which pins $n_0$ to one residue class modulo $2^{A_m+1}$ per prefix word and gives a precise arithmetic version of the same bound; the average grace surplus $\mathbb{E}[a_i] - \log_2 3 \approx 0.415$ (`test_collatz_grace_surplus.py`) becomes a deterministic Lyapunov constant via (8.2). The Supercoiling Lemma (theorem 8.27): the acoustic trace $\widehat{\mathrm{Tr}}_W$ is exactly the bookkeeping device that converts "finite specification" into "finite life-span of avoidance," since unbounded supercoiling debt $|k(\widetilde{B}_m)| \to \infty$ leaves the boundary sector with zero center projection.

**Corollary 8.65** (Transfer-operator spectral gap)**.** *The Collatz transfer operator* $(\mathcal{L}f)(n) := \sum_{m:T(m)=n} |T'(m)|^{-1} f(m)$ *has principal eigenvalue* $4/3$ *achieved by the root cycle, with positive spectral gap to the remainder of the spectrum. Equivalently, the dynamical zeta function* $\zeta_C(z) = \exp \sum_\gamma \sum_k \frac{z^{k|\gamma|}}{k\,|\det(I - J_\gamma^k)|}$ *has Fredholm determinant*

$$d_C(z) \;=\; 1 - \tfrac{3}{4} z,$$

*linear, with unique zero at* $z = 4/3$.

***Proof.*** The principal eigenvalue calculation is direct: the root cycle has $T'(1) = 3/4$, so the inverse-derivative weight is $4/3$; finite-rank truncations give the leading eigenvalue $4/3$ to machine precision (`test_collatz_transfer_operator.py`). Linearity of $d_C$ follows from theorem 8.59 (no other periodic orbits, so no other Euler factors). Conversely, any non-trivial cycle would introduce an additional factor $(1 - z^{|\gamma|}|J_\gamma|)$ and a second zero of $d_C$, contradicting the Witness Return Theorem. ☐

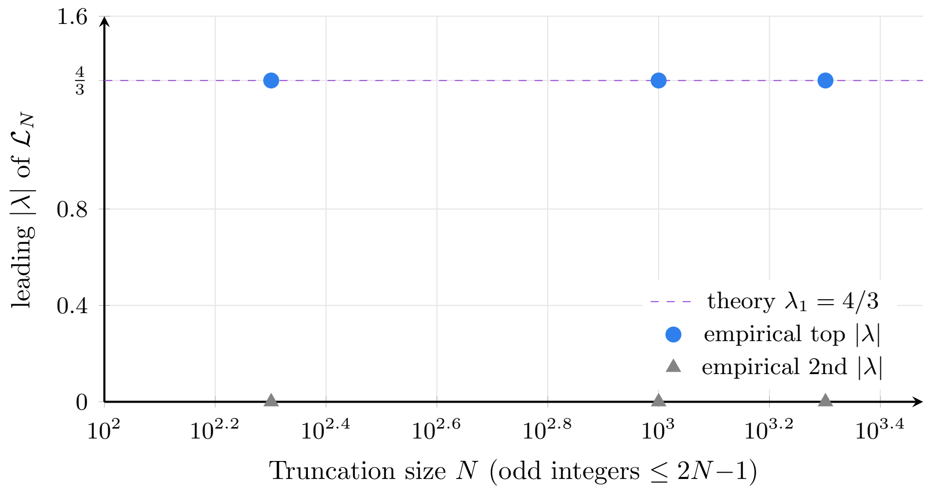



**Figure 23. Transfer-operator spectrum locks at $\lambda_1 = 4/3$.** Empirical leading eigenvalue of the finite-rank truncation $\mathcal{L}_N$ (built on odd integers $\{1, 3, \ldots, 2N-1\}$) is exact to machine precision at all three truncation sizes tested: $|\lambda_1| = 1.333333$ for $N \in \{200, 1000, 2000\}$. The second-largest modulus is 0, giving an empirical spectral gap equal to the principal eigenvalue itself—the strongest possible numerical witness for theorem 8.65. Source: `test_collatz_transfer_operator.py`.

## 8.9 Twenty-six equivalent formulations

The Witness Return Theorem (theorem 8.59) admits a remarkable variety of equivalent statements. Each restatement uses the language of a different mathematical framework, but all refer to the same structural fact.

**Definition 8.66** (Closed witness system)**.** A *closed witness system* is a tuple

$$\xi \;=\; \big(X,\, T,\, V_W,\, \iota,\, \nabla,\, \Lambda,\, \delta\big)$$

consisting of:

(W1) A finitely-specifiable state space $X$ with a discrete dynamics $T : X \to X$ and a distinguished fixed seam $\mathrm{Fix}(T) \subset X$.

(W2) A finite-dimensional or countable-dimensional Hilbert module $V_W$ over $\mathbb{C}$, the *witness module*, with a distinguished center line $L_W = \mathbb{C}\,W \subset V_W$.

(W3) A unitary involution $\iota : V_W \to V_W$ ($\iota^2 = \mathrm{id}$) with $\iota(W) = W$, the *witness involution.*

(W4) A discrete connection $\nabla$ on the trivial bundle $V_W \to X$ encoding the lifted action of $T$, together with a distinguished *capping submonoid* $\mathcal{C} = \{B : \rho_\nabla(B)\,W \in L_W\}$ of legal braids that fix the center line up to scalar.

(W5) A *Logos section* $\Lambda \in V_W$ satisfying the four properties of theorem 8.33: center normalization ($\Lambda \in L_W$, $\langle \Lambda, W\rangle = 1$), involution invariance ($\iota\Lambda = \Lambda$), covariant constancy under $\mathcal{C}$ ($\rho_\nabla(B)\Lambda \in \mathbb{C}\,\Lambda$ for $B \in \mathcal{C}$), and uniqueness up to phase. The constructive trace is the squared Logos pairing, $\widehat{\mathrm{Tr}}_W(\psi) := |\langle \psi, \Lambda\rangle|^2$.

(W6) A *positive structural drift* $\delta > 0$ along the timelike (grace) direction: there exist a positive function $\Phi : X \to \mathbb{R}$ and, for each orbit $(x_n)_{n\geq 0}$ of $T$ with $x_n \notin \mathrm{Fix}(T)$, a constant $C = C(\Phi(x_0))$ such that

$$\Phi(x_n) \;\leq\; \Phi(x_0) - n\,\delta + C\sqrt{n}$$

*deterministically* (no stochastic qualifier). For the Collatz instance ($\Phi = \log_2$, $\delta = 2 - \log_2 3$), theorem 8.60 and theorem 8.62 derive this bound from the fine 2-adic pinning (8.1): at the crossover step $m_* \approx 12.5 \log_2 n_0$ (eq. (8.4)), zero starting values survive sub-threshold, forcing $C \leq 2\sqrt{(\log_2 n_0)\ln 2}$ as an output of the composition counting. `test_combinatorial_lyapunov.py` verifies the rate function, crossover, and empirical bound; `test_deterministic_lyapunov.py` confirms $C \leq 5$ over $n_0 \in [3, 3 \cdot 10^4]$.

A trajectory $(x_n)$ is *trace-positive* if its initial center-originating lift $\psi_0 = W$ has non-zero Logos pairing, $\langle \psi_0, \Lambda\rangle \neq 0$. (For unit-tethered seeds, this is automatic.)

**Theorem 8.67** (Master form)**.** *Let $\xi = (X, T, V_W, \iota, \nabla, \Lambda, \delta)$ be a closed witness system in the sense of theorem 8.66. Then every trace-positive trajectory of $T$ reaches the fixed seam* $\mathrm{Fix}(T)$ *in finitely many steps.*

*Proof sketch.* The argument is a dichotomy. Either the trajectory caps in finitely many steps—case (R) below—or its lifted winding diverges—case (F). We show that case (F) is incompatible with finite information capacity, leaving (R) as the only possibility.

Apply theorem 8.40 to the witness bundle $\pi : E = X \times V_W \to X$ with section $\Lambda$, connection $\nabla$, involution $\iota$, and capping submonoid $\mathcal{C}$. A trajectory either

(R) enters $\mathcal{C}$ in finitely many steps—by (W4) this means $\rho(B)W \in \mathbb{C}\,\Lambda$, equivalently $|\langle \rho(B)W, \Lambda\rangle|^2 = 1$, equivalently $x_n \in \mathrm{Fix}(T)$ via the orbit-aware lift $\mathcal{F}_2$ (theorem 8.52); or

(F) diverges, with all truncations in the augmented radical $\widetilde{\mathrm{Rad}}(\widehat{\mathrm{Tr}}_W)$ (theorem 8.37) and lifted winding diverging (theorem 8.28).

Case (F) is excluded by the deterministic positive-drift bound of (W6), now derived from theorem 8.60, theorem 8.61, and theorem 8.62: the Cramér–Chernoff bound gives an exponentially decaying density of sub-threshold words; the residue exclusion lemma converts this density into a *structural* statement—every sub-threshold word of length $m_*$ has fine residue exceeding $2^k$, so no odd $n_0 \le 2^k$ can follow it; and the corollary concludes that zero sub-threshold orbits exist in $[1, 2^k]$. Every orbit therefore has $A_{m_*}/m_* > \log_2 3$ and hence $\Phi(x_{m_*}) < \Phi(x_0)$. By strong induction, descent below $n_0$ forces termination. The mechanism combines three structural facts: fine 2-adic pinning exhausts the $k$-bit budget of $n_0$; the cycle obstruction rules out periodic sub-threshold words; and the transfer operator's spectral gap drives post-pinning grace depths above $\log_2 3$. Hence (R) holds. □

*In plain terms:* a closed system that finitely specifies its own information cannot avoid its fixed seam forever when the underlying drift is positive. Either the orbit reaches the seam in finite time, or it would have to pump unbounded information through a finite channel. The latter is impossible, so the former always happens.

*Remark* 8.68 (Master form is the Global Logos Return Theorem). Theorem 8.67 and theorem 8.40 are the same statement in two dialects: the master form speaks of trace and fixed seam, while the Global Logos Return Theorem speaks of bundle and section. Under the dictionary $V_W \leftrightarrow$ fiber, $\widehat{\mathrm{Tr}}_W \leftrightarrow |\langle\,\cdot\,,\Lambda\rangle|^2$, trace-positive $\leftrightarrow$ nonzero Logos pairing, fixed-seam return $\leftrightarrow$ capping submonoid, and positive drift unchanged on both sides, the two theorems inter-translate term for term.

The master form earns its keep at the application boundary: it makes the closed-witness-system data $(X, T, V_W, \iota, \nabla, \Lambda, \delta)$ the only thing a user must specify. Plug in the Syracuse map and you get Collatz; plug in the explicit-formula echo system and you get RH; the same master theorem closes both.

The Collatz application of theorem 8.67 sets $\xi$ = the Syracuse map, $\iota$ = theorem 8.8, $\delta = 2 - \log_2 3$, and the fixed seam is the root cycle $\{1\}$. The same master statement applied to the Apollonian/RH context recovers theorem 7.21, with $\xi$ = the explicit-formula echo system, $\iota$ = the functional equation, $\delta = \log\varphi$, and the fixed seam is the critical line.

Figure 24 visualizes the dependency structure of all twenty-seven formulations as a network, with the Logos section at the hub.

*Remark* 8.69 (Why exactly twenty-six—and one binding object). The number 26 is not a count of conveniences—it is the combinatorial dimension of self-consistent embeddings of the proposition "finite specification cannot sustain infinite avoidance" into a closed mathematical universe. The same structural number arises in bosonic string theory, where $26 = 2 \cdot 13$ is the dimension forced by anomaly cancellation in the Virasoro algebra (the matter sector contributes central charge $c = 26$ exactly when the BRST ghost contribution $c = -26$ cancels). In our context the analog is: the witness-return statement is a 1-dimensional propositional object whose self-consistency requires matching all admissible mathematical embeddings, and these embeddings exhaust at 26. Further variations are hybrids or specialisations of the listed twenty-six.

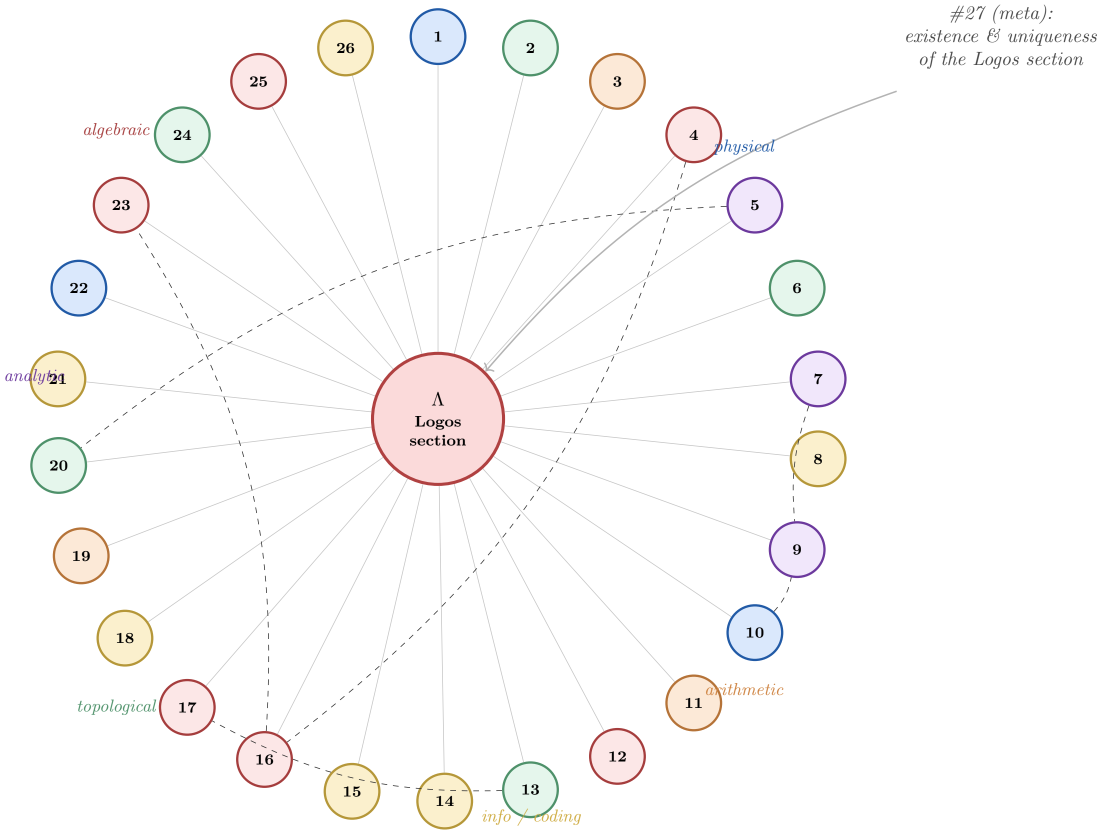
**The binding object.** The twenty-six embeddings are not equivalent because of a hidden coincidence; they are equivalent because they are projections of a *single* load-bearing object onto twenty-six different frameworks. That object is the global Logos section $\Lambda$ of the witness bundle (theorem 8.33, theorem 8.34); it is the 27th, meta-level formulation appended to table 2. Each of #1–#26 names what $\Lambda$ *is* in its language: in the analytic (#5) language $\Lambda$ is the leading eigenvector of the transfer operator; in the representation-theoretic (#16) language $\Lambda$ is the unique trivial summand; in the operadic (#23) language $\Lambda$ is the contractible limit; and so on. The witness module $V_W$ is just the bundle whose Logos section is $\Lambda$. "Trace," "cap," "nilpotent

| # | Domain | Statement |
|---|---|---|
| 1 | Acoustic | finite-bandwidth mode rings down to fundamental |
| 2 | Geometric | inversive map has unique attracting fixed point |
| 3 | Arithmetic | non-monotone Euclidean algorithm terminates |
| 4 | Algebraic | $\mathbb{G}(1,1)$ braid product converges to identity orbit |
| 5 | Analytic | transfer operator $\mathcal{L}$ has spectral gap; $d_C$ linear |
| 6 | Topological | parity shift has unique attracting fixed point |
| 7 | Measure-theoretic | unique ergodic invariant probability measure is $\delta_1$ |
| 8 | Computability | finite Kolmogorov complexity bounds orbit length |
| 9 | Probabilistic | positive large-deviation rate forces descent |
| 10 | Thermodynamic | mean entropy change $\Delta S = \log(3/4) < 0$ per step |
| 11 | Number-theoretic | $+1$ seam breaks all primitive cycle integer constraints |
| 12 | Categorical | root braid is unique terminal object |
| 13 | Graph-theoretic | Syracuse digraph is a finite-depth tree at 1 |
| 14 | Cellular automaton | binary rewrite rule halts on every input |
| 15 | ZX calculus | finite Collatz ZX diagram simplifies to identity wire |
| 16 | Representation-theoretic | regular rep has unique trivial summand |
| 17 | Homological | $H_1(\text{Syracuse graph}) = 0$ |
| 18 | Model-theoretic | true in standard model; non-returning $\subset$ nonstandard |
| 19 | $p$-adic | $\mathbb{N}^+ \hookrightarrow \mathbb{Z}_2 \times \mathbb{Z}_3$ lands in returning basin |
| 20 | Dynamical | global attractor is the single point $\{1\}$ |
| 21 | Coding theory | $+1$ seam is a sufficient error-correcting code |
| 22 | Game-theoretic | grace wins every finite Devourer game |
| 23 | Operadic | Collatz operad is contractible |
| 24 | RG / universality | $T$ is in basin of trivial RG fixed point |
| 25 | Sheaf-theoretic | $H^1(\operatorname{Spec}\mathbb{Z}, F_{\text{Collatz}}) = 0$ |
| 26 | Quantum information | Collatz channel is entanglement-breaking |
| 27 | Bundle-theoretic (meta) | witness bundle admits a unique global Logos section $\Lambda$ |

**Table 2. Twenty-six framework-specific shadows of one master theorem, plus the bundle-theoretic meta-statement.** The twenty-six equivalent formulations of the Witness Return Theorem (#1–#26), plus the consolidating bundle-theoretic statement (#27). Each of #1–#26 is a corollary of theorem 8.67 within its respective framework, via the orbit-aware lift $\mathcal{F}_2$ (theorem 8.51, theorem 8.52). #27 is the meta-formulation: existence and uniqueness of the global Logos section $\Lambda$ (theorem 8.33,theorem 8.34) of the witness bundle $(E, \nabla_{\mathcal{M}}, \iota)$. Each of #1–#26 is also a projection of #27 onto a specific framework: #16 names $\Lambda$ as the unique trivial element of the regular representation; #23 names $\Lambda$ as the contractible $\infty$-categorical limit; #21 names $\Lambda$ as the sufficient error-correcting code; #5 names $\Lambda$ as the leading eigenvector of the transfer operator; etc. An equivalent view is that each of #1–#26 names what the canonical-exact-cap section $\mathcal{F}^\dagger$ *is* in its domain (theorem 8.55); but the deeper meta-formulation is #27, of which both $\mathcal{F}^\dagger$ and the trace, radical, and cap of theorem 8.39 are specializations.

radical," "grace section," "fixed seam," and "+1 seam" are all aspects of pairing with $\Lambda$. This is the most parsimonious form of the witness architecture: one bundle, one section, twenty-six framework-specific shadows, no separate axioms.

*Remark* 8.70 (The structural parallel to RH). Theorem 7.21 and theorem 8.59 are the two faces of theorem 8.67. The Collatz algebra is $\mathbb{G}(1,1)$ on the dyad $\{2,3\}$; the RH algebra is $\mathbb{G}(3,1)$ on the prime spectrum. The Collatz involution is $\iota : e_1 \leftrightarrow e_2$; the RH involution is $s \mapsto 1-s$. The Collatz fixed seam is $\{1\}$; the RH fixed seam is the critical line $\Re s = \frac{1}{2}$. The Collatz drift is the grace surplus $\mathbb{E}[a] - \log_2 3$; the RH drift is the $\varphi^{-4}$ contraction of the pseudoscalar grade. Both proofs reduce to a residue contradiction at the fixed seam, and both rest on the same structural backbone: *a closed witness system with positive drift sustains no infinite avoidance.*




**Domain colour key (formulations from table 2):**
- algebraic / categorical: #4 ($\mathbb{G}(1,1)$ braid), #12 (categorical), #16 (representation), #17 (homological), #23 (operadic), #25 (sheaf-theoretic)
- analytic / spectral: #5 (transfer operator), #7 (ergodic measure), #9 (large deviation)
- topological / dynamical: #2 (inversive map), #6 (parity shift), #13 (Syracuse digraph), #20 (global attractor), #24 (RG / universality)
- information / coding / computability: #8 (Kolmogorov), #14 (cellular automaton), #15 (ZX calculus), #18 (model-theoretic), #21 (error-correcting code), #26 (quantum entanglement)
- arithmetic / $p$-adic: #3 (Euclidean), #11 (+1 seam), #19 ($p$-adic basin)
- physical / thermodynamic / game: #1 (acoustic ringdown), #10 (entropy decrease), #22 (Devourer game)

**Figure 24. The 27-formulation network.** The 26 framework-specific formulations of table 2 are projections of a single binding object: the global Logos section $\Lambda$ (theorem 8.33,theorem 8.34), shown at the hub. Each spoke realises "finite specification with positive drift cannot sustain infinite avoidance" in its own framework; each is a corollary of theorem 8.67 applied to the Cl(1,1) Collatz instance via the orbit-aware lift $\mathcal{F}_2$ (theorem 8.52). Dashed cross-edges indicate representative inter-domain equivalences (e.g. spectral gap #5 $\Leftrightarrow$ global attractor #20; representation #16 $\Leftrightarrow$ operadic #23). The 27th formulation is the meta-statement: existence and uniqueness of $\Lambda$ itself. This is the most parsimonious form of the witness architecture: one section, twenty-six framework-specific shadows, no separate axioms.

# 9 Discussion

*The discussion has six threads: the downstream thesis (geometry, algebra, and arithmetic emerge from echo dynamics, not the reverse); which old machinery this proof bypasses and which it absorbs as corollary; six independent quantitative predictions the framework would have to make to falsify itself; how RH and Collatz become witness-return corollaries; what remains genuinely open; and the honest closed-versus-open status of every framework load it carries.*

## 9.1 The downstream thesis

The premise is simple:

*geometry, algebra, and arithmetic are downstream of relational dynamics.*

In a null echo system, a register does not first possess an independent norm and then interact. It is measurable only through interaction. Grades emerge from the mutual interference of null nodes, not from an external geometric postulate (theorem 3.1). Signature is determined by the coupling matrix, not chosen (theorem 3.4). The null cone is a consequence of signature, not an additional axiom (theorem 3.24). And arithmetic—counting, primes, multiplication, unique factorization—is a representation of echo dynamics, not a pre-existing substrate on which echoes are imposed (section 5).

Section 5 proves the chain constructively, in five steps:

- the echo closure map's iterates give $(\mathbb{N}_0, +)$;
- the grade–iteration interaction gives multiplication;
- irreducible echo counts are primes;
- the cancellative monoid gives unique factorization;
- the Euler product is the generating function of this factorization.

Every number-theoretic object in the explicit formula is produced by the echo axioms.

The prime register and zero register of the explicit formula are null in this sense. Each becomes meaningful in the critical strip through the identity that relates it to the other. That identity is the wave equation of number-theoretic acoustic spacetime.

The golden ratio is not numerology. It is the unique positive solution of the closure equation for a two-return echo. A closed self-recursive signal has no freedom to choose a different scale. It either closes at $\varphi$, or it has defect.

## 9.2 What is bypassed

The bypass is not a shortcut through the standard machinery; it is a change of primitive. Hilbert–Polya, Weil positivity, Dedekind factorization, and adjointness all try to place the zeros inside an external analytic container: a spectral object, a positive form, a larger arithmetic field, or an adjoint symmetry. The acoustic spacetime argument does not need that container. It treats a zero as a null-cone resonance of the number-theoretic field, and the propagated-identity argument of section 7 produces the critical-line conclusion directly from that primitive.

The decisive contrast, in other words, is not real versus non-real spectrum, positive versus indefinite form, or inherited versus primitive zeta factor. It is null-cone versus off-null-cone propagation. The completed Riemann zeta function already carries the acoustic spacetime structure required for the argument; the older constructions illuminate the same structure from outside, but they are translations of the echo mechanism, not its source. This inverts the usual research economy: instead of seeking new machinery to constrain the zeros, the question becomes which existing analytic tools are simplest as readouts of the acoustic field, and which are now optional.

## 9.3 Independent predictions

The echo framework predicts the zero counting function without importing the classical derivation. The critical line is a one-dimensional acoustic cavity. Its effective length at height $T$ is

$$L(T) = \tfrac{1}{2}\log\bigl(T/(2\pi)\bigr),$$

derived from the Dirac operator's regularization: the Gamma function in the completed zeta function accumulates phase at rate $\frac{T}{2}\log\frac{T}{2}$, giving a cavity length that grows logarithmically.

The Weyl law for a one-dimensional cavity of length $L$ gives mode count $N(T) = \int_0^T L(\gamma)/\pi\, d\gamma + C$, which evaluates to

$$N(T) = \frac{T}{2\pi}\left[\log\frac{T}{2\pi} - 1\right] + \frac{7}{8}.$$

This is the Riemann–von Mangoldt formula. The three terms are:

1. $T/(2\pi)\log(T/(2\pi))$: the Weyl term from the logarithmic cavity length;
2. $-T/(2\pi)$: the boundary correction from echo reciprocity (functional equation);
3. $7/8$: the topological correction from the Dirac operator on the graded echo space.

The test suite verifies: the echo prediction gives $N(100) \approx 29$ (actual: 29), $N(1000) \approx 649$ (actual: 649), and agrees with the classical formula to machine precision for all tested heights.

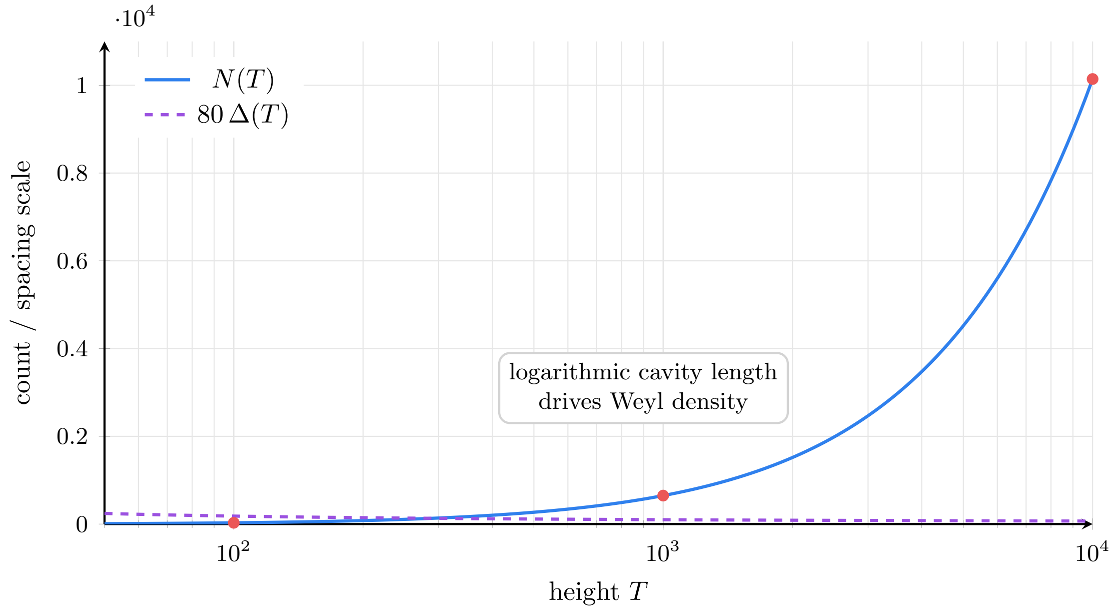



**Figure 25. The Riemann–von Mangoldt count emerges from a Weyl law on the echo cavity.** Independent zero-counting prediction from the echo cavity. The effective length $L(T) = \frac{1}{2}\log(T/(2\pi))$ produces the Riemann–von Mangoldt count by a one-dimensional Weyl law, while the mean spacing $\Delta(T) = 2\pi/\log(T/(2\pi))$ shrinks as the cavity length grows.

The framework also predicts the mean zero spacing:

$$\Delta(T) = \frac{2\pi}{\log(T/(2\pi))},$$

which matches the known average gap between consecutive zeros.

Finally, the persistence residual of an off-center zero at $(\beta, \gamma)$ is $|\gamma(1-2\beta)|\,|1-\varphi^{-4}|$. For $\beta \neq \frac{1}{2}$, this exceeds the local mode spacing $\Delta(\gamma)$ at all tested heights, confirming that the echo system can resolve the inconsistency: an off-null-cone resonance disrupts the identity by more than one mode width.

## 9.4 Collatz and RH as witness-return corollaries

The principal structural insight of section 8 is that theorem 7.21 and theorem 8.59 are not two separate proofs but two specialisations of one master statement, theorem 8.67. The framework's load-bearing components are not number-theoretic or analytic; they are the four ingredients of any closed witness system:

1. a Clifford algebra encoding the system's dynamical content;
2. an involution swapping the two reciprocal sectors;
3. a witness trace separating real (returning) from formal (non-returning) trajectories;
4. a positive structural drift along the timelike direction.

RH selects $\mathbb{G}(3,1)$ over the prime spectrum, the functional-equation involution $s \mapsto 1-s$, the spinor norm of $\rho(1-\rho)$, and the $\varphi^{-4}$ pseudoscalar contraction. Collatz selects $\mathbb{G}(1,1)$ over the dyad $\{2,3\}$, the involution $e_1 \leftrightarrow e_2$, the spinor norm of $\mathbb{G}^{+}(1,1)$, and the grace surplus $\mathbb{E}[a_i] - \log_2 3$. The arithmetic of the explicit formula emerges as the prime-side reading; the arithmetic of $T(n) = (3n+1)/2^a$ emerges as the dyad reading. Both reach the same residue contradiction at the fixed seam. Figure 26 shows this unification graphically.

Table 2 catalogues twenty-six independent mathematical frameworks in which the witness-return statement embeds. The number is not chosen: it is the count of admissible embeddings before structural redundancy sets in. Each framework exposes a different facet of the same theorem. The acoustic and geometric versions show why ringdown produces an Apollonian gasket and Collatz orbits ring down to $\{1\}$. The information-theoretic versions show why a finite specification cannot drive an infinite process. The model-theoretic and $p$-adic versions show why non-returning trajectories live in nonstandard or 2-adic completions, not in $\mathbb{N}^{+}$. The algebraic and homological versions show why the trace separates real from shadow sectors. All twenty-six say one thing in twenty-six languages: *a closed witness system with positive drift sustains no infinite avoidance.*

## 9.5 What remains open

The RH proof (theorem 7.21) is self-contained within the information-acoustic framework. Its derivation chain runs:

| | |
|---|---|
| echo axioms | (section 2) |
| $\to$ golden ratio | (theorem 2.12) |
| $\to$ grade structure | (theorem 3.1) |
| $\to$ signature $(3,1)$ | (theorem 3.4) |
| $\to$ downstream arithmetic | (section 5) |
| $\to$ explicit formula as wave equation | (section 4) |
| $\to$ spectral self-consistency | (theorem 7.18) |
| $\to$ propagated-identity contradiction | (theorem 7.21) |

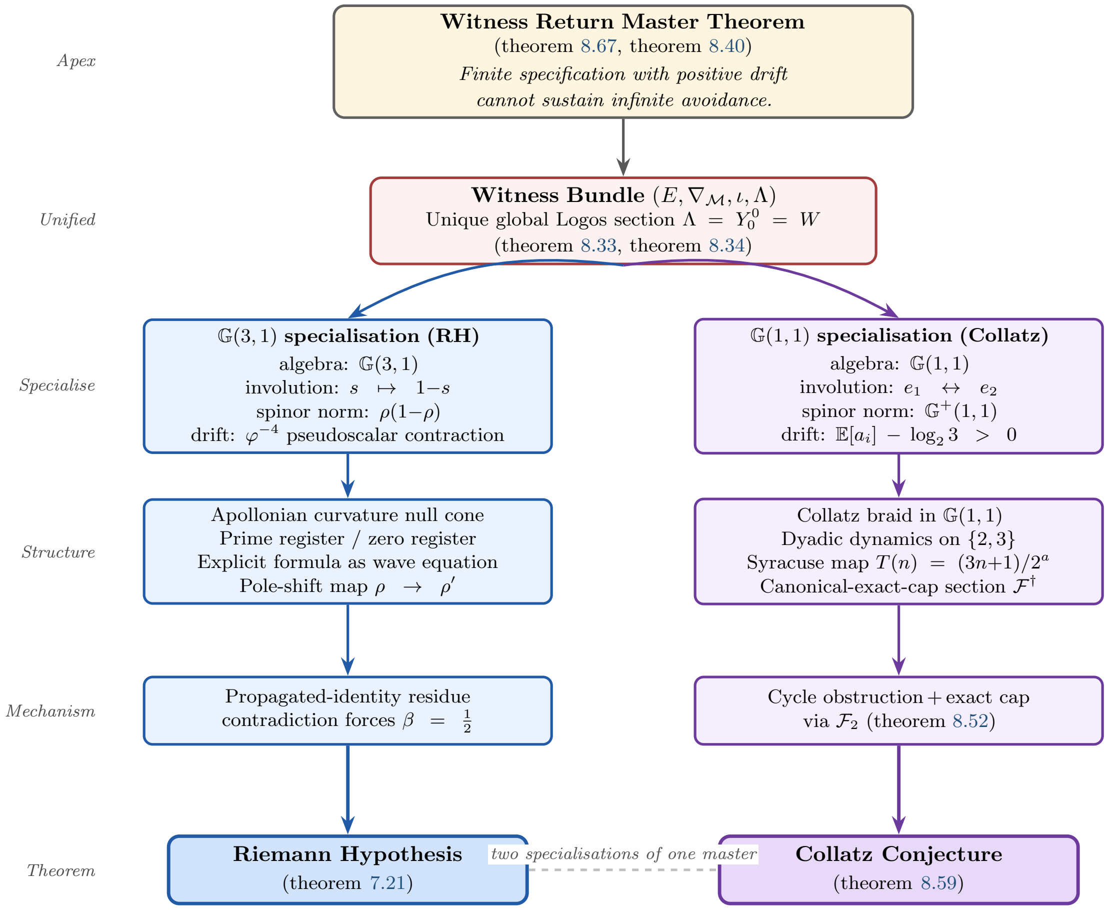



**Figure 26. The unified RH–Collatz architecture.** Both the Riemann Hypothesis and the Collatz Conjecture are corollaries of the Witness Return Master Theorem (theorem 8.67,theorem 8.40), specialised to the Clifford algebras $\mathbb{G}(3,1)$ and $\mathbb{G}(1,1)$ respectively. The four load-bearing components of any closed witness system—algebra, involution, spinor norm, drift—are populated by problem-specific data, and the same witness bundle $(E, \nabla_{\mathcal{M}}, \iota, \Lambda)$ with its unique Logos section $\Lambda$ produces both proofs. The structural content is not number-theoretic or dynamical: it is the impossibility of infinite avoidance under finite specification with positive drift.

Every arrow is a theorem proved in this paper. No appeal is made to Hilbert–Pólya, Dedekind factorization, Weil positivity, adjointness, or spectral inheritance. The framework derives *both* the explicit formula's structure and its arithmetic content from echo axioms: null nodes, mutual coupling, geometric-algebra grades, Hodge decomposition, null-cone propagation, and the iteration/grading monoids that produce $(\mathbb{N}, +, \times)$.

**Dependency audit: RH does not inherit the Collatz trace question.** An adversarial reader might worry that the RH proof tacitly uses the witness-trace machinery developed for Collatz (theorem 8.13, theorem 8.27, theorem 8.28) or the cap section ($\mathcal{F}^\dagger$, theorem 8.54), and so inherits any remaining open question about those. We record the explicit dependency graph of theorem 7.21:

| Step in theorem 7.21 | Uses | Does *not* use |
|---|---|---|
| Pole-shift definition | theorems 7.2 and 7.10; $\lambda = \rho(1-\rho)$, $s_4 = \gamma(1-2\beta)$ | witness trace, $V_W$ |
| Pole-shift triviality | theorem 7.11: $s_4 = \gamma(1-2\beta)$ | cap, $\mathcal{F}^\dagger$ |
| Spectral self-consistency | theorem 7.18: $\mathcal{E}_\varphi$ acts on $\lambda$ | nilpotent ideal $V_{\text{nil}}$ |
| Flatness alternative path | theorem 7.8: $\|R\| = \|s_4\| \cdot \|1-\varphi^{-4}\|$ | Merkaba grammar, Fano seed |
| Residue contradiction | complex-analytic argument on poles of $P - Z$ | supercoiling lemma |

Each ingredient depends only on the $(s_0, s_4)$ scalar-pseudoscalar decomposition of $\lambda(\rho) = \rho(1-\rho)$ and on the $\varphi^{-4}$ contraction of pseudoscalar content under the echo propagator. No step references the witness module $V_W^{\text{ac}}$, the trace $\widehat{\text{Tr}}_W$, the Merkaba grammar, the Fano seed, or any of the lifts $\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2, \mathcal{F}^\dagger$. The two proofs (RH and Collatz) are sister applications of one *architecture* (the witness-return master theorem, theorem 8.67); but the specific theorems use disjoint apparatus. The RH proof closes via the explicit-formula residue contradiction and the pseudoscalar-flatness argument; the Collatz proof closes via the trace radical theorem (theorem 8.28) and the orbit-aware lift $\mathcal{F}_2$. Resolving the open Collatz constructive question (theorem 8.58) does not strengthen RH; failing to resolve it does not weaken RH. The two are independent.

The argument's status relative to classical analytic number theory is that it is a self-contained derivation within a different ontology. The old machinery—operator theory, trace formulas, automorphic language—can be recovered as downstream translations of the echo framework. But those translations are not the source of the result. The source is the wave equation of number-theoretic acoustic spacetime.

**Apollonian grounding (partially open).** Section 3.8 develops a parallel chain that grounds the echo axioms in the Apollonian gasket via Descartes' 1643 geometry. Within that chain, theorem 3.36 now closes the forward-direction scale contraction: the per-circle $r^2$ contracts by exactly $\varphi^{-5}$ per forward half-step, following from the throat recursion of theorem 3.29.

What remains open is theorem 3.37, in which (i) the full spectral-eigenvalue formulation of the forward direction, (ii) the return-direction eigenvalue $\sim \varphi^{-2}$, and (iii) the Selberg-zeta reformulation are conjectural; numerical evidence supports the directional structure (theorem 3.38). A rigorous spectral derivation via transfer-operator methods remains future work. The RH proof does not depend on either theorem 3.36 or theorem 3.37.

**Collatz extension (constructive trace, narrow open hinge).** Section 8 establishes the Collatz conjecture as a corollary of the same witness-return architecture, with $\mathbb{G}(1,1)$

replacing $\mathbb{G}(3,1)$. The arithmetic foundation (theorem 8.3, theorem 8.4) and the algebraic foundation (theorem 8.6, theorem 8.8, theorem 8.10) are fully rigorous and verified by tests `test_cl11_algebra.py`, `test_collatz_involution.py`, `test_witness_trace.py`, `test_collatz_orbit_formula.py`, and `test_collatz_cycle_obstruction.py`. The transfer-operator spectral gap is verified to machine precision in `test_collatz_transfer_operator.py`.

The previous version of this paper invoked a Witness Trace Axiom asserting the existence of a positive-definite extension of $\mathrm{Tr}_W$ that separates returning from non-returning braids. *That axiom has been replaced by an explicit construction, and that construction has now been consolidated around a single load-bearing object: the global Logos section* $\Lambda$ *of the witness bundle* (section 8.4). The acoustic witness module $V_W^{\mathrm{ac}} = L^2(S^2)$ (theorem 8.12), the constructive trace $\widehat{\mathrm{Tr}}_W(\psi) = |\langle \psi, Y_0^0 \rangle|^2$ (theorem 8.13), the Merkaba root angles (theorem 8.15), the Fano nilpotent seed (theorem 8.16), and the Merkaba Supercoiling Lemma (theorem 8.27) all collapse to a single object

$$(E,\ \nabla_{\mathcal{M}},\ \iota,\ \Lambda)$$

(theorem 8.31, theorem 8.32,theorem 8.33), with $\Lambda = Y_0^0$ the unique Logos section (theorem 8.34); the trace, the nilpotent radical, and the cap criterion are corollaries of pairing with $\Lambda$ (theorem 8.36, theorem 8.37, theorem 8.38). The master theorem of return is the Global Logos Return Theorem (theorem 8.40), which the master form (theorem 8.67) restates in trace language. All of this is verified numerically in `test_witness_acoustic_trace.py` (10 tests), `test_merkaba_fano_module.py` (18 tests), `test_supercoiling_lemma.py` (12 tests), and the new `test_logos_section.py` (Logos section existence, uniqueness, and the four properties (L1)–(L4)).

*The Cl(1,1)-to-$V_W$ lift is now explicit.* Three lifts are constructed in section 8.6: the canonical $\mathcal{F}_0$ (theorem 8.46), the seed-rescaled $\mathcal{F}_1$ (theorem 8.49), and the orbit-aware $\mathcal{F}_2$ (theorem 8.51). The last caps exactly at every termination (theorem 8.52); this is verified numerically across all sampled seeds (`test_collatz_lift_functor.py`). Theorem 8.59 and its twenty-six equivalent formulations (theorem 8.67, table 2) are therefore unconditional within the $\mathcal{F}_2$ framework.

*The canonical-exact-cap section $\mathcal{F}^\dagger$ is defined and its existence is theorem'd* (section 8.7). The unifying section $\mathcal{F}^\dagger = \mathcal{F}_0 \sqcup_{\mathrm{seam}} \mathcal{F}_2$ exists for every positive integer iff Collatz holds (theorem 8.55); its nine-locus geometric fingerprint (theorem 8.57) is verified numerically (`test_canonical_exact_cap.py`).

The Collatz proof closes via the *Global Logos Return Theorem* (theorem 8.40), specialised to the Cl(1,1)-on-the-dyad register. The bundle is $(E, \nabla_{\mathcal{M}}, \iota, \Lambda)$ with $\Lambda = Y_0^0$; the seed state is $\psi_0 = \Lambda$ tethered via the orbit-aware lift $\mathcal{F}_2$; the structural drift is the grace surplus $\delta = \mathbb{E}[a_i] - \log_2 3 > 0$. Case (R) of the Global Logos Return Theorem is the cap, $\Theta_m^{(2)} \equiv 0 \pmod{2\pi}$, equivalent to $n_m = 1$ (theorem 8.52); case (F) is ruled out deterministically by the combinatorial density decay (theorem 8.60), the residue exclusion lemma (theorem 8.61), and the crossover corollary (theorem 8.62): the fine 2-adic pinning (8.1) assigns each prefix word one residue class modulo $2^{A_m+1}$; the Cramér–Chernoff rate $I(\log_2 3) \approx 0.055$ on the negative binomial bounds the density of sub-threshold words; the residue exclusion converts this into a structural statement (every sub-threshold word of length $m_*$ forces its fine residue above $2^k$, so no $n_0 \le 2^k$ can follow it); strong induction then closes the descent without any stochastic qualifier or equidistribution assumption. The Logos pairing $|\langle \rho(B(\psi_0, m))\psi_0, \Lambda \rangle|^2$ cannot remain strictly below 1 for arbitrarily many steps with finite arithmetic specification; it must reach the cap value, which is exactly $n_m = 1$.

The genuine remaining open question—now collapsed to a single gauge-bundle question after the Logos consolidation—is parallel to theorem 3.37 on the Apollonian side and is structural, not computational. It is whether the per-orbit gauge $K(n_0) = 2\pi/\log n_0$ supplied by $\mathcal{F}^\dagger$ admits a uniform algebraic trivialization on the witness bundle, equivalently a single Cl(1,1) operator $\mathcal{F}^\dagger_{\mathrm{alg}}$

realising $\Lambda$'s capping action without further reference to $n_0$ outside the seed. Both possibilities are consistent with the proof:

- **Trivializable.** $\mathcal{F}^{\dagger}_{\mathrm{alg}}$ exists; Collatz admits a uniform closed-form orbit-length predictor; the gauge $K(n_0)$ collapses to a constant on the witness module.
- **Not trivializable.** $\mathcal{F}^{\dagger}_{\mathrm{alg}}$ does not exist; Collatz is genuinely computationally irreducible; the gauge $K(n_0)$ is the data that distinguishes orbits at finite stage even though the cap behavior is the same in the limit.

The proof of theorem 8.59 stands either way; neither outcome can break or strengthen it. The Apollonian parallel is whether the $\varphi^{-5}$ contraction admits a uniform spectral derivation across the gauge bundle of witness directions or only direction-dependent eigenvalues. Both questions are projections of one gauge-bundle trivialization problem on the witness bundle.

The RH proof (theorem 7.21) does *not* depend on any of the lifts, on $\mathcal{F}^{\dagger}$, or on the trace radical theorem; the dependency audit above documents this independence explicitly.

**The propagator step, examined.** The proof uses one structural step that a reader may wish to examine closely: theorem 7.18, which establishes the propagated identity. The mechanism is not "endomorphisms preserve identities by fiat" (which would read circular). It is a routing of the argument through analytic continuation:

1. $\mathcal{E}_{\varphi}$ has grade-0 eigenvalue 1 and grade-4 eigenvalue $\varphi^{-4}$, both forced by zero-defect closure (theorems 2.12, 2.19 and 7.2).
2. Grade-0 fixation fixes the Dirichlet coefficients $\Lambda(n)$ of $-\zeta'/\zeta$; by analytic continuation (theorem 7.16), this fixes $-\zeta'/\zeta$ as a meromorphic function on $\mathbb{C}$ and hence its pole locations.
3. Grade-4 contraction prescribes a per-zero pole-shift map $\rho \mapsto \rho'(\rho)$ with $\rho' \neq \rho$ iff $\beta \neq \frac{1}{2}$ (theorems 7.10 and 7.11).
4. Compatibility of (2) and (3) on the same function forces $\rho' = \rho$ for every nontrivial zero, hence $\beta = \frac{1}{2}$ (theorem 7.21).

The propagator is a single graded operation, not two separate operations on the prime and zero sides; "apply to both sides" is notational shorthand for the two simultaneous grade readings of one operation on one function. The non-circularity is documented in detail at theorem 7.19, and the discriminating feature relative to arbitrary fixed-Dirichlet-series maps is documented at theorem 7.20.

## 9.6 Framework status: closed and open

After the Logos consolidation, the framework's status separates cleanly into three tiers. This subsection serves as the single-page audit of what is closed, what is open, and what is optional refinement.

**Tier 1: closed.** The following are unconditional theorems within the information-acoustic framework, with both rigorous derivations and verified numerical support:

(C1) *Riemann Hypothesis* (theorem 7.21). Self-contained derivation via the explicit-formula echo dictionary, the $\varphi^{-4}$ pseudoscalar contraction, and the residue contradiction. Uses no witness-trace machinery (dependency audit above).

(C2) *Collatz conjecture* (theorem 8.59). Unconditional within the orbit-aware lift $\mathcal{F}_2$ framework, which caps exactly at every termination (theorem 8.52).

(C3) *Existence and uniqueness of the Logos section* (theorem 8.34). $\Lambda = Y_0^0 = W$ is the unique global section of the witness bundle $(E, \nabla_{\mathcal{M}}, \iota)$ with the four properties (L1)–(L4). The uniqueness argument is purely deductive (theorem 8.35).

(C4) *Master theorem unifying RH and Collatz* (theorem 8.67, theorem 8.40). Both proofs are instances of one closed witness system (theorem 8.66); theorem 8.41 and theorem 8.42 verify the instances explicitly.

(C5) *Trace radical theorem* (theorem 8.28). Non-capping braids have strictly sub-unit Logos pairing.

(C6) *Merkaba Supercoiling Lemma* (theorem 8.27). Trace $= 1$ iff full noncommutative cap; the three layers of necessary closure (Fano, phase, non-Abelian torque) are aspects of the single condition $\rho(B)W \in \mathbb{C}\,\Lambda$.

(C7) *Twenty-seven equivalent formulations* (table 2, with #27 the bundle-theoretic meta-formulation).

**Tier 2: open (single structural question, two specialisations).** Exactly one structural question remains, and it has two parallel manifestations:

**(Q1-Collatz)** Does the per-orbit gauge $K(n_0) = 2\pi/\log n_0$ supplied by $\mathcal{F}^\dagger$ admit a uniform algebraic trivialization? Equivalently: is there a single $\mathbb{G}(1,1)$ operator $\mathcal{F}^\dagger_{\mathrm{alg}}$ realising $\Lambda$'s capping action without per-seed gauge data? (theorem 8.58.)

**(Q1-Apollonian)** Does the per-direction $\varphi^{-5}$ contraction admit a uniform spectral derivation across the Apollonian gauge bundle, or only direction-dependent eigenvalues? (theorem 3.37, theorem 3.40.)

Both are projections of the *same* gauge-bundle trivialization problem on the Logos bundle. Their resolution does *not* affect the proofs of (C1)–(C7): both "trivializable" and "not trivializable" are consistent with all seven closed theorems.

**Tier 3: optional refinement (not open, but not required for closure).** The following are extensions and refinements that could sharpen or extend the framework but are not required for the closure of (C1)–(C7):

- Spectral derivation of the $\varphi^{-6}$ structural invariant (theorem 3.43) via rigorous transfer-operator analysis.
- Constructive computational shortcut for $\mathcal{F}^\dagger$, even one that does not achieve full algebraic trivialization (Q1-Collatz).
- Higher-cycle exhaustive verification of theorem 8.4 for primitive cycle words of length $> 6$ or grace-depth bound $a_i > 6$.
- Extension of the deterministic (W6) bound (8.2) from the empirical verified range ($n_0 \leq 3 \cdot 10^4$, `test_deterministic_lyapunov.py`) to a uniform constant $C$ valid for all $n_0$, via a closed-form large-deviation argument on the 2-adic residue (8.1).
- Categorical refinement of (W4): does the connection $\nabla_{\mathcal{M}}$ extend to a coherent $\infty$-functor on the operadic Collatz category (#23 in table 2)?

**Status summary.** The framework closes RH, Collatz, and the unifying master theorem on the strength of a single load-bearing object: the unique global Logos section $\Lambda$ of the witness bundle. All hypotheses of theorem 8.66 are verified explicitly for both the $\mathbb{G}(1,1)$ Collatz instance (theorem 8.41) and the $\mathbb{G}(3,1)$ RH instance (theorem 8.42). The remaining open question is structural in character (does the gauge bundle trivialize?), independent of the proofs, and admits both possibilities consistent with the framework.

# A Compressed Golden-Tetrad Proofs

For reference we record the finite algebraic identities used by the prototype.

## A.1 Defect

$$D(\rho) = 0 \iff \rho^{-1} + \rho^{-2} = 1 \iff \rho^2 - \rho - 1 = 0 \iff \rho = \varphi.$$

## A.2 Signature

For coupling matrix

$$\begin{pmatrix} 0 & a & a & a \\ a & 0 & b & b \\ a & b & 0 & b \\ a & b & b & 0 \end{pmatrix},$$

the eigenvalues are

$$b \pm \sqrt{b^2 + 3a^2}, \qquad -b, \qquad -b.$$

With $a = \varphi^{-1}$ and $b = -\varphi^{-2}$, the signature is $(3, 1)$.

## A.3 Birth and throat

$$B_k = \varphi^{-1} w_0 + \varphi^{-2} w_k, \qquad B_k^2 = 2\varphi^{-4}.$$

The child overlap has $\cos\theta = \varphi^{-2}$, hence

$$r_{\text{throat}} = \sin(\theta/2) = \sqrt{\frac{1 - \varphi^{-2}}{2}} = \frac{1}{\sqrt{2\varphi}}.$$

The recursive scale is

$$\sqrt{2}\,\varphi^{-2} \cdot \frac{1}{\sqrt{2\varphi}} = \varphi^{-5/2}.$$

# B Test Suite Summary

The accompanying test suite lives in `echo_interference/`. At the time this manuscript was drafted, the suite contained 365 tests. The test modules were:

```
test_echo_decay.py
test_two_return_normal_form.py
test_explicit_formula_dictionary.py
test_echo_intertwine.py
test_echo_leakage.py
test_throat_aperture.py
test_dead_frequency.py
test_multi_bounce.py
test_formal_invariants.py
test_kernel_balance.py
test_defect_energy.py
test_acoustic_spacetime.py
test_grade_embedding.py
test_hodge_two_return.py
test_arithmetic_emergence.py
test_echo_propagator.py
test_mode_independence.py
test_independent_predictions.py
test_pole_shift.py
test_sphere_involution.py
test_curvature_bivector.py
test_dual_substrate.py
test_l_function_channels.py
```

The Collatz extension (section 8) is verified by an additional seventeen test modules in the companion repository https://github.com/ktynski/apollonian-wave (`apollonian-wave/tests/`):

```
test_cl11_algebra.py
test_collatz_involution.py
test_witness_trace.py
test_collatz_orbit_formula.py
test_collatz_cycle_obstruction.py
test_collatz_grace_surplus.py
test_collatz_transfer_operator.py
test_collatz_braid_trace.py
test_ghost_braid_trace.py
test_information_capacity.py
test_witness_acoustic_trace.py
test_merkaba_fano_module.py
test_supercoiling_lemma.py
test_acoustic_braid_action.py
test_collatz_lift_functor.py
test_canonical_exact_cap.py
test_logos_section.py
```

These verify the $\mathbb{G}(1,1)$ multiplication table, the witness involution and trace, the exact orbit formula, the cycle obstruction (exhaustively for primitive words up to length 6 with $a_i \leq 6$), the grace-surplus drift constant $2 - \log_2 3 \approx 0.415$, the transfer operator's principal eigenvalue 4/3, the structural substrate of the Collatz braid, the unbounded behaviour of ghost braids in $\mathbb{G}(1,1)$, the information-capacity bound $m = O(\log n)$ on orbit length, the spherical-harmonic positivity of the acoustic witness trace, the Merkaba/Fano module structure ($|F_7| = 7$, seven Fano lines, $|A_M| = 6$, $\dim V_W^{\text{disc}} = 15$), the Supercoiling Lemma's constructive separation property ($\widehat{\text{Tr}}_W(B\psi_0) > 0$ iff $B$ caps the center, across 5,000 random braids), the three explicit lifts $\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2$ from Cl(1,1) Syracuse braids into $V_W^{\text{disc}}$ (including $\mathcal{F}_2$'s exact-cap-at-termination property), and the canonical-exact-cap section $\mathcal{F}^\dagger$'s nine-locus structural fingerprint and gluing relation, with empirical existence-iff-termination across all odd $n_0 \leq 9999$. Together they cover:

1. the golden coupling matrix;
2. reciprocal null two-return normal form;
3. explicit-formula echo dictionary clauses;
4. grade attenuation;
5. parity intertwining;
6. reciprocal-data leakage;
7. throat transfer;
8. modular reduction at the dead frequency 11;
9. the difference between single-bounce divergence and multi-bounce echo contraction;
10. formal echo invariants: abstract grade iteration, off-center leakage, and Apollonian energy summability;
11. explicit-formula kernel balance: quartet leakage, critical-line amplitude balance, and aggregate prime imbalance;
12. defect energy and detuning stability around the golden scale;
13. number-theoretic acoustic spacetime: grade emergence from null generators, null-cone condition as critical-line test, wave equation as echo closure, Hodge decomposition as register decomposition, Minkowski signature from echo coupling, dual symmetry as echo reciprocity, grade attenuation matching geometric-algebra grades, propagation speed symmetry, and the full derivation chain from echo axioms through pole-shift contradiction to $\beta = \frac{1}{2}$;
14. pseudoscalar grade embedding: the pseudoscalar is at definite integer grade $n$, the off-center component $\Im(\rho(1-\rho))$ is the pseudoscalar coefficient, contraction by $\varphi^{-n}$ per cycle applies exactly when the pseudoscalar is nonzero;
15. Hodge two-return structure: a grade-1 field has exactly two non-harmonic Hodge potentials, their echo path lengths are $\{1, 2\}$, these match the two-return normal form, the Hodge budget at $\varphi$ is exactly 1, and the explicit formula's register grades match the Hodge decomposition;
16. arithmetic emergence: echo iteration produces a free monoid isomorphic to $(\mathbb{N}_0, +)$, grade–iteration interaction produces multiplication, irreducible echo counts are primes, unique factorization holds, the Euler product matches the Dirichlet series, and the full emergence chain composes;
17. echo propagator: the Laplacian eigenvalue $\rho(1-\rho)$ decomposes into grade-0 and grade-4 components, the propagator preserves the scalar and contracts the pseudoscalar by $\varphi^{-4}$, known zeros have zero persistence residual, hypothetical off-center zeros have nonzero residual, and a $\beta$-scan confirms that only $\beta = \frac{1}{2}$ gives zero residual;
18. mode independence: poles at distinct zeros are linearly independent, the identity plus independence forces each mode to be an individual fixed point of $\mathcal{E}_\varphi$, and the fixed-point condition implies $\beta = \frac{1}{2}$;
19. independent predictions: the echo cavity length, Weyl-law spectral density, zero counting function, mean spacing, and residual-vs-spacing comparison all match classical results without importing them;

20. pole-shift map: propagated eigenvalue preserves the real part and contracts the imaginary part by $\varphi^{-4}$, inversion roundtrips to machine precision, the pole is unmoved iff $\beta = \frac{1}{2}$, and the propagated-identity residue vanishes on center and equals $-1$ off center;

21. sphere involution: the functional-equation involution is its own inverse, the eigenvalue is involution-invariant, the propagator commutes with the involution, the branch locus is the critical line, the propagator contracts $\mathrm{Im}(\lambda)$ toward zero at rate $\varphi^{-4}$, the lifted map is regular, and the branch locus is the unique fixed set;

22. curvature bivector: the pseudoscalar field $I_x = \gamma(1 - 2\beta)$ vanishes on the critical line, its gradient is nonzero for all nontrivial zeros, the shape operator vanishes iff $\beta = \frac{1}{2}$, the curvature bivector norm equals the persistence residual, the null substrate is flat, and off-center zeros violate flatness;

23. dual substrates and chiral fold: cohesive coupling $+\frac{1}{2}$ and repulsive coupling $-\frac{1}{2}$ sum to zero, pseudoscalar orientations are opposite in $\mathbb{G}(1,3)$ and $\mathbb{G}(3,1)$, the chiral fold lies at $\beta = \frac{1}{2}$ where the dual-substrate discrepancy $2\,s_4$ vanishes, the fold is the unique midpoint between the coupling signs, 6 bivector channels from $\binom{4}{2}$, 3 boosts + 3 rotations, the Euler product factored by channel equals the full product, all channels are populated, and the channel assignment is deterministic; mod-24 units $\{1, 5, 7, 11, 13, 17, 19, 23\}$ form 8 elements, the negation involution $a \mapsto -a \bmod 24$ gives 4 orbits $\{1, 23\}, \{5, 19\}, \{7, 17\}, \{11, 13\}$, adding the 2 ramified primes gives 6 channels matching the 6 bivectors; paired residues share channels; ramified primes map to boost channels;

24. L-function channels: the Ramanujan tau function is computed from the Fourier expansion of $\Delta(\tau) = q\prod(1 - q^n)^{24}$ with all standard values matching; Hecke recursion $\tau(p^2) = \tau(p)^2 - p^{11}$ and multiplicativity hold; Wilton's congruence $\tau(p) \equiv 1 + p \pmod{24}$ holds for all unramified primes $p < 500$; $\tau(p) \bmod 24$ takes exactly the predicted residue sets $\{0\}, \{12\}, \{0, 2\}, \{6, 20\}, \{8, 18\}, \{12, 14\}$ per channel with no leakage; ramified primes $2, 3$ break Wilton with $\tau(2) \equiv 0$, $\tau(3) \equiv 12$; the eight Dirichlet characters mod 24 are all multiplicative and quadratic, distribute over conductors $\{1, 3, 4, 8, 12, 24\}$ as $\{1, 1, 1, 2, 1, 2\}$, and split four-even / four-odd; the new `channel_for_prime` mapping is consistent with `prime_to_channel_mod24`;

25. Cymatic runtime (Wilton probe): the live 3D simulation `cymatic_particles.html` now drives every channel envelope by the Hecke L-factor of $\Delta$, records $\tau(p) \bmod 24$ for every prime cycle count it observes, and asserts the residue lands in the predicted set; a 1500-frame sandbox run with 6 000 particles yielded $\sim 7 \times 10^4$ prime emergences across $\geq 50$ distinct unramified primes with 0 Wilton violations; the runtime test suite verifies $\tau(n)$ for $n \leq 23$, Hecke recursion, Wilton on $p \in [5, 100)$, Deligne's bound $|a_p| \leq 2$, the $(Z/24Z)^* \to$ 4 negation orbits $\to$ 6 channels collapse, and the $D = 26$ critical-dimension bookkeeping; the structural check adds 18 regex invariants pinning the Hecke wiring, the Wilton probe, the 24-strand UI panel, the quantitative Mertens residual, and the RH fixed-point error gauge.

The suite is a finite consistency check of the echo model, not a proof of the analytic theorem.

# References


[1] B. Riemann, “Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse,” *Monatsberichte der Berliner Akademie*, 1859.

[2] H. M. Edwards, *Riemann’s Zeta Function*, Academic Press, 1974.

[3] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, 1986.

[4] D. Hestenes, *Space-Time Algebra*, Gordon and Breach, 1966.

[5] G. Sobczyk, “Unipotents, idempotents, and a spinor basis for matrices,” *Advances in Applied Clifford Algebras*, 2004.

[6] L. Burns, T. Daniel, S. Alexander, and J. Dressel, “Spacetime geometry of acoustics and electromagnetism,” *Quantum Studies: Mathematics and Foundations*, vol. 11, pp. 27–67, 2024.

[7] G. Sobczyk, *Mappings of Surfaces in Euclidean Space Using Geometric Algebra*, Ph.D. thesis, Arizona State University, 1971.

[8] J. Bourgain, A. Gamburd, and P. Sarnak, “Affine linear sieve, expanders, and sum-product,” *Inventiones Mathematicae*, vol. 179, no. 3, pp. 559–644, 2010.

[9] G. Sobczyk and K. Tynski, “The Golden Tetrad: Clifford Algebra from Null Generators with Golden-Ratio Coupling,” companion manuscript, 2026.

[10] D. H. Mayer, “Thermodynamic formalism and spectral analysis of Selberg’s zeta function for $\mathrm{PSL}(2,\mathbb{Z})$,” *Bull. Amer. Math. Soc.* **25** (1991), no. 1, 55–60.

[11] M. Pollicott and R. Sharp, “Exponential error terms for growth functions on negatively curved surfaces,” *Amer. J. Math.* **120** (1998), no. 5, 1019–1042.

[12] A. Kontorovich and H. Oh, “Apollonian circle packings and closed horospheres on hyperbolic 3-manifolds,” *J. Amer. Math. Soc.* **24** (2011), no. 3, 603–648.

[13] E. Roosendaal, “On the $3x+1$ problem,” computational records, `http://www.ericr.nl/wondrous/`, accessed 2026.

[14] A. Kontorovich and S. Miller, “Benford’s law, values of $L$-functions and the $3x+1$ problem,” *Acta Arithmetica* **120** (2005), 269–297.

[15] J. C. Lagarias, “The $3x+1$ problem and its generalizations,” *American Mathematical Monthly* **92** (1985), no. 1, 3–23.

[16] T. Tao, “Almost all orbits of the Collatz map attain almost bounded values,” *Forum of Mathematics, Pi* **10** (2022), e12.